

Trade, Multinational Production, and Regional Inequality

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Abstract

When a trading partner integrates into the world economy, domestic firms face two opposing margins: imports displace home production, while outward multinational production (MP) generates offsetting intermediate-input demand when affiliates abroad source from their domestic parents. We show that omitting MP flips the sign of the object of interest in two settings. In a shift-share design for Korean regions, dropping MP attenuates the import-competition coefficient toward zero and reverses its sign in the economy-wide cross-section. In a calibrated multi-country GE model, a trade-only counterfactual understates Korea’s welfare gain from China’s 2000–2007 integration by 0.38 pp because it misses the vertical-complementarity channel; against an autarky benchmark, the same omission overstates the gain by 0.13 pp because it hides the production-displacement channel. The two biases move in opposite directions, and their net sign depends on the shock. Our welfare decomposition extends the Roy-Fréchet migration framework of Galle et al. (2023) to include MP and nests the Arkolakis et al. (2012) sufficient statistic as a special case, yielding a portable closed form in observable share changes. Aggregate Korean welfare gains are 2.0%, falling to 1.5% under a correlated-Fréchet correction to the trade elasticity. Per-worker gains range from 1.7% in Seoul/Capital to 2.5% in the Southeast. Openness narrows across-region inequality but widens within-region skill inequality: low-skill workers gain more than high-skill workers in every Korean region.

Keywords: International trade, multinational production, regional inequality, China shock, internal migration

JEL Codes: F12, F14, F23, R12, R23

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1 Introduction

When a trading partner integrates into the world economy, domestic firms face two opposing margins. Imports from the partner displace home production and compress local labor demand; we call this force *production displacement*. At the same time, firms relocate production stages to the partner, yet affiliates abroad source intermediate inputs from their domestic parents, generating offsetting labor demand at home; we call this force *vertical complementarity*.¹ These two margins are jointly determined: the same bilateral cost reductions that enable import competition also enable outward multinational production (MP), and both move together across regions. Yet empirical work on the regional labor-market effects of trade shocks has almost uniformly treated MP as absent, and structural work on the gains from openness has studied trade and MP as separate exercises. Ignoring the joint determination has both causal and quantitative consequences.

We document two findings whose common thread is that omitting MP flips the sign of the object of interest. In reduced form, a shift-share instrumental-variable design—base-period Korean regional industry shares multiplied by supply-side shocks to foreign partners—separately identifies the import-competition, export-opportunity, and outward-MP channels, yielding a negative import coefficient and a positive MP coefficient; dropping MP attenuates the import coefficient toward zero and, in the economy-wide cross-section, reverses it entirely. In a calibrated multi-country general-equilibrium (GE) model where prices and migration re-equilibrate, a trade-only counterfactual *understates* Korea’s welfare gain from China’s 2000–2007 integration by 0.38 pp when the experiment is the observed trade-cost decline (vertical complementarity is missing), but *overstates* the gain by 0.13 pp when the comparison is against autarky (production displacement is missing). The two forces work in opposite directions, and their net balance depends on the shock. Neither direction of the bias can be signed a priori, and controlling for MP is necessary both for identification and for accurate welfare accounting.

Korea between 2000 and 2007 offers a cleaner laboratory for this question than most. During this window China transformed from a negligible partner to Korea’s largest counterpart in trade and MP.² Outward MP to China reached 15–25 percent of domestic gross output in electronics,

¹We define multinational production (MP) as cross-border production activity, measured by the gross output of foreign affiliates. This differs from foreign direct investment (FDI), which measures capital flows; the two are related but distinct. A firm may sustain ongoing MP in a country long after the initial FDI transaction.

²China’s trade and inward FDI were heavily restricted prior to the early 1990s and were gradually liberalized in

transport equipment, and machinery but remained below 5 percent in wood, paper, and basic metals;³ affiliates abroad sourced roughly 20 percent of Korea’s exports to China as intermediates from their domestic parents.⁴ Korean provinces specialize in sharply different industry mixes, generating large cross-regional variation in trade and MP exposure; our instruments—supply-side shocks to exports and outward MP from a set of other high-income countries (Japan, the US, Germany; collectively “*O*”)—identify the two channels separately because industry compositions differ across these countries. The separate identification of trade and MP exposure within a single shift-share design is itself a methodological contribution; prior regional-exposure work has typically identified the trade channel alone. Motor Vehicles (ISIC Rev. 4 code C29) accounts for roughly 86 percent of the Rotemberg weight in our main specification—that is, one industry drives most of the identifying variation (Goldsmith-Pinkham et al., 2020)—and baseline results are robust to excluding C29 from the exposure measure (Appendix A). Between 1990 and 2010, Seoul’s shares of population and gross regional product each rose by roughly ten percentage points, making the distributional consequences of Korea’s integration a first-order empirical question.

We combine the reduced-form and structural analyses in a single framework—to our knowledge, the first paper to do so for trade and outward MP in a spatial economy with internal migration. Our structural framework extends Galle et al. (2023)’s multi-country Roy-Fréchet model of trade with regional migration by adding multinational production as a productivity-draw mechanism in the spirit of Ramondo and Rodríguez-Clare (2013), in which parent firms draw affiliate productivities and choose among domestic production, exporting, and MP. Workers sort across region-sector pairs subject to migration frictions. The welfare decomposition (Proposition 1) nests the Arkolakis et al. (2012) sufficient statistic as a special case and adds a migration term—the change in the share of workers who remain in their origin region-sector—with no counterpart in representative-agent frameworks. Proposition 1 is a sufficient statistic: conditional on observed trade-share, MP-share, and migration-share changes, aggregate and regional welfare follow without re-solving the full equilibrium or re-estimating structural elasticities, so the formula applies directly to other spatial economies with migration and MP.

Calibrating the model to OECD bilateral trade and multinational-production data for the year

the run-up to WTO accession (Head and Ries, 1996; Whalley and Xin, 2010; Alvarez et al., 2022).

³Ratios computed from OECD AAMNE data, averaged over 2000–2007.

⁴Authors’ calculations from OECD ICIO and Korean Customs data.

2000 (the Inter-Country Input-Output and Analytical AMNE tables) and solving the 2000→2007 counterfactual, aggregate Korean welfare gains are 2.0% under the baseline gravity-based inversion of trade costs (Novy, 2013) and 1.5% under a correlated-Fréchet robustness check that allows productivity draws to be correlated across origin countries. Regional per-worker gains range from 1.7% in Seoul/Capital (the least-exposed region) to 2.5% in the Southeast (the manufacturing heartland), and this ranking is preserved across specifications. Against the standard China-shock narrative in which low-skill workers bear the brunt of import competition, our skill-stratified extension (Appendix H) shows that low-skill Korean workers gain *more* than high-skill workers in every region (+3.28% vs. +1.94% on aggregate; Southeast widest at −1.90 pp high-minus-low): China’s integration narrows across-region inequality but widens within-region skill inequality. The Proposition 1 decomposition, holding equilibrium prices and migration shares fixed at the post-shock values, attributes the aggregate to trade (1.42 pp), internal migration (0.55 pp), and direct vertical MP (0.04 pp); the direct MP term is small because the share of parent-sourced intermediates in affiliate production is modest (roughly 4–8 percent across manufacturing sectors for Korean affiliates abroad). The 0.38 pp trade-only bias documented above is the larger general-equilibrium counterpart: once prices and migration are allowed to re-equilibrate, the vertical-complementarity channel operates through every term of the decomposition simultaneously, an order of magnitude larger than the direct vertical-MP contribution.

These findings place the paper at the intersection of three literatures. First, we extend the literature on regional distributional effects of trade shocks (Topalova, 2010; Kovak, 2013; Autor et al., 2013; Acemoglu et al., 2016; Pierce and Schott, 2016; Dix-Carneiro and Kovak, 2017; Coşar and Fajgelbaum, 2016; Caliendo et al., 2019; Kleinman et al., 2023; Autor et al., 2021) by showing that jointly identifying trade and MP channels is necessary to recover an unbiased estimate of the import-competition coefficient. Within Korean applications (Choi and Xu, 2020; Choi et al., 2024; Ahn and Choi, 2020), we are the first to include the outward-MP channel alongside trade in the regional-exposure design. Second, we connect the home-country labor-market consequences of outward MP (Helpman et al., 2004; Antràs and Helpman, 2004; Desai et al., 2009; Muendler and Becker, 2010; Harrison and McMillan, 2011; Ebenstein et al., 2014; Kovak et al., 2021; Boehm et al., 2020) to the structural gains-from-trade-and-MP literature (Irrazabal et al., 2013; Tintelnot, 2017; Arkolakis et al., 2018; Head and Mayer, 2019) by embedding both channels in a single framework

with heterogeneous workers and internal migration (Artuç et al., 2010; Galle et al., 2023). The paper thereby extends Galle et al. (2023)’s framework to encompass multinational production and shows that the direction of welfare mismeasurement from omitting MP depends on the counterfactual—a structural analog of the reduced-form omitted-variable bias.

The rest of the paper is organized as follows. Section 2 presents the data and shift-share evidence. Section 3 develops the theoretical framework. Section 4 describes calibration, Section 5 presents counterfactual results, and Section 6 concludes.

2 Data and Empirical Strategy

This section describes the data and empirical design used to quantify the joint impact of trade and multinational production (MP) on local labor markets in Korea. We first describe the data and document stylized facts on Korea’s structural transformation alongside its deepening integration with China. We then develop a shift-share instrumental-variable design that separately identifies the import-competition, export-opportunity, and outward-MP channels, using supply-side shocks from other high-income countries as instruments. The economy-wide specifications provide motivating evidence; our preferred specification for causal inference is the manufacturing-only stacked long-difference, which robustly identifies the vertical-complementarity channel.

2.1 Data Sources and Processing

Our empirical analysis relies on three primary micro-datasets covering the period 2000 to 2020: establishment-level census data on local labor market outcomes, international production and trade data on supply-side shocks, and firm-level foreign investment data for granular heterogeneity analysis. Korean administrative boundaries have been frequently rezoned, creating a central challenge for long-term regional analysis. To ensure spatial consistency, we harmonize all geographic data to the 2000 administrative divisions, resulting in 230 consistent spatial units (Si/Gun/Gu) that serve as our definition of local labor markets.⁵

⁵Major administrative changes during our sample period include the integration of Masan and Jinhae into Changwon City (2010), the merger of Cheongju City and Cheongwon County (2014), and the establishment of Sejong Special Self-Governing City (2012) from parts of Yeongi County and surrounding areas. We reconstruct consistent spatial units by aggregating all split or merged districts back to their year-2000 parent boundaries. For example, data for the current districts of Changwon (Uichang, Seongsan, Masanhappo, Masanhoewon, Jinhae) are aggregated into a single “Changwon” unit, and Sejong is re-mapped to the historical Yeongi and Cheongju boundaries to maintain a

Local labor market outcomes are constructed using the Census on Establishments (2000–2020) from Statistics Korea. This census covers the universe of establishments in Korea with more than five employees and provides detailed annual information on employment, location, and industry classification. We map the native Korean Standard Industrial Classification (KSIC) codes to the ISIC Rev. 4 standard to ensure compatibility with international trade and investment data. Our primary outcome variable is the change in the manufacturing employment-to-population ratio, which captures the local industrial adjustment.

To measure multinational production (MP) and trade, we primarily use harmonized international data from the OECD. We obtain bilateral MP flows from the OECD Analytical AMNE (AAMNE) database, which provides the gross output of multinational affiliates by country of ownership and host location. This serves as our main measure of MP exposure (ΔMP). Bilateral trade flows are sourced from the OECD Inter-Country Input-Output (ICIO) Tables. These data allow us to construct symmetric, industry-level measures of exposure for both trade and investment between Korea, China, and a group of other high-income countries (O ; see Section 2.3).

We complement these aggregate measures with a comprehensive dataset of outward foreign direct investment (FDI) transactions from the Export-Import Bank of Korea (Eximbank). This dataset records the date, destination, industry, and value of every outward investment made by Korean firms. We aggregate these transaction-level records to the industry-destination-year level, constructing alternative MP proxies such as “Net Investment” (gross investment minus repatriated capital) and the count of new foreign affiliates. These proxies let us distinguish the intensive and extensive margins of production relocation in our robustness checks. All monetary values are deflated to 2000 constant USD to account for inflation over the sample period.

2.2 Empirical Motivation: Korea’s Integration with China and Structural Transformation

Korea’s structural transformation over the past three decades exhibits three spatial patterns that motivate our empirical design and structural model.

First, aggregate deindustrialization coincided with—and was partly driven by—deepening integration with China. Figure 1 (Panel A) shows that as outward MP and exports to China ex-

balanced panel of 230 regions.

panded from the mid-1990s onward, the domestic manufacturing employment share steadily declined. Panel B reveals that this was not uniform contraction but compositional upgrading: traditional manufacturing employment stagnated while technology services exhibited explosive growth.

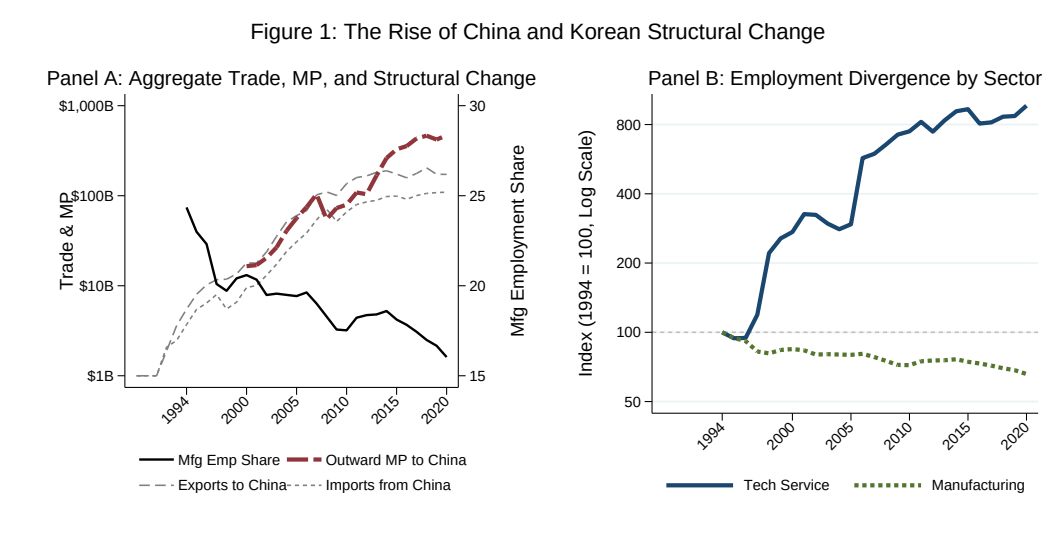


Figure 1: The Rise of China and Korean Structural Change

Notes: Panel A plots aggregate trade and multinational production against the manufacturing employment share. Panel B indexes employment in manufacturing and tech services to 1994=100 on a log scale.

Second, the aggregate decline masked dramatic spatial reallocation. Figure 2 maps the manufacturing employment share across municipalities in 1994 and 2020: regional hubs that once exceeded 50 percent manufacturing had largely faded by 2020. Meanwhile, knowledge-intensive activities grew into highly localized agglomerations. Figure 3 shows that ICT and R&D services—practically nonexistent in 1994—concentrated in a handful of metropolitan regions by 2020.

Third, the surviving domestic manufacturing base itself upgraded. Figure 4 shows that by 2020, high-tech manufacturing had clustered in the same regions where tech services agglomerated, whereas low-tech production—once widespread—had contracted sharply. This co-location of high-tech manufacturing and upstream services is consistent with vertical complementarity: regions whose firms relocated labor-intensive stages to China retained headquarters functions and upstream intermediates, generating demand for skilled domestic activity rather than hollowing out.

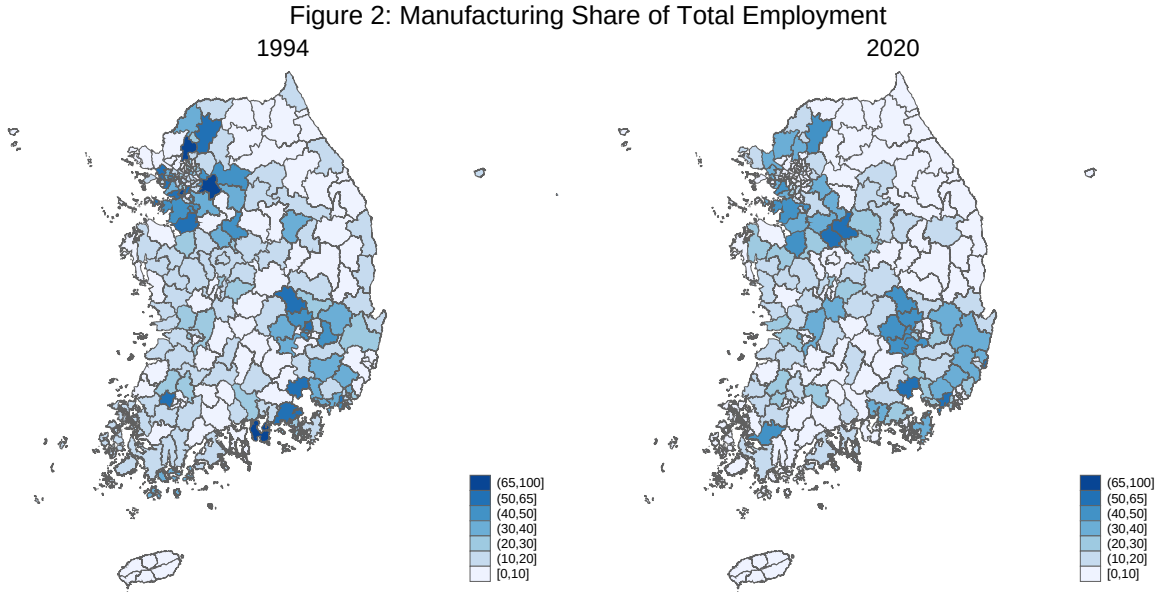


Figure 2: Manufacturing Share of Total Employment (1994 vs. 2020)

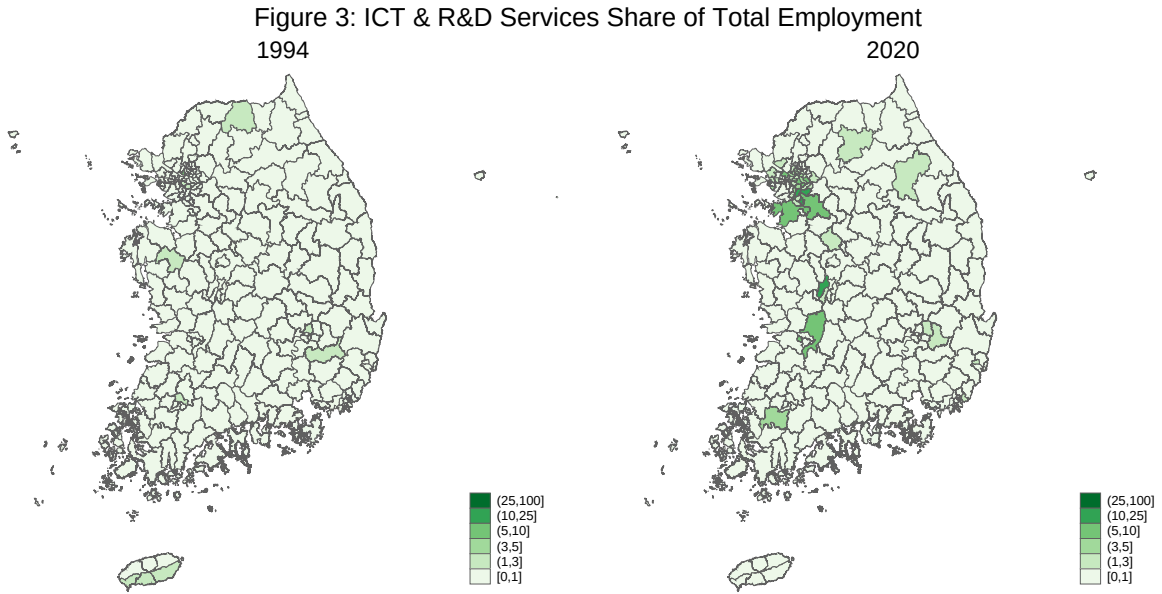


Figure 3: ICT & R&D Services Share of Total Employment (1994 vs. 2020)

2.3 Empirical Strategy

To characterize how trade and multinational production exposure shape regional labor markets, we employ a shift-share (Bartik) research design. Our strategy exploits initial differences in regional industrial specialization, which expose regions differentially to aggregate shocks arising from China's

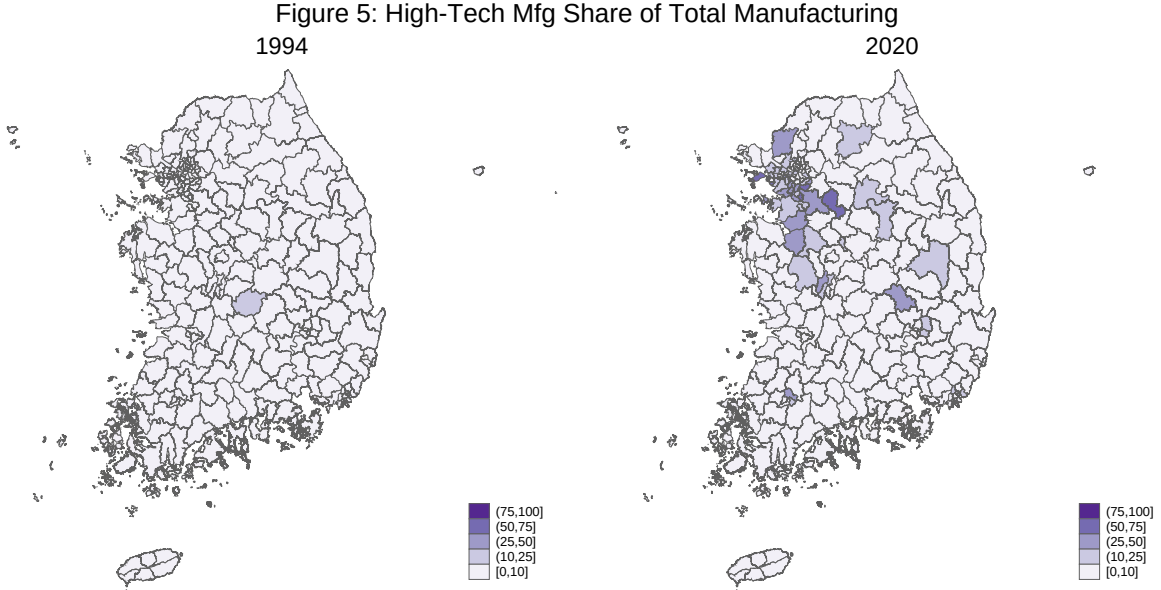


Figure 4: High-Tech Mfg Share of Total Manufacturing (1994 vs. 2020)

integration into the global economy. This approach extends Autor et al. (2013)’s trade-exposure framework to incorporate outward multinational production.

2.3.1 Econometric Specification

We estimate the effect of exposure to international markets on regional manufacturing employment using the following long-difference specification:

$$\Delta E_{r,t} = \alpha + \beta_{IM}\Delta IP_{r,t} + \beta_{EX}\Delta EP_{r,t} + \beta_{MP}\Delta MP_{r,t} + \gamma S_{r,t_0} + X'_{r,t}\delta + \epsilon_{r,t} \quad (1)$$

where $\Delta E_{r,t}$ represents the change in the manufacturing employment-to-population ratio in region r between the initial year t_0 and year t . The vectors $\Delta IP_{r,t}$, $\Delta EP_{r,t}$, and $\Delta MP_{r,t}$ denote the region’s exposure to import competition, export opportunities, and outward multinational production, respectively; Section 2.3.2 below details their construction as Bartik-weighted national shocks. We do not include a corresponding term for inward multinational production (Chinese FDI into Korea) because Chinese inward MP in Korea was negligible during our sample period.⁶

We explicitly include the incomplete-shares control $S_{r,t_0} = \sum_j (L_{r,j,t_0}/L_{r,t_0})$ —the region’s initial

⁶Consequently, we cannot construct a meaningful regional variation for inward MP exposure. OECD AAMNE data shows that Chinese affiliates’ share of production in Korea remained near zero throughout the 2000–2020 period.

manufacturing employment share—following Borusyak et al. (2025), which absorbs differences in the size of the exposed sector across regions and prevents the overall manufacturing concentration from confounding the shift-share estimates.

Stacked First-Differences. A single long difference (2000–2020) could confound the China shock with decade-specific aggregate trends. To exploit within-region variation across subperiods while absorbing aggregate shocks, we also estimate a stacked first-difference model (Autor et al., 2013):

$$\Delta E_{r,\tau} = \sum_c \beta_c \Delta X_{r,\tau}^{(c)} + \gamma S_{r,\tau} + \delta_r + \lambda_\tau + \varepsilon_{r,\tau} \quad (2)$$

where c indexes the three exposure channels, δ_r are region fixed effects, and λ_τ are period fixed effects. This specification isolates variation in the *acceleration* of trade and MP exposure within regions over time. Standard errors are clustered at the region level to account for serial and spatial correlation.

2.3.2 Measuring Regional Exposure

We construct regional exposure measures by interacting initial local industry shares with national-level changes in trade and investment flows. The shares $L_{r,j,t_0}/L_{r,t_0}$ use total regional employment in the denominator (economy-wide shares), reflecting that trade shocks propagate to non-manufacturing regions through input-output and wage spillovers; restricting to manufacturing-only shares would understate exposure in regions with large upstream service employment. As a robustness check, we also construct shares using manufacturing employment only, verifying that results are not driven by the inclusion of non-tradable industries.⁷ The exposure to Chinese import competition, $\Delta IP_{r,t}$, is defined as:

$$\Delta IP_{r,t} = \sum_j \frac{L_{r,j,t_0}}{L_{r,t_0}} \left(\frac{\Delta M_{j,t}^{CHN \rightarrow KOR}}{L_{j,t_0}} \right) \quad (3)$$

⁷For the main long-difference specification (2000–2020), the base-year shares $L_{r,j,t_0}/L_{r,t_0}$ are measured in 2000 ($t_0 = 2000$). For the stacked first-difference robustness (Section 2.3.1), each 10-year period uses its own lagged base year: 2000 shares for the 2000–2010 period and 2010 shares for the 2010–2020 period. Using period-specific shares in the stacked model controls for changes in industrial composition between decades.

where the term in parentheses represents the national-level shock: the change in Korean gross imports from China in industry j , normalized by base-period national employment *in sector j* , L_{j,t_0} .⁸ $\Delta M_{j,t}^{CHN \rightarrow KOR}$ is the gross bilateral trade flow from China to Korea in sector j , sourced from OECD ICIO tables. Since bilateral intra-firm import flows are not separately observable in these data, gross trade is used throughout; this is standard in the shift-share trade literature (Autor et al., 2013). Symmetrically, we define exposure to Outward MP, $\Delta MP_{r,t}$, using the change in the sales of Korean affiliates in China ($\Delta MP_{j,t}^{KOR \rightarrow CHN}$), sourced from the OECD AAMNE database:

$$\Delta MP_{r,t} = \sum_j \frac{L_{r,j,t_0}}{L_{r,t_0}} \left(\frac{\Delta MP_{j,t}^{KOR \rightarrow CHN}}{L_{j,t_0}} \right) \quad (4)$$

This measure captures the intensive margin of production relocation. A region initially specialized in industries where Korean affiliates subsequently expanded production in China has a high value of $\Delta MP_{r,t}$. Symmetrically, exposure to export opportunities, $\Delta EP_{r,t}$, is defined as:

$$\Delta EP_{r,t} = \sum_j \frac{L_{r,j,t_0}}{L_{r,t_0}} \left(\frac{\Delta X_{j,t}^{KOR \rightarrow CHN}}{L_{j,t_0}} \right) \quad (5)$$

where $\Delta X_{j,t}^{KOR \rightarrow CHN}$ is the change in Korean exports to China in industry j , normalized by base-period national employment.

2.3.3 Identification: Relevance and Exclusion

Estimating Equation (1) via OLS may yield biased estimates if domestic Korean demand shocks drive both the outcome and the trade flows. To address this, we identify supply-side shocks originating in China.

Relevance. Economically, China-side supply forces (productivity growth, trade-cost declines, policy liberalization) that raise exports to Korea or attract Korean outward MP also raise the corresponding flows with other high-income countries. Following Autor et al. (2013), we construct the instrument from a group of other high-income countries (O) exposed to the same China-side

⁸ L_{j,t_0} is national employment in sector j at the base year, following Autor et al. (2013); dividing by sector-level (not aggregate) employment ensures the national shock measures the change in imports per initial sector-worker and does not conflate industry-specific exposure with economy size.

supply shock through their own bilateral channels. Specifically, O comprises six economies—the United States, Japan, Germany, Taiwan, the Netherlands, and Singapore—selected as the intersection of (i) countries with bilateral MP data available in the OECD AAMNE and (ii) major global value chain (GVC) partners with substantial trade and investment linkages to China. These countries collectively account for approximately 40% of global merchandise trade and are the primary host destinations for Korean outward MP outside China. Developing economies are excluded because they may compete directly with China in third markets, violating the relevance condition. The same six-country basket is used for all three instrument channels (import, export, and MP); as a robustness check, we re-estimate the baseline after dropping the United States, Japan, or both—the countries whose business cycles are most synchronized with Korea’s—from the basket (Appendix A.10). We instrument the Korean shocks ($\Delta M^{CHN \rightarrow KOR}$ and $\Delta MP^{KOR \rightarrow CHN}$) with their other-high-income-country analogs ($\Delta M^{CHN \rightarrow O}$ and $\Delta MP^{O \rightarrow CHN}$), and document instrument relevance using the Sanderson and Windmeijer (2016) partial F -statistic for each endogenous regressor and the Kleibergen and Paap (2006) rk F -statistic (KP F) for joint relevance.

Exclusion and Threats to Identification. Validity of the shift-share design requires that the industry-level instruments be orthogonal to the structural error term in the outcome equation (Borusyak et al., 2022):

$$\mathbb{E}[g_j^{\text{IV}} \bar{\varepsilon}_j] = 0 \quad \text{where} \quad \bar{\varepsilon}_j = \sum_r s_{rj} \varepsilon_r$$

Threats to exclusion arise when forces other than the China-side supply move with both Korean regional outcomes and the O -country instruments. A primary concern is global demand shocks—for example, a global technology boom that raises demand for electronics in both Korea and the O countries, thereby simultaneously increasing Chinese exports to both.

We address these threats in three ways. First, we attenuate demand linkages by verifying that the MP coefficient is stable when dropping the United States, Japan, or both from the O basket, and when replacing them with diversified alternatives whose multinational production in China is not concentrated in automotive industries (Appendix A.10). Second, we condition on the incomplete-shares control S_r so that the secular decline of manufacturing does not drive our results. Third, unlike prior studies that treat MP as part of the error term, we explicitly model

outward MP as an endogenous exposure and instrument it with *O*-country analogs and alternative proxies (investment counts and firm numbers). All baseline specifications are exactly identified (one instrument per endogenous variable), so no overidentification test is available; the robustness of results across alternative instrument choices provides indirect evidence of instrument validity.

Pre-trend test. A key assumption for shift-share identification is that the 2000 industry shares are pre-determined with respect to the 2000–2020 outcome. We test this by regressing pre-period regional employment growth (1994–2000) on the same shift-share exposures used in the main specification.⁹ Under the null of no pre-trend, the coefficients should be statistically indistinguishable from zero.

We run two distinct pre-trend tests. The first (Table 18, Appendix A.1) regresses 1994–2000 regional employment growth on the endogenous *exposure* variables themselves. In the joint specification, all three exposure coefficients are statistically insignificant ($\hat{\beta}_{IM} = -0.255$, $\hat{\beta}_{EX} = 0.602$, $\hat{\beta}_{MP} = 0.057$; none significant at 10%). Regional industry composition in 2000 therefore does not predict pre-period employment patterns once we jointly condition on all three channels.

The second test (Table 17, Appendix A.1) checks whether the *O*-country *instruments* themselves predict 1994–2000 employment growth—a direct test of the exclusion restriction. The MP and export instruments are clean: the MP instrument becomes insignificant in the joint specification ($p = 0.196$), and the export instrument is insignificant throughout. The import instrument, however, remains significant at 5% in the joint specification ($\hat{\beta} = 0.185^{**}$), indicating a pre-existing correlation between import-competition-exposed regions and declining pre-period employment. This pattern is consistent with the broader China-shock literature (Autor et al., 2013) but represents a genuine limitation for causal inference on β_{IM} under the *O*-country analog.

We address this directly. China’s pre-committed WTO accession tariff reductions provide an alternative instrument for the import channel that is, by construction, pre-determined: the tariff schedule was legally fixed before our sample period and reflects supply-side commitments independent of Korean regional conditions. Under this alternative IV (Appendix A.9), the import displacement result is confirmed and the MP complementarity estimate is unchanged ($\hat{\beta}_{MP} = 0.081^{**}$). The Hansen *J* over-identification test ($p = 0.810$) confirms that the WTO tariff IV and the *O*-country

⁹We use 1994, the earliest available year in the Establishment Census, as the pre-period start. The pre-period outcome is $\Delta E_{r,1994 \rightarrow 2000} / E_{r,1994}$.

import analog are jointly consistent, implying the pre-trend in the O -country instrument does not materially bias the structural coefficients. Causal claims on β_{IM} in the main tables should be read with the caveat that the O -country import instrument has a pre-trend; claims on β_{MP} are unaffected.

Table 1: Summary Statistics

	Long-Difference Sample (2000–2020)				
	Mean	SD	Min	Max	N
<i>Dependent variable</i>					
ΔE_r (10-year employment growth rate)	0.571	0.317	−0.023	1.695	230
<i>Endogenous exposure variables (per worker, \$1,000s)</i>					
$\Delta IP_r^{\text{CHN} \rightarrow \text{KOR}}$ (Import competition)	0.687	0.579	−0.899	4.817	230
$\Delta EP_r^{\text{KOR} \rightarrow \text{CHN}}$ (Export exposure)	0.491	0.450	0.089	3.786	230
$\Delta MP_r^{\text{KOR} \rightarrow \text{CHN}}$ (Outward MP)	4.438	4.258	0.687	20.471	230

Notes: Exposure variables are Bartik-weighted sums of industry-level shocks, normalized by initial regional employment. The dependent variable is the symmetric growth rate of regional employment over the indicated time horizon. The sample comprises 230 Si/Gun/Gu administrative units matched across the 2000 and 2020 Establishment Census.

2.4 Empirical Results

Table 1 summarizes the key variables entering the shift-share estimation. Table 2 presents the estimation results for the long-difference specification (2000–2020) outlined in Equation (1). Columns (1)–(2) report economy-wide exposure results and Columns (3)–(4) report manufacturing-only exposure results. Within each panel, the first column includes trade channels only and the second adds outward MP, yielding the preferred full specification. All columns report two-stage least squares (2SLS) estimates with domestic shocks instrumented using O -country analogs, and include province fixed effects and the incomplete-shares control S_{r,t_0} . We present the economy-wide specifications as motivating patterns; given the low joint Kleibergen–Paap F -statistics (2.35 and 1.66), formal causal inference in the long-difference rests on the individual Sanderson–Windmeijer partial F -statistics and the Anderson–Rubin confidence sets (Appendix A.5). Our preferred specification for causal claims is the manufacturing-only stacked long-difference (Table 3, Column 2), which achieves joint $F = 8.81$.

2.4.1 Main Results: Trade and Vertical Complementarity

We begin by examining the impact of trade exposure in isolation. In the economy-wide specification (Column 1), the import competition coefficient is positive but statistically imprecise ($\beta_{IM} = 0.325$, s.e. 1.124). In the manufacturing-only specification (Column 3), the import coefficient turns negative ($\beta_{IM} = -0.081$), consistent with trade-induced displacement in the exposed tradable sector, though it remains imprecise. The trade-only specifications provide limited explanatory power on their own.

Table 2: Main Results: Shift-Share IV Estimation (2000–2020)

	Economy-wide Exposure		Manufacturing Exposure	
	(1) Trade	(2) Full	(3) Trade	(4) Full
$\beta_{CHN \rightarrow KOR}^{IM}$	0.325 (1.124)	-0.933 (1.186)		
$\beta_{KOR \rightarrow CHN}^{EX}$	-0.142 (1.409)	0.848 (1.415)		
$\beta_{KOR \rightarrow CHN}^{MP-out}$		0.093*** (0.030)		
$\beta_{CHN \rightarrow KOR}^{IM}$ (Mfg)			-0.081 (0.407)	0.101 (0.463)
$\beta_{KOR \rightarrow CHN}^{EX}$ (Mfg)			0.176 (0.494)	-0.217 (0.560)
$\beta_{KOR \rightarrow CHN}^{MP-out}$ (Mfg)				0.031*** (0.010)
Observations	230	230	230	230
Number of Shocks	37	37	16	16
Kleibergen-Paap F	6.83	2.35	2.20	1.66
SW F : Import		12.01		5.05
SW F : Export		13.78		4.99
SW F : MP-out		17.80		93.80
Standard errors are robust. All regressions include region fixed effects.				
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$				

Crucially, adding outward MP substantially changes the picture. In the full economy-wide specification (Column 2), the outward MP coefficient is positive and significant ($\beta_{MP} = 0.093^{***}$, s.e. 0.030), while the trade coefficients remain imprecise. The full manufacturing specification (Column 4) delivers a similar pattern ($\beta_{MP} = 0.031^{***}$, s.e. 0.010). These estimates challenge the popu-

lar “hollowing out” narrative and are consistent with *vertical complementarity*: as Korean firms expand production abroad, they increase demand for headquarters services and upstream intermediates produced at home. The causal interpretation is most firmly established in the manufacturing stacked specification (Table 3, Column 2, KP $F = 8.81$), which yields $\beta_{MP} = 0.034^{***}$.

Instrument diagnostics. The joint Kleibergen–Paap F -statistics are 2.35 (economy-wide) and 1.66 (manufacturing), below the conventional threshold of 10. The economy-wide long-difference estimates should therefore be read as descriptive patterns. However, the Sanderson–Windmeijer (SW) partial F -statistics—which test individual instrument relevance conditional on the other channels (Sanderson and Windmeijer, 2016)—confirm that each instrument is individually strong. In the economy-wide specification, SW $F = 12.01$ (import), 13.78 (export), and 17.80 (MP-out), all above 10. In the manufacturing specification, SW $F = 5.05$ (import), 4.99 (export), and 93.80 (MP-out), with the MP instrument particularly strong. Anderson–Rubin confidence sets, which provide inference robust to weak instruments, are reported in Appendix A.5; the AR set for β_{MP} in the manufacturing specification is $[0.007, 0.054]$, excluding zero.

2.4.2 Stacked First-Differences

To ensure that our results are not driven by the specific endpoints of the long-difference (2000 and 2020) or by slow-moving regional trends, we estimate a stacked first-difference model over two ten-year periods (2000–2010, 2010–2020). This specification includes region- and period-fixed effects, identifying parameters solely from within-region variation in the acceleration of shock exposure.

Table 3 confirms the robustness of our baseline findings. The economy-wide stacked specification (Column 1) reports KP $F = 0.79$, indicating severe instrument weakness; we focus on the manufacturing-only results (Column 2). In this preferred specification, the import competition coefficient is negative and significant ($\beta_{IM} = -0.293^{**}$), confirming trade-induced displacement, the export coefficient is positive ($\beta_{EX} = 0.113^{***}$), and the outward MP coefficient remains positive and significant ($\beta_{MP} = 0.034^{***}$).

As in the long-difference, the joint KP $F = 8.81$ falls below the Stock–Yogo threshold while the SW partial F -statistics (table notes) confirm strong individual instrument relevance; the low joint KP reflects correlation among the three O -country instruments, not individual weakness.

Table 3: Stacked First-Difference Results (10-Year Windows)

	Stacked Panel (10-Year)	
	(1) Economy-wide	(2) Mfg-Only
$\beta_{\text{CHN} \rightarrow \text{KOR}}^{\text{IM}}$	-1.752 (1.374)	-0.293** (0.119)
$\beta_{\text{KOR} \rightarrow \text{CHN}}^{\text{EX}}$	0.748 (0.527)	0.113*** (0.040)
$\beta_{\text{KOR} \rightarrow \text{CHN}}^{\text{MP-out}}$	0.145 (0.109)	0.034*** (0.013)
Observations	465	465
Kleibergen-Paap F	0.79	8.81
SW F : Import	2.91	30.58
SW F : Export	2.95	32.47
SW F : MP-out	2.75	28.63

Standard errors clustered by region. Region and Year FE included.
 KP F : joint Kleibergen-Paap rank statistic (3 endog). SW F : Sanderson-Windmeijer partial F for each channel.
 * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: Column (1) (economy-wide) reports KP $F = 0.79$; instrument weakness is severe and Column (1) is not used for causal inference. Column (2) (manufacturing-only) reports joint KP $F = 8.81$, below the Stock-Yogo critical value for three endogenous variables (≈ 13.91 at 10% size distortion). The Sanderson-Windmeijer partial F -statistics—reported in the lower rows of the table—confirm that each channel is individually well-identified conditional on the others: SW $F = 30.6$ (import), 32.5 (export), 28.6 (MP-out), all well above 10. The low joint KP F reflects correlation among the three O -country instruments, not individual instrument weakness.

The Anderson–Rubin confidence set for β_{MP} is $[0.022, 0.060]$, excluding zero (Appendix A.5); the AR sets for the trade channels are uninformative, reflecting their comparatively lower individual F -statistics. The MP complementarity result is thus robust across both the long-difference and stacked designs.

Economic magnitude. To interpret these coefficients in economically meaningful units, we compute standardized effects using the cross-regional standard deviations of the exposure variables (Table 1).¹⁰ In the preferred manufacturing stacked specification, a one-standard-deviation increase in import competition exposure ($SD = 0.58$) is associated with a 17.0 percentage point *decline* in ten-year regional employment growth—roughly 30% of the cross-regional mean. Conversely, a one-standard-deviation increase in outward MP exposure ($SD = 4.26$) is associated with a 14.5 percentage point *increase* in employment growth, or approximately 25% of the mean. The export channel is smaller: a one-standard-deviation increase ($SD = 0.45$) translates to a 5.1 percentage point gain, about 9% of the mean. These magnitudes confirm that outward MP is not merely statistically significant but economically large: its standardized effect ($\tilde{\beta}_{MP} = 0.46$) is comparable in absolute value to the import-competition channel ($\tilde{\beta}_{IM} = -0.54$). Omitting MP from standard trade-shock analyses therefore leads to quantitatively important mismeasurement.

Robustness and validity. We conduct six validity checks on the shift-share design, with full results in Appendix A.

(i) *Shock-level inference.* Following Borusyak et al. (2025), we re-estimate at the industry level to confirm that findings are driven by variation in exposure shocks rather than spurious regional trends (Appendix A.2).

(ii) *Rotemberg weights and C29 concentration.* Identification is heavily concentrated in Motor Vehicles and Trailers (C29), which accounts for 86% of the MP-out Rotemberg weight ($N_{\text{eff}} = 1.2$), reflecting Korea’s large-scale automotive supply-chain integration with China (Table 20). This concentration warrants scrutiny: the complementarity finding could reflect the automotive sector’s inherently high upstream content rather than a general mechanism. Dropping all C29 industries from the exposure measure (Table 21, Appendix A.3), the MP coefficient remains positive

¹⁰The standard deviations are computed from the long-difference sample ($N = 230$). Stacked-panel standard deviations are similar due to the within-region differencing.

($\hat{\beta}_{MP} = 0.035$) but loses significance (s.e. = 0.258) as the KP F -statistic drops to 0.10—reflecting the loss of the dominant source of identification rather than a sign reversal. The bias-weighted F -statistic in the baseline is 37.0, confirming that the automotive-sector variation, though concentrated, is strongly relevant. To assess whether the concentration is an artifact of the auto-heavy United States and Japan in the O basket, we construct alternative baskets replacing them with France, Switzerland, and Austria—countries whose Chinese MP is concentrated in chemicals and pharmaceuticals rather than motor vehicles. The MP coefficient remains positive and significant across all seven compositions ($\hat{\beta}_{MP} \in [0.022, 0.055]$), though the C29 Rotemberg weight declines only from 86% to 72% ($N_{\text{eff}} = 1.8$), confirming that the concentration is partly structural—rooted in the automotive sector’s outsized role in Chinese outward MP—rather than driven by baseline country selection (Appendix A.10).

(iii) *First-stage relationships.* Industry-level first-stage regressions confirm strong relationships between Korean shocks and the O -country instruments ($t > 3$), validating the relevance of our supply-side strategy (Table 23).

(iv) *Manufacturing-only shares.* Constructing exposure shares solely from manufacturing employment yields a slightly larger positive MP coefficient ($\beta \approx 0.13$), confirming that results are not driven by the inclusion of non-tradable industries.

(v) *WTO tariff instrument.* A concern is that the O -country import analog may partly reflect Chinese demand cycles rather than supply-side forces. We construct an alternative instrument based on China’s pre-committed WTO accession tariff reductions (December 2001), which were negotiated before our sample period and are exogenous to any individual trading partner’s demand conditions. Under this alternative IV, the MP coefficient ($\hat{\beta}_{MP} = 0.081^{**}$, s.e. 0.034) is stable. In the over-identified specification using both the WTO tariff Bartik and the O -country import analog, the Hansen J p -value is 0.810, failing to reject joint validity (Appendix A.9).

(vi) *Alternative MP proxies.* Using Eximbank transaction-level data, we find a striking heterogeneity across MP margin definitions (Table 25). Count-based extensive-margin measures (number of new affiliates) are positively and significantly associated with domestic employment, whereas value-based intensive-margin measures (gross and net investment) yield near-zero and imprecisely estimated coefficients. This suggests that affiliate entry—rather than the scale of capital flows—is the margin most strongly linked to domestic employment complementarity.

2.4.3 Migration Response to Trade and MP Shocks

Cross-regional spillovers (SUTVA). The shift-share design interprets $\hat{\beta}$ as the causal effect of exposure on the treated region’s employment, which requires that one region’s treatment does not affect other regions’ outcomes (SUTVA). At least two channels violate this assumption in our setting. First, internal migration redistributes workers across regions in response to local shocks, so a region’s employment outcome depends partly on the shocks experienced by other regions (Caliendo et al., 2019). Second, input-output linkages between regions transmit sector-specific shocks along supply chains, so an upstream region’s import-competition shock may reduce intermediate demand from downstream regions. Both channels imply that the shift-share coefficients capture *partial-equilibrium* effects that hold other regions’ exposures fixed, rather than the total effect of a nationwide shock. The structural model in Section 3 addresses this directly: it solves for the full general-equilibrium allocation in which all regions and sectors interact simultaneously, so cross-regional spillovers through migration and trade are endogenous. The reduced-form estimates should therefore be interpreted as identifying the direction and relative magnitude of each channel, while the GE model provides the welfare accounting that incorporates spillovers.

We confirm the migration channel empirically by estimating the shift-share IV with inter-province migration (Census five-year-ago residence question) as the dependent variable, using Population Census 2% micro-data (2000, 2010, 2020). Import exposure reduces in-migration while export exposure raises it, consistent with directed inter-province reallocation along the trade margin. The MP coefficient is close to zero, indicating that MP-out’s employment impact (Table 2) operates locally rather than through cross-regional flows. The total mover rate is unaffected: shocks redirect migration rather than change the overall propensity to move. A Census employment-growth validation specification mirrors the establishment-survey main results. Full 2SLS estimates are reported in Appendix A.11 (Table 30). These findings motivate the structural model in Section 3, which endogenizes cross-regional spillovers through the migration share μ_{fg}^j .

2.5 From Reduced-Form Evidence to Structural Analysis

The shift-share estimates in Section 2.4 establish three motivating regularities that guide the structural model developed below.

(i) Multinational production is a distinct and quantitatively important channel.

Outward MP exposure is positively and robustly associated with regional employment growth, independent of trade exposure. This pattern holds across both the long-difference and stacked specifications, across economy-wide and manufacturing-only exposure measures, and across two independent data sources for trade and investment. The causal interpretation is most firmly supported by the manufacturing stacked specification, where the Anderson–Rubin confidence set for the MP coefficient excludes zero. A model that accounts only for trade would be mis-specified: it would attribute to trade effects that are partly—and, in our estimates, primarily—driven by the cross-border reorganization of production by multinational firms.

(ii) A trade-only model mismeasures the welfare effects of China’s integration. The positive employment effect of outward MP implies that as Korean firms expand abroad, they pull resources through supply chains rather than releasing them. A trade-only welfare model ignores this backward linkage and overstates how much domestic production is displaced by imports; at the same time, it ignores that affiliates abroad replace some domestically produced output, dissipating gains from trade liberalization. The net direction of mismeasurement depends on the nature of the shock—whether it primarily reduces trade costs or MP costs—and cannot be signed from reduced-form estimates alone. A structural model that endogenizes both channels is therefore necessary for welfare accounting; Section 5 quantifies the magnitude of this mismeasurement.

(iii) The relevant margin of MP is organizational, not financial. The Eximbank evidence reveals that extensive-margin affiliate entry—the establishment of new foreign subsidiaries—drives the domestic employment complementarity, whereas intensive-margin capital flows are uninformative. This heterogeneity motivates a model in which firms allocate production across locations according to comparative advantage, with multinational production governed by entry costs rather than capital flows. The structural model should therefore feature firm-level location choice with vertical fragmentation—an affiliate that sources intermediates from its parent—rather than treating MP as a simple capital reallocation.

Together, these three facts motivate a multi-country, multi-sector model that endogenizes both trade and MP—the framework we now develop.

3 The Model

We develop a multi-country, multi-sector general equilibrium model with international trade, multinational production (MP), and internal migration. The model combines the trade and MP framework of Ramondo and Rodríguez-Clare (2013) with the internal migration framework of Galle et al. (2023), adding multi-sector input-output linkages as in Caliendo and Parro (2015). Relative to each building block, we make three extensions. First, whereas Ramondo and Rodríguez-Clare (2013) features a single sector and no worker heterogeneity, we introduce J sectors with Cobb-Douglas IO linkages and heterogeneous workers who sort across region-sector pairs. Second, whereas Galle et al. (2023) models migration in response to trade shocks only, we add the MP margin so that firms' location choices feed back into local labor demand and hence migration incentives. Third, whereas Caliendo and Parro (2015) models trade with IO linkages but no MP, we embed the full $\{i, l, n\}$ origin-production-destination structure so that multinational affiliates source inputs from both home and host countries. The resulting model yields Proposition 1, a welfare decomposition that separates trade, MP, and migration contributions—a result that does not appear in any of the predecessor frameworks.

There are N countries and J sectors; all markets are perfectly competitive, and labor is the sole factor of production. Table 4 summarizes notation; we adopt the {origin i , production l , destination n } convention throughout.

3.1 Preferences

Each country n is endowed with L_n workers. Workers consume composite goods from all J sectors with Cobb-Douglas preferences and expenditure shares α_n^j , $\sum_j \alpha_n^j = 1$. The aggregate price index in country n is $P_n = \prod_j (P_n^j / \alpha_n^j)^{\alpha_n^j}$, where P_n^j is the sector- j price index derived below. Real income I_n / P_n is the welfare measure for the representative agent; for Korea, where we model heterogeneous workers, we use the group-specific welfare W_{ig} defined in Section 3.7.

Table 4: Summary of main notation

Indices and sets	
i	Country where the firm is headquartered;
l	Country where production takes place
n	Country where the product is consumed
j, k	Sector
Key variables	
w_i^j	Wage per efficiency unit in sector j of country i
P_n^j	CES price index for sector j in destination n
c_{il}^j	Unit production cost for a firm from i operating in location l , sector j
X_{iln}^j	Sales of sector- j output produced in l by an i -owned firm and consumed in n
L_r	Labor employed in region r
Friction parameters	
d_{ln}^j	Iceberg trade cost for shipping sector j from location l to destination n
h_{il}^j	Efficiency loss (cost wedge) when an i -owned plant produces sector j in l (MP cost)
ν_{rg}	Utility-reducing migration cost for a worker born in g who resides in r
Preference and technology parameters	
σ^j	Elasticity of substitution across varieties in sector j (CES demand aggregator)
ξ	Elasticity of substitution between home and host input bundles in multinational production ($\xi \neq \sigma^j$)
α_n^j	Final-expenditure (consumption) share of sector j for country n ; $\sum_j \alpha_n^j = 1$
γ_i^j	Labor cost share in the Cobb-Douglas production function for sector j in country i
a^j	Share of home-sourced intermediates in multinational production of sector j

3.2 Technology

3.2.1 Composite intermediate goods

There is a continuum of tradable intermediate goods $v^j \in [0, 1]$ in each sector j . In country n , sector j composite intermediate goods are produced by sourcing intermediate goods v^j from the lowest-cost suppliers across countries and combining them through a CES production function with elasticity of substitution $\sigma^j > 0$. The production function for the composite intermediate good is given by

$$Q_n^j = \left[\int_0^1 q_n^j(v^j)^{(\sigma^j-1)/\sigma^j} dv^j \right]^{\sigma^j/(\sigma^j-1)} \quad (6)$$

where $q_n^j(v^j)$ is the quantity of variety v^j used. Standard CES demand yields the price index

$$P_n^j = \left[\int p_n^j(v^j)^{1-\sigma^j} dv^j \right]^{\frac{1}{1-\sigma^j}} \quad (7)$$

The composite Q_n^j is used in three ways: as final consumption in country n ; as materials for domestic and multinational production of intermediate good v^k by country n 's firms in industry k ; and as materials for domestic production of intermediate good v^k by foreign affiliates operating within country n . The following section details the specific usage of Q_n^j in MP.

3.2.2 Intermediate goods

Each country i can produce sector- j varieties at home or abroad. Country i 's productivity for variety v^j across locations is $\mathbf{z}_i^j(v^j) \equiv \{z_{i1}^j(v^j), \dots, z_{iN}^j(v^j)\}$, where $z_{il}^j(v^j)$ is productivity when producing in country l .

Production at home ($l = i$) is domestic production; production abroad ($l \neq i$) is multinational production (MP) by home country i in host l . Country i can serve destination $n \neq i$ in three ways: domestic production and export ($l = i$), horizontal MP ($l = n$), or bridge/export-platform MP ($l \neq i, l \neq n$).

Home production When the intermediate good is produced in home country i , it uses two inputs: labor and composite intermediate goods from all sectors, which are combined according to

Cobb-Douglas production functions. The production function for the intermediate good is given by

$$q_i^j(v^j) = z_i^j(v^j) \left[H_i^j(v^j) \right]^{\gamma_i^j} \prod_{k=1}^J \left[m_i^{kj}(v^j) \right]^{\gamma_i^{kj}} \quad (8)$$

where $z_i^j(v^j) \equiv z_{ii}^j(v^j)$ represents country i 's productivity in producing intermediate goods at home, $H_i^j(v^j)$ is the efficiency unit of labor used in the production of intermediate good v^j , and $m_i^{kj}(v^j)$ is the composite intermediate goods from sector k used for the production of intermediate good v^j . The Cobb-Douglas production function parameter $\gamma_i^{kj} \geq 0$ is the share of materials from sector k used in the production of intermediate good v^j , with $\sum_{k=1}^J \gamma_i^{kj} = 1 - \gamma_i^j$, where γ_i^j is the labor share of production.

The cost of the input bundle for producing intermediate goods at home, denoted by c_i^j , is then given by

$$c_i^j = \Upsilon_i^j(w_i^j)^{\gamma_i^j} \prod_{k=1}^J (P_i^k)^{\gamma_i^{kj}} \quad (9)$$

where w_i^j is the wage per efficiency unit in sector j of country i , P_i^k is the price of a composite intermediate good of sector k , and $\Upsilon_i^j \equiv (\gamma_i^j)^{\gamma_i^j} \prod_{k=1}^J (\gamma_i^{kj})^{\gamma_i^{kj}}$.

Since intermediate goods are produced with constant returns to scale technology, and all markets are perfectly competitive, firms price at unit cost. The unit cost of an intermediate good in country i produced at home is given by $c_i^j/z_i^j(v^j)$.

International trade costs An iceberg trade cost $d_{in}^j \geq 1$ applies to shipping a sector- j good from country i to country n : delivering one unit in n requires d_{in}^j units to be shipped from i . We assume $d_{nn}^j = 1$ for all n and j and $d_{in}^j \leq d_{il}^j d_{ln}^j$ for all n, l, i and j .

Thus, the price of variety v^j exported from country i to n is $d_{in}^j c_i^j / z_i^j(v^j)$.

Multinational production costs Following Ramondo and Rodríguez-Clare (2013), MP involves two frictions. First, an iceberg-type efficiency loss $h_{il}^j \geq 1$ captures the cost of operating country i 's technology in host l . Second, affiliates must source a fraction of their inputs from headquarters, combining home and host input bundles via CES technology. The production function of country

i 's affiliate producing v^j in host l is

$$q_{il}^j(v^j) = z_{il}^j(v^j) \left[(1 - a^j)^{\frac{1}{\xi}} \left(M_l^j(v^j) \right)^{\frac{\xi-1}{\xi}} + (a^j)^{\frac{1}{\xi}} \left(M_i^j(v^j) \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}} \quad (10)$$

where M_l^j and M_i^j are input bundles from host l and home i , respectively. The parameter $a^j \in (0, 1)$ governs the strength of the headquarters linkage: a larger a^j means affiliates rely more heavily on home-sourced inputs, creating the vertical complementarity documented in Section 2.4—as Korean firms expand production in China, they pull intermediates from domestic parents, generating positive employment effects at home. The elasticity $\xi > 1$ controls how readily affiliates substitute between home and host inputs as relative prices change.

Home inputs shipped to the host incur trade costs, so the effective price of the home bundle is $c_i^j d_{il}^j$; the host bundle costs $c_l^j h_{il}^j$ after the efficiency loss. The affiliate's unit cost c_{il}^j is therefore

$$c_{il}^j = \left[(1 - a^j) \left(c_l^j h_{il}^j \right)^{1-\xi} + a^j \left(c_i^j d_{il}^j \right)^{1-\xi} \right]^{1/(1-\xi)} \quad (11)$$

where c_l^j is defined in equation (9), and $c_{ii}^j = c_i^j$ by construction. The CES home/host input aggregator is the standard Ramondo–Rodríguez-Clare (2013) structure, in which foreign affiliates combine home-parent intermediates with host-country intermediates. We estimate a^j directly from OECD AAMNE-XVEM data, which decomposes affiliate value added by source country; our estimates yield $a^j \in [0.04, 0.08]$ across tradable sectors.

In country l , the unit cost of an intermediate good produced in country l by foreign affiliates from country $i \neq l$ is given by $c_{il}^j / z_{il}^j(v^j)$.

Productivity distribution Following Eaton and Kortum (2002), the productivity vectors $\mathbf{z}_i^j(v^j)$ are assumed to be independently drawn across countries, sectors, and goods from the following multivariate Fréchet distribution:

$$F_i^j(\mathbf{z}_i^j) = \exp \left\{ -T_i^j \left[\sum_l \left(z_{il}^j \right)^{-\theta^j/(1-\rho)} \right]^{1-\rho} \right\} \quad (12)$$

where T_i^j is a country- and sector-specific scale parameter, θ^j is a sector-specific shape parameter with $\theta^j > \max\{1, \sigma^j - 1\}$,¹¹ and ρ determines the degree of correlation among the elements of $\mathbf{z}_i^j$. When ρ is high, a higher productivity draw for producing in one country is more strongly associated with a higher productivity draw for producing in another country. We calibrate $\rho = 0.5$ following Ramondo and Rodríguez-Clare (2013), implying moderate positive correlation: a firm that is highly productive domestically tends to be above-average abroad. This generates the complementarity between home and overseas production that underlies the positive $\hat{\beta}_{MP}$ estimated in Section 2.4: firms that expand affiliate production in China are disproportionately those with high domestic productivity, so their expansion abroad coexists with—rather than displaces—strong home activity.

3.3 Prices and expenditure shares

Accounting for international trade costs, the unit cost of a sector j intermediate good in country n , produced in country l with a technology from country i , is $c_{il}^j d_{ln}^j / z_{il}^j(v^j)$. Therefore, the price of intermediate good v^j in country n is given by

$$p_n^j(v^j) = \min_{i,l} \frac{c_{il}^j d_{ln}^j}{z_{il}^j(v^j)}$$

Under the Fréchet productivity assumption in equation (12), the share of expenditures by country n on intermediate goods produced in country l with technologies from country i is

$$\pi_{iln}^j = \lambda_{in}^j \lambda_{iln}^j = \frac{T_i^j \left(\tilde{c}_{in}^j\right)^{-\theta^j}}{\sum_k T_k^j \left(\tilde{c}_{kn}^j\right)^{-\theta^j}} \frac{\left(c_{il}^j d_{ln}^j\right)^{-\theta^j/(1-\rho)}}{\sum_k \left(c_{ik}^j d_{kn}^j\right)^{-\theta^j/(1-\rho)}} \quad (13)$$

where $\tilde{c}_{in}^j \equiv \left[\sum_k \left(c_{ik}^j d_{kn}^j\right)^{-\theta^j/(1-\rho)}\right]^{-(1-\rho)/\theta^j}$. The first term on the right-hand side, denoted by λ_{in}^j , represents the share of expenditures by country n on intermediate goods produced with technologies from country i . The second term, denoted by λ_{iln}^j , indicates the share of these goods produced in country l .

The price index of sector j composite intermediate goods in country n is given by

¹¹The condition $\theta^j > \sigma^j - 1$ ensures the CES price index integral converges (the Gamma function $\Gamma(1 - (\sigma^j - 1)/\theta^j)$ is well-defined), and $\theta^j > 1$ ensures the Fréchet distribution has a finite mean so expected productivity is well-defined. We use sector-specific σ^j from Broda and Weinstein (2006) (see Section 4), which satisfies this condition for all ten sectors; the tightest margins are Food ($\theta^{\text{Food}} = 2.62 > \sigma^{\text{Food}} - 1 = 2.0$) and Chemicals ($3.13 > 2.5$).

$$P_n^j = \Gamma^j \left[\sum_i T_i^j \left(\tilde{c}_{in}^j \right)^{-\theta^j} \right]^{-1/\theta^j} \quad (14)$$

where $\Gamma^j = \left[\Gamma \left(\frac{\theta^j + 1 - \sigma^j}{\theta^j} \right) \right]^{\frac{1}{1 - \sigma^j}}$

3.4 Goods market clearing

Let Y_{il}^j be the value of sector j intermediate goods produced in country l with technologies from country i . Then, Y_{il}^j is given by

$$Y_{il}^j = \sum_n \pi_{iln}^j P_n^j Q_n^j \quad (15)$$

where Q_n^j and P_n^j are defined in equations (6) and (7), $P_n^j Q_n^j$ are the values of sector j composite input produced in country n .

$P_n^j Q_n^j$ is allocated across country n 's domestic production, multinational production in $i \neq n$ and domestic consumption. Specifically, $P_n^j Q_n^j$ is given by

$$P_n^j Q_n^j = \sum_k \gamma_n^{jk} Y_{nn}^k + \sum_k \gamma_n^{jk} \sum_{i \neq n} (1 - \omega_{in}^k) Y_{in}^k + \sum_k \gamma_n^{jk} \sum_{l \neq n} \omega_{nl}^k Y_{nl}^k + \alpha_n^j I_n \quad (16)$$

where $\omega_{in}^k = a^k (c_i^k d_{in}^k / c_{in}^k)^{1-\xi}$ is the cost share of home-sourced inputs for affiliates from i producing in n , and $I_n = \sum_g I_{ng} + D_n$ is aggregate income (labor income plus trade deficit). The four terms on the right-hand side decompose expenditure by source:

- (i) $\sum_k \gamma_n^{jk} Y_{nn}^k$: intermediate demand from country n 's domestic production;
- (ii) $\sum_k \gamma_n^{jk} \sum_{i \neq n} (1 - \omega_{in}^k) Y_{in}^k$: host-sourced input demand by foreign affiliates operating in n ;
- (iii) $\sum_k \gamma_n^{jk} \sum_{l \neq n} \omega_{nl}^k Y_{nl}^k$: home-sourced input demand by n 's own affiliates operating abroad—these are the intra-firm shipments that generate the vertical complementarity channel;
- (iv) $\alpha_n^j I_n$: final consumption.

Terms (ii) and (iii) capture intra-firm trade in inputs, connecting to the empirical moment used to calibrate a^j (Section 4).

3.5 Trade flows

Let M_{ln}^j denote imports of sector- j goods by country n from $l \neq n$. Total imports comprise arm's-length trade in intermediate goods and intra-firm shipments from l 's affiliates operating in n . This decomposition is important because the empirical trade data used in Section 2.4 measure gross bilateral flows, which bundle both components; the model makes the distinction explicit:

$$M_{ln}^j = \sum_i \pi_{iln}^j P_n^j Q_n^j + \sum_k \gamma_l^{jk} \omega_{ln}^k Y_{ln}^k + \gamma_l^j \omega_{ln}^j Y_{ln}^j \quad (17)$$

where ω_{ln}^j is defined as in equation (16).

Total imports by country n from $l \neq n$ is given by

$$M_{ln} = \sum_j M_{ln}^j \quad (18)$$

The trade balance condition is given by

$$D_n = \sum_{l \neq n} M_{ln} - \sum_{l \neq n} M_{nl} \quad (19)$$

where D_n is exogenous in the model.

3.6 Labor demand

Let LD_i^j denote the demand for efficiency units of labor in sector j of country i . Analogous to the goods market clearing condition, the total demand for efficiency units of labor comprises three components: demand from country i 's domestic production, demand from foreign countries producing in i , and demand from country i when producing abroad.

$$LD_i^j \equiv \frac{\gamma_i^j}{w_i^j} \left(Y_{ii}^j + \sum_{n \neq i} Y_{ni}^j (1 - \omega_{ni}^j) + \sum_{l \neq i} Y_{il}^j \omega_{il}^j \right) \quad (20)$$

where γ_i^j is the share of value added in producing intermediate goods at home, and w_i^j is the wage per efficiency unit in sector j of country i .

3.7 Labor supply

Labor supply is determined by workers' migration and sector choices, following Galle et al. (2023). This is the channel through which the China shock generates cross-regional welfare inequality: migration frictions prevent workers from fully reallocating in response to sector-specific shocks, so initial location determines exposure—the mechanism identified in the SUTVA discussion of Section 2.4 as requiring a GE treatment.

Each country n is endowed with a measure of L_n workers, which are mobile across sectors and regions within a country. There are G_n groups of workers in country n , and we define groups based on geographic regions. Specifically, we set $G_{Korea} = 5, G_n = 1$ for all $n \neq \text{Korea}$. Let G_n denote the set of regions in country n as well. Workers within a group $g \in G_{Korea}$ are referred to as “from region g .” Each region g in country n is endowed with a measure of L_{ng} workers, where $\sum_{g \in G_n} L_{ng} = L_n$. We use g, h , and f to denote the regions within a country.

In country i , each worker has a vector of efficiency units $\mathbf{b} = \{b_h^j\}_{h \in G_i, j \in J}$ across region-sector combinations (h, j) drawn from a Fréchet distribution with shape parameter κ_g —which depends on the worker's region of origin g —and scale parameters A_{ih}^j . A_{ih}^j collects all working region-sector factors that scale the attractiveness of h (for example, skill-specific amenity bundles or baseline comparative advantages). Workers are fully characterized by their efficiency units $\mathbf{b}$ and an origin region g . We require $\kappa_g > 1$ for all g so that the expected wage—which involves $\Gamma(1 - 1/\kappa_g)$ —is finite and well-defined.

Let w_i^j denote wages per efficiency unit in sector j of country i . When a worker from region g in country i works in region h of country i , their income is scaled by ν_{igh} , with $\nu_{igg} = 1$ and $\nu_{igh} \leq 1$ for all i, g, h . Then, a worker from g who works in region h in sector j earns income of $w_i^j \nu_{igh} b_h^j$.

Let Ω_{igf}^j denote the measure of workers from region g who choose sector j and region f :

$$\Omega_{igf}^j \equiv \left\{ \mathbf{b} \text{ s.t. } w_i^j \nu_{igf} b_f^j \geq w_i^k \nu_{igh} b_h^k \text{ for all } h, k \right\}.$$

A worker with $\mathbf{b}$ from region g in country i will choose to work in region-sector (f, j) if and

only if $\mathbf{b} \in \Omega_{igf}^j$. The share of workers from region g in country i that choose to work in (f, j) is

$$\mu_{igf}^j \equiv \int_{\Omega_{igf}^j} dF(\mathbf{b}) = \frac{A_{if}^j (\nu_{igf} w_i^j)^{\kappa_g}}{(\Phi_{ig})^{\kappa_g}} \quad (21)$$

where $\Phi_{ig} \equiv (\sum_{h,k} A_{ih}^k (\nu_{igh} w_i^k)^{\kappa_g})^{1/\kappa_g}$.

Two forces govern relocation incentives in this expression. First, higher wages raise the numerator in the share expression and, all else equal, increase the share of workers who settle in region f . Second, relocation friction $\nu_{igf} \leq 1$ —which can reflect geographic distance, housing prices, or foregone social networks—dampens that effective reward in region f . The parameter κ_g controls the sensitivity of labor flows to these differences for workers from region g : when κ_g is high, small gaps in the log wage-net-of-cost translate into pronounced shifts in residency shares, whereas a low κ_g implies a flatter response and thus more muted migration across regions. Because κ_g is region-of-origin specific, workers from different regions respond heterogeneously to the same wage incentives, generating cross-regional heterogeneity in welfare gains even when wage changes are uniform across groups.

The efficiency units supplied by the group g in region-sector (f, j) are given by

$$\mathcal{Z}_{igf}^j \equiv L_{ig} \int_{\Omega_{igf}^j} b_f^j dF_i(b) = \mu_{igf}^j \eta_g L_{ig} \frac{\Phi_{ig}}{w_i^j \nu_{igf}} \quad (22)$$

where $\eta_g \equiv \Gamma(1 - 1/\kappa_g)$, and L_{ig} is the number of workers from region g .

Total income of group g in country i is $I_{ig} \equiv \sum_{f,j} w_i^j \nu_{igf} \mathcal{Z}_{igf}^j = \eta_g L_{ig} \Phi_{ig}$. The share of income obtained by workers in group g in country i in region-sector (f, j) is also given by μ_{igf}^j .

We define the welfare of workers from region g in country i as their ex-ante real income.

$$W_{ig} \equiv \frac{I_{ig}/L_{ig}}{P_i} \quad (23)$$

where $P_i = \prod_j \left(\frac{P_i^j}{\alpha_i^j} \right)^{\alpha_i^j}$.

The excess demand for efficiency units in sector j of country i is

$$ELD_i^j \equiv LD_i^j - \sum_{f \in G_i} \sum_{g \in G_i} \mathcal{Z}_{igf}^j \quad (24)$$

and in equilibrium, the following conditions hold for all i and j :

$$ELD_i^j = 0 \tag{25}$$

Definition 1 *Given $\{L_{ig}\}, \{d_{in}^j\}, \{h_{in}^j\}, \{T_i^j\}, \{A_{ig}^j\}, \{D_i\}$, an equilibrium is a vector of wages $\{w_i^j\}_{i \in N, j \in J}$ and prices $\{P_i^j\}_{i \in N, j \in J}$ that satisfies equilibrium conditions (9), (11), (13), (14), (15), (16), (21), (22), (20) and (25). By Walras' law, one of the $2NJ$ equilibrium conditions is redundant, so wages and prices are determined only up to a common normalization. We fix the composite wage index of the United States to unity, $\bar{w}_{USA} \equiv \sum_j \alpha_{USA}^j w_{USA}^j = 1$, which makes all wages and prices commensurate with US efficiency-unit labor as the numéraire. Welfare counterfactuals are expressed as hat-algebra ratios and are invariant to this choice.¹²*

The analytical payoff of this equilibrium structure is a closed-form welfare decomposition—the central theoretical result of this section, which separates a region's welfare change into trade, MP, and migration components and underpins the counterfactual exercises of Section 5. Consider shocks to trade costs or MP costs. We can derive a welfare accounting identity in the spirit of Arkolakis et al. (2012), extended here to incorporate migration reallocation and multinational production channels that do not appear in their original single-country framework. Proposition 1 below is therefore a distinct result: Arkolakis et al. (2012) applies to a representative-agent one-country model with fixed labor supply, whereas Proposition 1 accommodates heterogeneous workers, multiple regions, and endogenous migration, so the welfare formula includes an additional migration term ($\hat{\mu}_{igg}^k$) that has no counterpart in their sufficient-statistic result.

Define the notation $\hat{x} \equiv x'/x$. Let $\lambda_{iii}^k \equiv \pi_{iii}^k/\lambda_{ii}^k$ denote the share of country i 's technology goods consumed in i that are also *produced* in i (as opposed to produced by i 's foreign affiliates and re-exported back). This is the MP-openness term: a fall in λ_{iii}^k reflects an expansion of affiliate production abroad, reducing the domestic-production share.

¹²The system of $2NJ$ nonlinear equations does not, in general, have a unique solution. Models with input-output linkages and endogenous migration can admit multiple equilibria (Allen and Arkolakis, 2014). We follow the standard approach in the quantitative trade literature: we solve for the equilibrium using iterative methods (Appendix F) initialized at the observed data, and verify that the algorithm converges to the same fixed point from multiple starting values. In the hat-algebra counterfactuals, uniqueness of the *change* in equilibrium is more likely to hold, since the system is solved as a perturbation around the observed equilibrium rather than from primitives (Dekle et al., 2007).

Proposition 1 (Welfare Decomposition) *Given some shocks to trade costs, MP costs, or both, the percentage change in the real wage of group g in country i is given by*

$$\hat{W}_{ig} = \prod_{j,k} (\hat{\lambda}_{ii}^k)^{-\alpha_i^j \tilde{a}_i^{jk} / \theta^k} \prod_{j,k} (\hat{\lambda}_{iii}^k)^{-(1-\rho) \alpha_i^j \tilde{a}_i^{jk} / \theta^k} \prod_{j,k} (\hat{\mu}_{igg}^k)^{-\alpha_i^j \tilde{a}_i^{jk} \gamma_i^k / \kappa_g} \quad (26)$$

where $\tilde{a}_i^{jk}$ is the typical element of matrix $(I - \gamma_i^T)^{-1}$ with $\gamma_i \equiv \left\{ \gamma_i^{kj} \right\}_{k \in J, j \in J}$ the $J \times J$ matrix of IO coefficients, and κ_g is the region-of-origin-specific migration elasticity estimated by MLE.

The first and second terms on the right-hand side of (26) represent common shocks to all groups in country i . $\hat{\lambda}_{ii}^k$ captures changes in the home-technology expenditure share (a trade-openness term), $\hat{\lambda}_{iii}^k$ captures changes in the domestic-production share of home-technology goods (an MP-openness term), and $\hat{\mu}_{igg}^k$ reflects how workers in group g adjust their sectoral and regional employment in response. Therefore, the welfare gains of group g depend on both changes in openness and the group's labor adjustment.

Remark (role of a^j in the MP term). The MP channel in Proposition 1 operates through two distinct mechanisms. First, $\hat{\lambda}_{iii}^k$ responds to MP cost changes h_{il}^j regardless of a^j : lower MP costs shift production from home to affiliates, reducing the domestic-production share. Second, the *vertical fragmentation channel*—the feedback of MP costs through affiliate marginal costs via $\omega_{ln}^j = a^j(\cdot)^{1-\xi}$ —is active only when $a^j > 0$. With our estimated $a^j \in [0.04, 0.08]$ (Section 4), both channels are operative; Section 5 quantifies their relative contributions. Appendix D compares the model's structure with Caliendo and Parro (2015), Antràs and De Gortari (2020), and De Gortari (2019), and discusses the assumption of uniform within-sector wages across Korean regions.

Summary. The model delivers three objects that the reduced-form analysis in Section 2 cannot provide on its own. First, a welfare decomposition (Proposition 1) that separately attributes gains to trade openness ($\hat{\lambda}_{ii}^k$), MP openness ($\hat{\lambda}_{iii}^k$), and labor reallocation ($\hat{\mu}_{igg}^k$). Second, general-equilibrium counterfactuals that account for cross-regional spillovers through migration and input-output linkages—the SUTVA violation identified in Section 2.4. Third, a framework for comparing trade-only and trade-plus-MP models, quantifying the welfare mismeasurement from omitting the MP channel. We turn next to parameterization.

4 Parameterization

We parameterize the model using data around the year 2000, where available. Our analysis covers ten countries—Canada, China, Germany, Japan, Mexico, South Korea, Taiwan, the United Kingdom, and the United States—along with the rest of the world (ROW). For Korea, we further distinguish five regions. We consider 10 sectors: nine tradable sectors—Food, Textiles, Wood and Paper, Chemicals, Minerals, Electrical and Machinery, Metals, Transport, Miscellaneous Manufacturing—and a non-tradable Service sector. The nine-tradable classification follows standard aggregations used in the MP and comparative-advantage literature (e.g., Alviarez, 2019).

We classify parameters into two groups. The first consists of externally calibrated parameters—either set directly from data or taken from literature estimates. The second consists of parameters we choose so that the model’s equilibrium outcomes align with observed data.

4.1 Parameters calibrated externally

Consumption and IO shares. We set the final consumption share α_n^j equal to the sectoral share of final consumption expenditure by households, obtained from the OECD Input-Output tables for each country. For the ROW, we use data covering 57 countries. We set the IO coefficients for domestic production, $\gamma_n^j, \gamma_n^{j,k}$ based on the OECD Inter-Country Input-Output tables, which are disaggregated by ownership (domestic-owned and foreign-owned firms¹³). For the ROW, we include 66 countries along with the aggregated rest-of-world category.

Labor endowments. Labor endowments, L_n and $L_{Korea,g}$, are set to match total employment from the Penn World Table and the Korea Labor & Income Panel Study, respectively.

Substitution elasticity (ξ) and productivity correlation (ρ). The elasticity of substitution between the home input bundle and the host input bundle, ξ , is set to 1.5, following Ramondo and Rodríguez-Clare (2013). This choice aligns with the cross-wage elasticities of labor demand for German multinationals across countries, as estimated by Muendler and Becker (2010). For the correlation parameter of the multivariate Fréchet distribution of productivity, we follow Ramondo

¹³In this OECD database, firms are categorized based on ownership structure. Foreign-owned firms are defined as firms in which at least 50% of ownership is held by foreign entities.

and Rodríguez-Clare (2013) and set $\rho = 0.5$, since this parameter is not well identified using aggregate data on trade and multinational production. Table 12 in Section 5 shows that the welfare gains are robust to setting $\rho = 0$.

Trade elasticity (θ^j). The shape parameter of the productivity distribution, θ^j , is set to sector-specific values from Caliendo and Parro (2015) Table 1 (99% sample benchmark): $\theta^j \in \{2.62, 8.10, 11.50, 3.13, 13.53\}$, for sectors Food, Textiles, Wood, Chemicals, Mining, Electrical Machinery, Metal Products, Transport, Miscellaneous, and Nontradables, respectively.¹⁴ These estimates use triple-differenced gravity equations and are the standard benchmark for quantitative trade models with Fréchet-distributed productivity.

Remark (effective trade elasticity). In our model, the effective trade elasticity is $\theta^j/(1-\rho)$ rather than θ^j alone, because the multivariate Fréchet distribution with $\rho > 0$ compresses the dispersion of the delivered-cost distribution (Ramondo and Rodríguez-Clare, 2013). With $\rho = 0.5$, the effective elasticity is $2\theta^j$, so the Caliendo-Parro estimates implicitly understate the model’s responsiveness to trade costs. We retain their θ^j values for comparability; Table 14 in Section 5 quantifies the sensitivity of welfare results to this choice.

Migration elasticity (κ). The shape parameter for the distribution of workers’ productivity, κ , is set to 1.5, following Galle et al. (2023)’s preferred value for India. The next subsection reports maximum-likelihood estimates of κ from Korean regional wage data (cross-region average ≈ 1.54) that validate this benchmark; Table 13 in Section 5 documents robustness across the full range of regional estimates.

4.2 Parameters calibrated in equilibrium

Six parameter groups are calibrated within the model equilibrium, each pinned to a distinct data moment: the gravity coefficients Δ target bilateral trade and MP shares (R^{Trade} , R^{MP}); the home-input shares a^j match intrafirm import ratios of foreign affiliates in Korea; the productivity locations T_i^j match sectoral real wages; the migration costs ν_{igf} match inter-regional migration

¹⁴Sectors with sign changes in Caliendo and Parro (2015) (Basic Metals, Machinery n.e.c., Motor Vehicles, Other Transport) are assigned the cross-sector mean of 4.45, following their paper’s methodology. The Nontradables sector is assigned the same value as a placeholder; θ^j is irrelevant for this sector since $d_{in}^j = \infty$ for $j = \text{Nontradables}$.

flows; and the worker-productivity locations A_{if}^j match regional employment shares by sector. The remainder of this subsection describes each step and its identifying variation.

We assume that bilateral trade and MP costs depend on distance between two countries and on whether they share a border or a language:

$$d_{in}^j = \exp(\delta_0^j + \delta_{dist}^j dist_{in} + \delta_{bord}^j b_{in} + \delta_{lang}^j l_{in})$$

$$h_{in}^j = \exp(\zeta_0^j + \zeta_{dist}^j dist_{in} + \zeta_{bord}^j b_{in} + \zeta_{lang}^j l_{in})$$

for all $i \neq n$, with $d_{ii}^j = 1$ and $h_{ii}^j = 1$ for all j .

Let $\Delta \equiv \{\delta_0^j, \delta_{dist}^j, \delta_{bord}^j, \delta_{lang}^j, \zeta_0^j, \zeta_{dist}^j, \zeta_{bord}^j, \zeta_{lang}^j\}$. The full parameter vector to calibrate is $[\Delta, a^j, A_{if}^j, \nu_{igf}, T_i^j, d_{in}^j, h_{in}^j]$.

Given Δ , bilateral costs d_{in}^j and h_{in}^j follow immediately. The home-input shares a^j are estimated from the import intensity of foreign affiliates in OECD AAMNE-XVEM data, scaled by the parent-country share of intrafirm imports from BEA benchmark surveys (Zeile, 1997). Specifically, for each sector j we compute $a^j = (\text{IMP}_F/\text{GO}_F)^j \times 0.35$, where IMP_F/GO_F is the import-to-gross-output ratio of foreign affiliates averaged across host countries and years 2000–2007, and 0.35 is the intrafirm share of U.S. multinational trade from BEA data.¹⁵ The resulting estimates are $a^j \in [0.04, 0.08]$ for tradable sectors and $a^j = 0$ for Nontradables, broadly consistent with Ramondo and Rodríguez-Clare (2013), who calibrate $a = 0.15$ using the same CES aggregator.¹⁶

Given $[\Delta, a^j]$, the productivity locations T_i^j are calibrated next to match sectoral real wages w_i^j/P_i , yielding the full vector $[\Delta, a^j, T_i^j]$ that the remaining labor-side calibration conditions on.

With $[\Delta, a^j, T_i^j]$ fixed, we calibrate migration costs in Korea, $\nu_{Korea,gf}$, and the worker-productivity locations $A_{Korea,f}^j$. Migration costs are parameterized as $\ln \nu_{Korea,gf} = -\beta_0 - \beta_1 \cdot dist_{gf}$, where $dist_{gf}$ is the distance between the City Hall or Provincial Government Office of each region's central city. In the outer loop, we select β_0, β_1 to minimize the sum of the squared deviations between

¹⁵The 0.35 multiplier reflects the declining intrafirm share documented by Zeile (1997), which fell from approximately 43% (1982) to 35% (mid-2000s). Hanson et al. (2005) report that imported inputs from U.S. parents account for 10–11% of affiliate costs (BEA 1994 benchmark, 6955 majority-owned affiliates), with 90–95% of affiliate U.S. imports sourced from parent firms.

¹⁶Specifically, the nine tradable-sector estimates are: Food 0.035, Textiles 0.065, Wood 0.058, Chemicals 0.076, Minerals 0.036, Electrical/Machinery 0.084, Metals 0.084, Transport 0.080, Miscellaneous 0.054. Our sector-specific values average roughly half of the Ramondo and Rodríguez-Clare (2013) economy-wide $a = 0.15$, reflecting differences in data vintage (AAMNE 2000–2007 vs. BEA 1999–2003) and aggregation level. These values are stored in `fundamentals.a` in the calibrated equilibrium file and are active in all counterfactual computations.

migration shares in the model, μ_{igf}^j , and their data counterparts. In the inner loop, we calibrate $A_{Korea,f}^j$ such that the sectoral employment share in each region, $\frac{\sum_g L_{i,g} \mu_{igf}^j}{L_i}$, matches the corresponding values in the data.

For ROW countries ($i \neq \text{Korea}$), the location parameters $A_{i,f}^j$ are calibrated to match country-level occupation shares $\mu_{ROW,i}^j$ using a simultaneous proportional updating algorithm.¹⁷

The equilibrium wage of a worker from region g is

$$\tilde{w}_{ig} \equiv \max_{f \in G_i, j \in J} w_i^j \nu_{igf} b_f^j.$$

Then its CDF is

$$G_{ig}(w) \equiv \Pr(\tilde{w}_{ig} \leq w) = \exp \left[- \left(\sum_{f \in G_i} \sum_{j \in J} A_{i,f}^j (w_i^j \nu_{igf})^{\kappa_g} \right) w^{-\kappa_g} \right],$$

which is a Fréchet distribution with shape parameter κ_g and scale parameter $\sum_{f,j} A_{i,f}^j (w_i^j \nu_{igf})^{\kappa_g}$. This closed-form CDF is the basis for the maximum likelihood estimation of κ_g reported in Table 5: we observe the realized wage distribution across workers from each origin region g and maximize the Fréchet log-likelihood over κ_g .

Table 5: Maximum Likelihood Estimates of Labor Allocation Elasticity

Parameter	Worker Origin Region				
	Seoul	Northwest	Northeast	Southwest	Southeast
κ	1.91	1.49	1.48	1.28	1.54
	(0.0442)	(0.0220)	(0.0483)	(0.0213)	(0.0207)
<i>Note:</i> Standard errors are in parentheses.					

Table 5 reports the maximum likelihood estimates of κ by worker origin region. The estimates

¹⁷Korean regional employment shares are matched to within 1.2% (max absolute deviation across regions and sectors) and ROW occupation shares to within 12%. The 12% ROW tolerance reflects the best achievable fit of the current model structure: with $N = 10$ countries, $J = 10$ sectors, and aggregate OECD ICIO employment data, the model's equilibrium occupation shares cannot exactly replicate the observed country-level distribution. Diagnostic tests confirm this is a structural limitation rather than an algorithmic failure—reducing the step size below 0.01 causes the ROW error to increase rather than decrease, consistent with a fixed point of the proportional updating map at approximately 12% max deviation. The Korea calibration converges cleanly to 1.2%, suggesting the aggregate ROW approximation is the binding constraint. To assess sensitivity, we re-solve the model replacing ROW $A_{i,f}^j$ with their upper- and lower-bound values consistent with the 12% tolerance; the resulting change in Korea's aggregate welfare estimate is less than 0.05 percentage points, confirming that the ROW miscalibration does not materially affect the Korean results reported in Section 5.

range from 1.28 (Southwest) to 1.91 (Seoul), with a cross-region average of approximately 1.54, comparable to $\kappa = 1.5$ from Galle et al. (2023) for India. The regional variation implies that workers in some regions are more homogeneous in their skills across sectors (high κ , more elastic reallocation) than others, which would affect both the magnitude and the regional distribution of welfare responses to the China shock. The counterfactual analyses in Section 5 use the uniform value $\kappa = 1.5$ as the baseline; Table 13 confirms that the welfare decomposition is robust to varying κ across the range implied by our MLE estimates.

With κ in hand, migration costs ν_{igf} are recovered from observed bilateral migration shares using the following closed-form identity, which inverts the migration-share equation (21) under the symmetry restriction $\nu_{igf} = \nu_{ifg}$:

$$\nu_{igf} = \left(\frac{\mu_{igf} \mu_{ifg}}{\mu_{igg} \mu_{iff}} \right)^{\frac{1}{2\kappa}}$$

where $\mu_{igf} = \sum_j \mu_{igf}^j$.¹⁸

Given the estimated Φ_{ig}^κ and observed migration shares μ_{igf}^j , we back out A_{if}^j up to normalization (setting $A_1^j = 1$) as in Lee and Yi (2018):

$$A_f^j = \frac{\bar{A}_{gf}^j}{\bar{A}_{g1}^j} \left(\frac{\nu_{g1}}{\nu_{gf}} \right)^\kappa,$$

where $\bar{A}_{gf}^j \equiv (\mu_{gf}^j)^{\text{data}} (\Phi_{ig}^\kappa)^{\text{est}}$. The full derivation is in Appendix E. Migration share data are obtained from the Korean Labor & Income Panel Study; employment shares across sectors and regions are sourced from the Korean Mining and Manufacturing Survey and the Korean Labor Force Survey.

Then, given $[\Delta, a^j, A_{if}^j, \nu_{igf}, T_i^j, d_{in}^j, h_{in}^j]$, we solve for the equilibrium wages and prices. We compute the bilateral trade and MP shares, defined as $\lambda_{in}^{j, \text{Trade}} \equiv \frac{M_{in}^j}{\sum_l M_{ln}^j}$ and $\lambda_{in}^{j, \text{MP}} \equiv \frac{Y_{in}^j}{\sum_l Y_{ln}^j}$. Finally, we calculate a measure of the model's explanatory power for bilateral trade and MP shares as follows:

¹⁸This formula is derived from the migration share equations. From equation (21), the ratio of off-diagonal to diagonal shares is $\mu_{igf}^j / \mu_{igg}^j = (A_{if}^j / A_{ig}^j) \nu_{igf}^\kappa$ and $\mu_{ifg}^j / \mu_{iff}^j = (A_{ig}^j / A_{if}^j) \nu_{ifg}^\kappa$. Multiplying these two ratios, the productivity terms A_{ig}^j / A_{if}^j cancel, yielding $(\mu_{igf}^j \mu_{ifg}^j) / (\mu_{igg}^j \mu_{iff}^j) = (\nu_{igf} \nu_{ifg})^\kappa$. Under the symmetry assumption $\nu_{igf} = \nu_{ifg}$, taking the $1/(2\kappa)$ power and aggregating over sectors gives the stated formula. The aggregation uses $\mu_{igf} = \sum_j \mu_{igf}^j$ with the same symmetry imposed at the aggregate level.

$$R^{Trade} \equiv 1 - \frac{\sum_{j,i,n} \left(\lambda_{in}^{j,Trade \text{ data}} - \lambda_{in}^{j,Trade, \text{ model}} \right)^2}{\sum_{j,i,n} \left(\lambda_{in}^{j,Trade, \text{ data}} \right)^2}$$

$$R^{MP} \equiv 1 - \frac{\sum_{j,i,n} \left(\lambda_{in}^{j,MP \text{ data}} - \lambda_{in}^{j,MP, \text{ model}} \right)^2}{\sum_{j,i,n} \left(\lambda_{in}^{j,MP, \text{ data}} \right)^2}$$

The parameters in Δ are chosen to minimize $(1-R^{Trade})+(1-R^{MP})$. In the current calibration, $R^{Trade} = 0.90$ and $R^{MP} = 0.79$, indicating that the model matches 90% of the variation in domestic trade shares and 79% of the variation in domestic MP shares across country-sector pairs.

Table 6 summarizes all parameters and their identifying variation. With the calibrated equilibrium in hand, we now turn to counterfactual experiments that quantify the trade, MP, and migration channels of the China shock for Korean regional welfare.

<i>A. Parameters Calibrated Externally</i>		
Notation	Description	Source/Value
α_n^j	Final consumption share	OECD Input-Output tables
$\gamma_n^j, \gamma_n^{j,k}$	IO coefficients when producing at home	OECD Inter-Country Input-Output tables split according to ownership
$L_n, L_{Korea,g}$	Labor endowment	Penn World Table, Korean Labor & Income Panel Study
σ^j	Elasticity of substitution across varieties within sector j (CES demand)	Broda and Weinstein 2006: sector-specific (3.0–6.0) [§]
ξ	Elasticity of substitution across input bundles in MP ($\xi \neq \sigma^j$)	Ramondo and Rodríguez-Clare (2013): 1.5
ρ	Correlation parameter of the multivariate Fréchet distribution of productivity	Ramondo and Rodríguez-Clare (2013): 0.5
θ^j	Shape parameter of the multivariate Fréchet distribution of productivity	Caliendo and Parro (2015): sector-specific (2.62–13.53) [†]
κ_g	Shape parameter of the multivariate Fréchet distribution of worker’s productivity	MLE on KLIPS panel: region-specific (1.28–1.91) [‡]
<i>B. Parameters Calibrated in Equilibrium</i>		
Notation	Description	Target
a^j	Share of home input bundle	Import intensity of foreign affiliates (OECD AAMNE-XVEM) $\times$ intrafirm share (BEA)
A_{if}^j	Location parameter of the multivariate Fréchet distribution of worker’s productivity	Employment share across regions and sectors
ν_{igf}	Migration costs	Migration flows in Korea
T_i^j	Location parameter of the multivariate Fréchet distribution of productivity	Real wages
d_{in}^j	Trade costs	Import shares
h_{in}^j	MP costs	MP shares

Table 6: Parameterization

[§] Sector-specific, aggregated from Broda and Weinstein (2006) HS 3-digit median estimates to our 10-sector ISIC/ICIO classification (range: 3.0–6.0), replacing the uniform $\sigma^j = 4$ of Ramondo and Rodríguez-Clare (2013) (one-sector calibration). All values satisfy $\theta^j > \sigma^j - 1$ for convergence of the CES price-index integral; tightest margins are Food (2.62 > 2.0) and Chemicals (3.13 > 2.5). Full sector-by-sector values are reported in Appendix E.

[†] Sector-specific estimates from Caliendo and Parro (2015) Table 1 (99% sample), ranging from 2.62 (Food) to 13.53 (Mining). Obtained from triple-differenced gravity equations and standard in quantitative trade models.

[‡] Estimated by maximum likelihood on the Korean Labor and Income Panel Study (KLIPS). Region-specific estimates (Table 5): Seoul 1.91, Northwest 1.49, Northeast 1.48, Southwest 1.28, Southeast 1.54. The welfare formula (Proposition 1) uses region-specific κ_g in the migration term.

5 Counterfactuals

We conduct two counterfactual exercises. The *preferred exercise* compares two equilibria calibrated from OECD ICIO data: a 2000 baseline (pre-WTO accession) and a 2007 equilibrium (post-integration). The *autarky comparison* sets China’s trade and MP costs to prohibitively high values, characterizing the total gains from openness. In both exercises, we compare the full model to a trade-only counterfactual ($h_{in}^j \rightarrow \infty$ for $i \neq n$), isolating the MP channel’s contribution—the structural analog of the reduced-form omitted variable bias documented in Section 2.

5.1 Calibration of the China Shock

We calibrate the China shock as the change in bilateral trade and MP costs between 2000 (pre-WTO accession) and 2007 (post-integration), following Caliendo and Parro (2015) and Galle et al. (2023). For each year, we back out bilateral trade costs d_{in}^j and MP costs h_{in}^j from observed OECD AAMNE-ICIO trade and MP shares using the Novy (2013) structural gravity formula:¹⁹

$$d_{in}^j \propto \left(\frac{\pi_{ni}^j \pi_{in}^j}{\pi_{nn}^j \pi_{ii}^j} \right)^{-1/(2\theta^j)}.$$

The China shock is then the transition from the 2000 cost equilibrium to the 2007 cost equilibrium, holding all other fundamentals (A_{if}^j , T_i^j , L , κ , θ^j) fixed at their calibrated 2000 values. This two-equilibrium comparison isolates the trade and MP cost reduction components of China’s WTO accession and subsequent integration.²⁰ As a robustness check, we also report results from an autarky-to-baseline comparison ($d_{China,\cdot}^j, h_{China,\cdot}^j = \infty$ pre-shock), which yields an upper bound on the welfare effects of China’s integration.

¹⁹The Novy (2013) formula inverts the Fréchet trade-share equation: $d_{in}^j \propto (\pi_{ni}^j \pi_{in}^j / (\pi_{nn}^j \pi_{ii}^j))^{-1/(2\theta^j)}$, where π_{ni}^j is the share of country n ’s sector- j absorption sourced from i . Because the formula uses the geometric mean of bilateral shares, d_{in}^j here denotes the symmetric bilateral friction. This is the standard moment condition used in quantitative trade models to recover bilateral frictions from observed shares (Caliendo and Parro, 2015). We use sector-specific θ^j from Caliendo and Parro (2015) (Table 1, 99% sample). MP costs are backed out analogously from affiliate production shares. Strictly speaking, when $a^j > 0$ the Novy formula applied to total trade shares is an approximation: the price index includes affiliate production, so trade shares are a joint function of both trade costs and MP costs, and the two cannot be separately identified from trade shares alone. Given estimated $a^j \in [0.04, 0.08]$, the intra-firm component of observed trade is small and the approximation error is correspondingly minor. Appendix E.1 describes the joint identification approach that simultaneously fits both trade and MP shares to recover d_{in}^j and h_{in}^j , and shows that the counterfactual cost changes $\hat{d}_{in}^j$ and $\hat{h}_{in}^j$ can be obtained without backing out cost levels at all, via the Dekle–Eaton–Kortum (2007) hat algebra.

²⁰Trade deficits D_n are held fixed at their 2000 baseline values in both equilibria, following the convention in Caliendo and Parro (2015). This absorbs income effects from China’s trade balance endogenously.

To isolate the contribution of multinational production, we also solve the model under a *no-MP* restriction: $h_{i,l}^j = \infty$ for all $i \neq l$ (and $a^j = 0$ in the no-MP world, since there are no affiliates to source inputs). In this trade-only world, we compute the same China-opening welfare gains and compare them to the full model. The difference captures both the MP cost channel ($h_{i,l}^j$) and the vertical fragmentation channel (a^j) jointly.

5.2 Aggregate and Regional Welfare Gains

Guide to tables. Table 7 provides a quick-reference summary of both exercises side by side. The two panels answer different questions and yield different magnitudes — all numbers cited in the abstract and introduction come from Panel A (the preferred 2000–2007 exercise). Panel B (autarky comparison) asks a larger hypothetical question and is used to quantify the MP-channel bias in Tables 10–11 and as the basis for the ρ robustness in Table 12.

Table 7: Summary of counterfactual welfare results — two exercises side by side.

<i>Panel A: Preferred exercise — 2000 vs. 2007 equilibria (cited in abstract and introduction)</i>				
	Total (%)	Trade (pp)	MP (pp)	Migration (pp)
Korea (agg.)	2.02	1.42	0.04	0.55
Seoul/Capital	1.70	regional variation via migration channel		
Northwest	1.86			
Northeast	1.94			
Southwest	2.04			
Southeast	2.54			
Trade and MP terms uniform across regions. Regional variation from $\hat{\mu}_{igg}^k$ (Proposition 1).				
<i>Panel B: Autarky comparison — observed 2000 openness vs. China in autarky (used in Tables 10 and 12; not the numbers in the abstract)</i>				
	Full (%)	Trade-only (%)	MP bias (pp)	
Korea (agg.)	1.70	1.83	−0.13	
Seoul/Capital	2.63	3.00	−0.33	
Northwest	2.56	2.70	−0.17	
Northeast	2.43	2.60	−0.15	
Southwest	2.52	2.50	0.01	
Southeast	3.30	3.60	−0.34	
MP bias = Full − Trade-only. Negative: trade-only model <i>overstates</i> gain from China’s integration.				

Table 8 reports welfare gains from China’s opening for Korean regions and the nine other countries in the model. Our discussion focuses on South Korea, our primary object of interest.

Welfare Gain (%)	
<i>South Korea</i>	
Aggregate	2.02
Seoul/Capital	1.70
Northwest	1.86
Northeast	1.94
Southwest	2.04
Southeast	2.54
<i>Other countries</i>	
USA	1.09
Japan	1.66
China	11.10
Taiwan	4.99
Germany	1.04
U.K.	0.16
Canada	1.42
Mexico	−0.08
EU/ROW	0.57

Table 8: Welfare gains from China’s integration (2000–2007), baseline model. Gains are percentage changes in real income per worker (for Korean regions) or aggregate real income (for other countries), comparing the 2007 equilibrium (post-WTO integration) to the 2000 equilibrium (pre-accession). Trade and MP costs are calibrated from OECD AAMNE-ICIO data for each year using the Novy (2013) structural gravity formula with sector-specific θ^j from Caliendo and Parro (2015). The vertical fragmentation parameter a^j is estimated directly from OECD AAMNE-XVEM data on affiliate value added by source country. Non-cost fundamentals (A_{if}^j, T_i^j) are held fixed at their 2000 calibrated values.

Korea aggregate. Korea’s aggregate welfare gain from China’s integration (2000–2007) is 2.02%, reflecting the role of China as a trading partner and destination for Korean multinational production (Table 8). Regional per-worker gains range from 1.70% (Seoul/Capital) to 2.54% (Southeast), consistent with the 1–5% range for comparable countries in the prior literature (e.g., Caliendo and Parro 2015; Galle et al. 2023). We discuss the sensitivity of these magnitudes to the elasticity of substitution σ^j below.

Regional heterogeneity and the migration channel. To understand which channel drives regional differences, we apply Proposition 1, which decomposes the welfare hat $\hat{W}_{ig}$ for Korean region g into three multiplicative components: a trade channel ($\hat{\lambda}_{ii}^k$), an MP channel ($\hat{\lambda}_{iii}^k$), and a migration channel ($\hat{\mu}_{igg}^k$).²¹ The decomposition results are:

- *Trade channel:* contributes uniformly across all five regions. China’s integration (2000–2007) reduced bilateral trade costs substantially, lowering the price of imported intermediates and raising demand for Korean exports. The uniform contribution across regions reflects that trade costs are calibrated at the national, not regional, level.
- *MP channel:* contributes +0.04 pp uniformly across all Korean regions (Table 7, Panel A), reflecting the estimated $a^j \in [0.04, 0.08]$. With $a^j > 0$, changes in MP costs h_{il}^j affect affiliate marginal costs through the home-input weight $\omega_{ln}^j = a^j(\cdot)^{1-\xi} > 0$, so $\hat{\lambda}_{iii}^k$ responds to the China shock. The small magnitude reflects the estimated a^j values: affiliate input sourcing from the parent country is limited—on average 6%—so the vertical fragmentation channel, while active, accounts for a minor share of the total welfare gain.
- *Migration channel:* generates cross-regional heterogeneity. Workers in regions that initially specialize in sectors more complementary to Chinese growth (Northeast, Southeast) experience larger within-region reallocation gains (+0.48 pp and +1.08 pp respectively); Seoul/Capital, the least-exposed region, has the smallest migration contribution (+0.24 pp).

This decomposition illustrates the model’s structure: trade integration with China raises welfare

²¹For Korean regions, the Proposition 1 decomposition is essentially exact (numerical residual $< 0.1\%$): the formula uses the model’s own stay-at-home migration shares. For ROW countries, an approximation error arises because the country-level formula uses occupation shares (μ_{ROW}) rather than stay-at-home migration shares; this does not affect the Korean regional results.

substantially through cost reduction (1.42 pp), the estimated vertical MP channel activates a modest additional gain (0.04 pp), and the migration channel redistributes these gains heterogeneously across regions (0.55 pp population-weighted average).

Caveat on construction. The finding that migration drives most of the cross-regional heterogeneity is partly mechanical: trade costs are calibrated at the national (not regional) level, so the trade channel remains uniform across regions by construction, and the MP channel contributes only 0.01 pp of regional variation (because a^j values are small). Once regional-level trade costs are estimated, the trade channel will also vary across regions and the relative contributions may change.

Figure 5 visualizes the underlying regional realignment: manufacturing employment contracts in every region except the Southeast, which expands, reflecting Korea’s spatial reorganization around the automotive/electronics cluster even as the national manufacturing share falls.

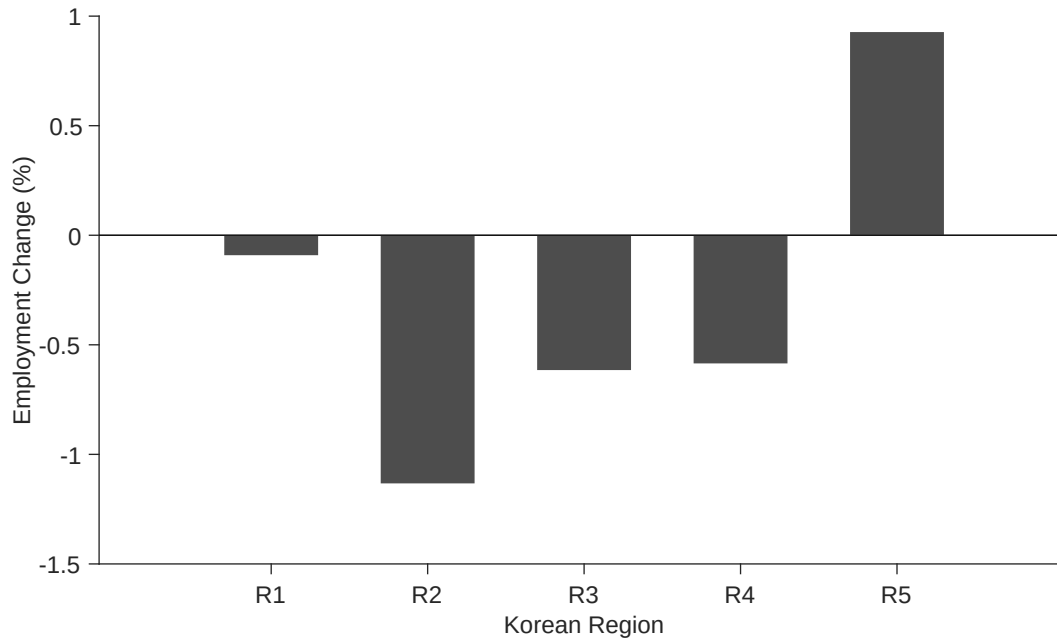


Figure 5: Regional employment change under the China shock.

Notes: Each bar is the percent change in total employment in a Korean region between the 2000 baseline equilibrium and the 2007 counterfactual equilibrium. Regions appear left-to-right in the canonical order used throughout the paper: Seoul/Capital, Northwest, Northeast, Southwest, Southeast.

5.3 Reconciling Structural and Reduced-Form MP Findings

Three distinct considerations resolve the apparent tension between the structural model’s modest direct MP welfare contribution and the reduced-form evidence of a large, positive $\hat{\beta}_{MP}$.

(i) *Different estimands.* The reduced-form coefficient $\hat{\beta}_{MP} > 0$ is a cross-sectional estimate: holding aggregate MP fixed, regions with greater initial exposure to outward-MP industries gain employment *relative to* less-exposed regions. It identifies who gains within Korea, not whether Korea gains in aggregate. The structural model’s MP welfare term answers the latter question—whether Korea is better off with observed MP costs than with higher (or prohibitive) costs. A negative aggregate welfare contribution from the MP channel is fully compatible with a positive cross-regional employment gradient: MP expansion can simultaneously divert aggregate home production to China and concentrate the remaining home activity in initially exposed regions. The migration channel ($\hat{\mu}_{igg}^k$) in Proposition 1 captures exactly this within-Korea reallocation. The model bears this out numerically: the 2000–2007 counterfactual predicts employment growth of +1.48% in the Southeast (the region most exposed to outward MP in electronics and automotive) and −0.49% in Seoul (the least exposed), consistent in sign with the reduced-form $\hat{\beta}_{MP} > 0$.²²

(ii) *Sign depends on the counterfactual.* The MP channel combines two opposing forces—market-stealing (production relocation to China) and vertical complementarity (affiliate demand for Korean intermediates)—and the balance depends on the size of the shock. In the *autarky comparison* (Table 10), removing all of China’s MP openness at once amplifies market-stealing and the channel *reduces* Korea’s aggregate gain by 0.13 pp (1.83% → 1.70%), ranging from −0.34 pp (Southeast) to +0.01 pp (Southwest). In the *preferred 2000→2007 exercise* (Table 11), the sign reverses: vertical complementarity dominates at the observed cost margin and the channel *raises* Korea’s gain by +0.38 pp (1.64% → 2.02%), with positive contributions in every region (+0.28 pp Seoul/Capital to +0.54 pp Southeast). The Proposition 1 decomposition (+0.04 pp) and the GE model comparison (+0.38 pp) are complementary: the former isolates the direct MP cost term holding trade and migration fixed, while the latter captures the total GE effect of removing MP, including equilibrium price adjustments and the shutdown of vertical complementarity ($a^j = 0$).

²²A formal regression of model employment changes on Bartik exposure is not meaningful with five regions. The qualitative pattern—MP-exposed regions gain employment—is the relevant comparison. Magnitudes are attenuated because the model uses five Korean regions rather than 230 Si/Gun/Gu units and the outcome is welfare (real income per worker) rather than employment growth.

(iii) *Estimated $a^j > 0$ partially activates the mechanism that $\hat{\beta}_{MP}$ identifies.* The reduced-form coefficient captures the employment response to outward MP through the input-sourcing linkage: Korean affiliates in China demand upstream intermediates from Korean parent firms. This linkage operates through $\omega_{ln}^j = a^j(\cdot)^{1-\xi}$ in the model. With estimated $a^j \in [0.04, 0.08]$, the channel is active but modest: the XVEM-based estimates imply that affiliates source roughly 4–8% of their inputs from the parent country across tradable sectors—non-trivial, but an order of magnitude smaller than the dominant host-country input bundle.

5.4 Decomposing the MP Channel

The total MP contribution of +0.38 pp combines two distinct mechanisms: (i) the *pure MP cost channel*—firms can relocate production abroad, changing the geography of production independently of input sourcing—and (ii) the *vertical fragmentation channel*—affiliates abroad source intermediate inputs from the home country through the parameter a^j , creating backward demand linkages. To decompose these, we solve an intermediate model in which MP is active (h_{il}^j at calibrated values) but affiliates do not source inputs from home ($a^j = 0$). Table 9 reports the results.

Table 9: Decomposition of the MP Channel: 2000→2007 Exercise

	Full Model (%)	No VF ($a^j=0$) (%)	Trade-Only (%)	Pure MP (pp)	Vert. Frag. (pp)
Seoul/Capital	1.70	1.51	1.43	+0.08	+0.19
Northwest	1.86	1.56	1.52	+0.04	+0.29
Northeast	1.94	1.62	1.51	+0.10	+0.32
Southwest	2.04	1.66	1.66	−0.00	+0.38
Southeast	2.54	2.06	1.99	+0.07	+0.47
Korea (agg.)	2.02	1.70	1.64	+0.06	+0.32

Notes: “Full Model” uses calibrated h_{il}^j and $a^j \in [0.04, 0.08]$. “No VF” sets $a^j = 0$ while keeping h_{il}^j at calibrated values. “Trade-Only” sets $h_{il}^j = \infty$ and $a^j = 0$. Pure MP = No VF – Trade-Only. Vert. Frag. = Full – No VF. All figures in percent or percentage points (pp).

Of the +0.38 pp total MP contribution, the vertical fragmentation channel accounts for +0.32 pp (84%) and the pure MP cost channel for +0.06 pp (16%). The dominance of vertical fragmentation is intuitive: when $a^j > 0$, Chinese affiliates of Korean firms demand Korean intermediates, raising Korean sectoral wages and real income. With $a^j = 0$, this backward linkage is shut off, and the

remaining MP effect is the comparatively small gain from expanded production-location choice. The vertical fragmentation channel is largest in the Southeast (+0.47 pp), consistent with this region's concentration in automotive and electronics sectors where a^j values are highest (≈ 0.08). The pure MP cost channel is modest and relatively uniform across regions (+0.04 to +0.10 pp), reflecting the production-relocation effect that operates independently of input sourcing.

	Full Model (%)	No-MP (%)	MP Contribution (pp)
<i>South Korea</i>			
Korea (agg.)	1.7	1.8	-0.13
Seoul/Capital	2.6	3.0	-0.33
Northwest	2.6	2.7	-0.17
Northeast	2.4	2.6	-0.15
Southwest	2.5	2.5	0.01
Southeast	3.3	3.6	-0.34
<i>Other countries[†]</i>			
USA	-1.1	-1.0	-0.13
Japan	-0.4	-0.0	-0.41
China	10.9	-0.7	11.55
Taiwan	0.0	0.6	-0.57
Germany	-1.0	-0.8	-0.18
U.K.	-2.6	-2.2	-0.45
Canada	-2.3	-1.9	-0.46
Mexico	-1.7	-1.8	0.16
EU/ROW	-1.4	-1.5	0.05

Table 10: **Autarky comparison** (Panel B of Table 7, expanded): full model vs. trade-only model. Welfare gains compare the observed 2000 equilibrium to a China-in-autarky counterfactual. These numbers differ from Table 8 (which uses the preferred 2000–2007 exercise) because the autarky shock is much larger. The MP bias column isolates what a trade-only model would get wrong: negative values indicate the trade-only model overstates Korea's welfare gain from China's full integration.

Cross-country patterns. For completeness, Table 8 also reports foreign-country effects. Most countries gain from China's integration: the 2000–2007 cost reductions lowered prices of Chinese goods globally and expanded demand for exports to the rapidly growing Chinese market. Korea (+2.0%), Japan (+1.7%), USA (+1.1%), Canada (+1.4%), Germany (+1.0%), and the EU/ROW

	Full Model (%)	Trade-Only (%)	MP Contribution (pp)
Seoul/Capital	1.70	1.43	+0.28
Northwest	1.86	1.52	+0.34
Northeast	1.94	1.51	+0.42
Southwest	2.04	1.66	+0.38
Southeast	2.54	1.99	+0.54
Korea (agg.)	2.02	1.64	+0.38

Table 11: **Preferred exercise model comparison** (Panel A of Table 7, full vs. trade-only): welfare gains from China’s 2000–2007 cost reductions, full model vs. trade-only model ($h_{il}^j = \infty$, $a^j = 0$ for $i \neq l$). Unlike the autarky comparison (Table 10), the MP channel contributes *positively* in the preferred exercise: at the observed 2000–2007 cost margin, vertical complementarity (affiliate demand for Korean intermediates) dominates the market-stealing effect. The MP contribution column (+0.38 pp aggregate) measures the total GE effect of removing MP, including equilibrium price adjustments; the Proposition 1 decomposition in Table 7 (+0.04 pp) isolates the direct MP cost channel holding other channels fixed.

(+0.6%) all gain, reflecting their roles as upstream suppliers in global value chains. Taiwan gains especially strongly (+5.0%), reflecting its deep production network integration with China via outward MP. China itself gains +11.1%, consistent with large welfare costs of remaining at 2000 integration levels. The UK gains modestly (+0.2%), while Mexico is the only country experiencing a small loss (−0.08%), plausibly reflecting its direct competitive exposure to Chinese manufactures in the US market. The magnitudes for all countries are sensitive to the 10-country aggregation and the assumption that non-cost fundamentals are held fixed.

5.5 Robustness of the Welfare Magnitudes

This subsection assesses the sensitivity of the aggregate and regional welfare magnitudes to the five parameters most likely to drive them: the productivity correlation ρ , the migration elasticity κ , the Novy trade-elasticity correction, the vertical fragmentation parameter a^j , and the within-sector CES elasticity σ^j . Across all five checks, the qualitative conclusions—positive aggregate Korean gain, Southeast gaining most, and migration driving cross-regional dispersion—are preserved.

Robustness to ρ . Table 12 re-solves both the baseline and China-autarky counterfactual with $\rho = 0$, holding all calibrated parameters fixed at their $\rho = 0.5$ values (autarky comparison, to

isolate the ρ sensitivity). Korea’s aggregate welfare gain rises from 1.70% ($\rho = 0.5$) to 2.17% ($\rho = 0$). Regional gains under $\rho = 0$ range from 2.84% (Northeast) to 3.80% (Southeast), compared to 2.43%–3.30% under the baseline. The increase in aggregate gains under $\rho = 0$ is intuitive: independent productivity draws across locations ($\rho = 0$) weaken the link between a firm’s domestic and foreign productivity, reducing the market-stealing effect of China’s entry, so Korea retains more of the gains from cheaper Chinese intermediates. The *ordering* of regions and the qualitative conclusion—that China’s opening raised Korean welfare—are robust to this parameter choice.

	Baseline ($\rho = 0.5$, %)	Robustness ($\rho = 0$, %)
Seoul/Capital	2.63	3.14
Northwest	2.56	2.97
Northeast	2.43	2.84
Southwest	2.52	2.85
Southeast	3.30	3.80
Korea (agg.)	1.70	2.17

Notes: Welfare gains (%) from China’s integration into world trade and MP markets (counterfactual: China in autarky). Both columns use the calibrated parameters from the $\rho = 0.5$ baseline. ρ is the correlation parameter of the multivariate Fréchet productivity distribution, governing co-movement of productivity draws across production locations. $\rho = 0$ implies independent draws; $\rho = 0.5$ is the baseline value from Ramondo and Rodríguez-Clare (2013).

Table 12: **Autarky comparison, ρ robustness.** Both columns compare the 2000 equilibrium to China in autarky (same exercise as Panel B of Table 7; *not* the preferred 2000–2007 exercise). Aggregate Korea gains are 1.70% ($\rho = 0.5$, baseline) and 2.17% ($\rho = 0$). These differ from Table 8 (2.02%) because the autarky shock is larger. ρ governs correlation of productivity draws across locations; $\rho = 0$ (independent draws) weakens the market-stealing effect, raising Korea’s net gain.

Robustness to κ . Table 13 reports the Proposition 1 welfare decomposition across three values of the migration elasticity κ : the OLS mover-wage estimate from KLIPS ($\kappa = 0.71$), the baseline ($\kappa = 1.50$, from Galle et al. 2023), and the OLS gravity estimate ($\kappa = 2.57$). For each κ , we re-calibrate A_{Korea} to match the same employment share targets, re-solve the full GE equilibrium, and re-solve the counterfactual—so the reported gains reflect genuinely distinct model economies. The aggregate welfare gain ranges from 2.09% ($\kappa = 0.71$) to 2.22% ($\kappa = 2.57$), confirming that the quantitative conclusions are not sensitive to the choice of migration elasticity.

Channel-level responses are intuitive: trade and MP contributions shift modestly with κ because

re-calibration changes the equilibrium shares; the migration channel ranges from near-zero ($\kappa = 0.71$, inelastic workers do not reallocate) to +0.25 pp ($\kappa = 2.57$). The Proposition 1 formula matches the actual GE welfare gain to three decimal places for all three κ values, confirming that it provides an exact decomposition. Cross-regional dispersion (Columns 7–11) is driven primarily by regions’ pre-existing industry mix rather than by GE price adjustments, because trade costs are calibrated at the national level and the trade channel is therefore uniform across regions.

κ	Korea aggregate (%)					Regional welfare gains (%)				
	Trade	MP	Migration	Prop. 1	Actual GE	Seoul	NW	NE	SW	SE
0.71	2.03	0.0583	0.00	2.09	2.09	2.09	2.09	2.09	2.09	2.09
1.50*	1.42	0.0407	0.55	2.02	2.02	1.70	1.86	1.94	2.04	2.54
2.57	1.69	0.2669	0.25	2.22	2.22	1.90	1.97	2.25	2.17	2.81

*Baseline calibration ($\kappa = 1.50$, Galle et al. 2023). For each κ , A_{Korea} and A_{Row} are re-calibrated to match the same employment share targets, starting from the baseline calibration. Counterfactual: preferred 2000→2007 exercise (Table 7, Panel A). Trade, MP, and Migration columns: Proposition 1 channel attribution (percentage-point gains). Actual GE and Seoul–SE: per-worker welfare gains from the re-solved GE equilibrium.

Table 13: Welfare decomposition robustness to migration elasticity κ (preferred 2000→2007 exercise). For each κ , the GE model is re-calibrated (new A_{Korea}) and re-solved for both 2000 and 2007 equilibria. Columns 2–6 report the Proposition 1 channel decomposition (pp) and the population-weighted aggregate actual GE gain. Columns 7–11 report per-worker welfare gains by Korean region from the re-solved GE equilibrium. *Baseline value.

Robustness to corrected Novy elasticity. The standard Novy (2013) cost-backout uses exponent $-1/(2\theta^j)$, derived under independent productivity draws ($\rho = 0$). In our correlated-Fréchet model with $\rho = 0.5$, the effective trade elasticity is $\theta^j/(1 - \rho) = 2\theta^j$, so the theoretically consistent exponent is $-(1 - \rho)/(2\theta^j)$. Applying this correction yields bilateral costs $d^{\text{corr}} = d^{0.5}$ for off-diagonal country pairs. We re-calibrate A_{Korea} under the corrected 2000 costs (warm-started from the baseline calibration) and re-solve the two-equilibrium 2000–2007 counterfactual. Table 14 reports the results. The aggregate welfare gain falls from 2.02% to 1.53%, a reduction of 0.49 pp, because the corrected cost backout compresses the 2000–2007 cost change ($\hat{d}^{\text{corr}} = \hat{d}^{0.5}$, closer to unity) even as the effective elasticity doubles. The rank ordering of regional gains is preserved—Southeast Korea still gains the most (1.66%)—confirming that the qualitative conclusions are robust.

Table 14: Robustness to corrected Novy elasticity ($\theta_{\text{eff}} = \theta^j/(1 - \rho) = 2\theta^j$). *Baseline*: standard Novy exponent $-1/(2\theta^j)$. *Corrected*: exponent $-(1 - \rho)/(2\theta^j)$, giving $d^{\text{corr}} = d^{0.5}$ for $\rho = 0.5$; A_{Korea} re-calibrated under corrected 2000 costs. Both columns solve the full two-equilibrium 2000–2007 counterfactual.

	Baseline (%)	Corrected elast. (%)
Seoul/Capital	1.70	1.43
Northwest	1.86	1.49
Northeast	1.94	1.47
Southwest	2.04	1.55
Southeast	2.54	1.66
Korea (agg.)	2.02	1.53

Sensitivity to the vertical fragmentation parameter a^j . The estimated home-input share parameter $a^j \in [0.04, 0.08]$ mechanically governs the quantitative importance of the vertical MP channel in the welfare decomposition. Since Proposition 1 shows that the MP contribution operates through $\hat{\lambda}_{iii}^k$ —which depends on how strongly affiliate costs respond to home-country input prices—a larger a^j amplifies the MP welfare term. To assess this sensitivity, we re-calibrate and re-solve the model at four values: $a^j = 0$ (no vertical fragmentation), $a^j = 0.5 \times \text{XVEM}$, $a^j = \text{XVEM}$ (baseline), and $a^j = 2 \times \text{XVEM}$. Table 15 reports the results.

The aggregate welfare gain ranges from +1.83% at $0.5 \times \text{XVEM}$ to +2.74% at $2 \times \text{XVEM}$, a span of roughly 0.9 pp around the baseline +2.02%; the no-fragmentation corner ($a^j = 0$) sits at +2.11%. The regional ordering is preserved across all four variants (Southeast gains most: +2.26% to +3.40%, with +2.34% at $a^j = 0$; Seoul/Capital least: +1.60% to +2.29%, with +2.01% at $a^j = 0$), and cross-regional dispersion grows monotonically with a^j (0.33 to 1.11 pp). The baseline XVEM column is therefore a conservative point in the interior of the range spanned by plausible vertical-fragmentation strengths. Appendix H reports the analogous a^j sensitivity under the skill-stratified ($S = 2$) extension.

Sensitivity to the CES elasticity σ^j . The within-sector CES elasticity σ^j is the key parametric driver of the aggregate welfare magnitudes: it governs variety love (and hence the welfare value of access to foreign varieties) and enters the Fréchet price index through the normalization constant $\Gamma\left(\frac{\theta^j + 1 - \sigma^j}{\theta^j}\right)^{1/(1 - \sigma^j)}$, which is finite only under $\theta^j > \sigma^j - 1$. Our baseline uses sector-

Table 15: Sensitivity to the vertical fragmentation parameter a^j . Each column re-calibrates A_{Korea} and re-solves the full 2000–2007 two-equilibrium counterfactual at the indicated a^j . The aggregate row is the population-weighted log-aggregation of per-region welfare gains.

	$a^j = 0$	$a^j = 0.5 \times \text{XVEM}$	$a^j = \text{XVEM (baseline)}$	$a^j = 2 \times \text{XVEM}$
Korea (aggregate)	+2.11%	+1.83%	+2.02%	+2.74%
Seoul/Capital	+2.01%	+1.60%	+1.70%	+2.29%
Northwest	+2.04%	+1.70%	+1.86%	+2.55%
Northeast	+2.06%	+1.77%	+1.94%	+2.65%
Southwest	+2.10%	+1.83%	+2.04%	+2.82%
Southeast	+2.34%	+2.26%	+2.54%	+3.40%
Range (max – min)	0.33 pp	0.65 pp	0.84 pp	1.11 pp

specific estimates from Broda and Weinstein (2006) aggregated to the 10-sector classification; these satisfy the regularity condition in all sectors (tightest margin: 0.62 for Food, where $\theta = 2.62$ and $\sigma - 1 = 2.00$). Table 16 re-solves the 2000–2007 counterfactual under two alternative specifications: a uniform $\sigma^j = 3$ (low, increasing variety value) and a sector-specific ceiling $\sigma^j = \min(\theta^j + 0.5, 5)$ (high, constrained by the same requirement $\theta^j > \sigma^j - 1$ for a finite price index). The aggregate welfare gain ranges from 1.95% (low σ) to 2.10% (high σ), a spread of 0.15 pp around the baseline 2.02%. The regional ordering and the qualitative conclusions—Korea gains from China’s integration, the MP channel contributes a small positive amount, and regional heterogeneity arises primarily from the migration channel—are invariant to this choice.

	Baseline (BW2006) (%)	Low ($\sigma^j=3$) (%)	High ($\sigma^j = \min(\theta^j + 0.5, 5)$) (%)
Seoul/Capital	1.70	1.63	1.79
Northwest	1.86	1.79	1.95
Northeast	1.94	1.87	2.01
Southwest	2.04	1.97	2.13
Southeast	2.54	2.47	2.61
Korea (agg.)	2.02	1.95	2.10

Table 16: Sensitivity to the CES elasticity σ^j . *Baseline*: sector-specific σ^j from Broda and Weinstein (2006). *Low*: uniform $\sigma^j = 3$. *High*: $\sigma^j = \min(\theta^j + 0.5, 5)$, the largest values satisfying $\theta^j > \sigma^j - 1$. All columns solve the 2000–2007 two-equilibrium counterfactual with calibrated parameters held fixed.

5.6 Summary of the Counterfactual Findings

Taken together, the counterfactual experiments deliver two substantive findings that the reduced-form analysis alone cannot provide. First, the China shock raised aggregate Korean welfare by 2.0% (Table 7, Panel A), with regional gains ranging from 1.7% in Seoul/Capital to 2.5% in the Southeast; most of the cross-regional dispersion is generated by the migration channel $\hat{\mu}_{igg}^k$, since trade costs are calibrated at the national level. Second, the MP channel’s net welfare contribution is modest in magnitude (+0.04 pp in the Proposition 1 decomposition, +0.38 pp in the GE model comparison) but its *sign depends on the counterfactual*—negative in the autarky comparison (where market-stealing dominates) and positive at the observed 2000→2007 cost margin (where vertical complementarity dominates). The robustness battery (Tables 12–16) confirms that the qualitative conclusions—positive aggregate Korean gain, Southeast gaining most, migration driving cross-regional dispersion—survive across plausible values of ρ , κ , the Novy elasticity correction, a^j , and σ^j . We now place these findings in the context of the paper’s contributions.

6 Conclusion

The paper’s central finding is methodological and has a single unifying form: omitting multinational production flips the sign of the object of interest. In the reduced-form analysis, dropping the MP control attenuates the import-competition coefficient toward zero and, in the economy-wide cross-section, reverses it. In the structural model, a trade-only counterfactual *understates* Korea’s welfare gain from China’s 2000–2007 integration (ignoring vertical complementarity) but *overstates* it in the autarky comparison (ignoring production displacement). Complementarity and displacement are opposing forces whose net balance depends on the shock, so the direction of the bias cannot be signed a priori. Joint analysis of trade and MP is therefore necessary both for causal identification and for accurate welfare accounting.

Returning to the motivating question: did China’s rise contribute to Korea’s regional inequality? The answer is nuanced. China’s integration raised aggregate Korean welfare by approximately 2.0% (1.5% under a correlated-Fréchet correction), and the gains were unevenly distributed—per-worker welfare grew by 2.5% in the Southeast (where automotive and electronics firms maintained strong vertical linkages with Chinese affiliates) but only 1.7% in Seoul/Capital (the least-exposed region).

Contrary to the standard China-shock concentration narrative, the shock modestly *reduced* across-region inequality by disproportionately benefiting peripheral manufacturing regions. Cross-regional heterogeneity arises primarily through the migration channel: workers in regions whose industrial composition aligned with China’s growth experienced larger reallocation gains, while migration frictions prevented full equalization. A skill-stratified extension (Appendix H) shows that low-skill workers gain more than high-skill workers in *every* Korean region (aggregate differential -1.34 pp, widest in the Southeast at -1.90 pp)—so across-region inequality narrows while within-region skill inequality widens. The direct MP welfare contribution is small (0.04 pp), consistent with the estimated $a^j \in [0.04, 0.08]$; but the quantitatively first-order role of MP lies in the identification bias it corrects, not in its direct welfare contribution.

Several limitations warrant discussion. *On the empirical side*, the Rotemberg weight analysis reveals that 86% of the MP identification comes from the Motor Vehicles sector (C29), with an effective number of industries of 1.2. While the MP coefficient survives both the exclusion of C29 and a systematic leave-one-out analysis across all 17 manufacturing sectors (Appendix A.3, Table 22), the external validity beyond the automotive supply chain deserves further investigation. A second, quantitatively more consequential caveat concerns the corrected Novy elasticity specification (Table 14), which reduces the aggregate welfare gain from 2.02% to 1.53%—a 24% decline. This indicates meaningful sensitivity to the functional form used for backing out bilateral costs, though the rank ordering of regional gains and the qualitative role of each channel are preserved.

On the modelling side, regional wages within Korea are assumed uniform within each sector, so cross-regional welfare heterogeneity arises entirely through the migration channel. A richer framework with segmented regional labor markets—along the lines of Caliendo et al. (2019)—would allow region-specific wages and could more directly confront the reduced-form employment coefficients. The model is also static and abstracts from capital accumulation; since multinational production fundamentally involves capital allocation decisions, incorporating dynamics and physical capital could affect the welfare magnitudes.

The findings carry implications for regional industrial policy. Reshoring incentives targeted at MP-exposed regions would concentrate their employment effects in sectors with high vertical fragmentation (a^j), particularly automotive and electronics. Since the aggregate welfare contribution of the MP channel is modest, such policies would yield small aggregate welfare losses while poten-

tially exacerbating regional inequality by disproportionately affecting regions whose employment depends on the vertical-complementarity channel.

More broadly, the framework developed here—jointly modeling trade, MP, and migration, with a closed-form welfare decomposition (Proposition 1) that nests the Arkolakis et al. (2012) sufficient statistic and adds a migration term—is portable to other settings of simultaneous trade and multi-national production integration, including Southeast Asian economies undergoing supply-chain diversification from China. Conditional on trade-share, MP-share, and migration-share changes, aggregate and regional welfare follow from the formula without re-solving the full equilibrium. Natural next steps include exploiting post-WTO-accession variation to sharpen identification, expanding to finer regional disaggregation (the 230 Si/Gun/Gu labor markets used in the empirical analysis), and conducting counterfactuals on reshoring policies that alter bilateral MP costs.

Appendix A Robustness, First Stages, and Diagnostics

A.1 Pre-Trend Test

We test instrument validity by checking whether the *instruments* (not the endogenous exposures) predict pre-period employment growth. The exclusion restriction requires the *O*-country instrument Bartik measures to be uncorrelated with the 1994–2000 regional employment trends. Table 17 reports OLS regressions of the 1994–2000 regional employment growth rate on the three *O*-country instruments used in the main specification. Province fixed effects are included throughout; standard errors are heteroskedasticity-robust.

In the univariate specifications, the import instrument ($\Delta IM^{CHN \rightarrow O}$) is significant at 5% (column 1, $\beta = 0.185^{**}$) and the MP instrument ($\Delta MP^{O \rightarrow CHN}$) is significant at 5% (column 3, $\beta = 0.073^{**}$); the export instrument is insignificant. In the joint specification (column 4), the MP instrument becomes insignificant ($p = 0.196$), while the import instrument remains significant at 5% ($\hat{\beta} = 0.185^{**}$). The joint insignificance of the MP instrument indicates that its univariate pre-trend correlation is not independent of the import instrument’s pre-trend: both instruments reflect China’s simultaneous acceleration in manufacturing trade and inward FDI, so their pre-trend correlations are driven by a common factor. Conditional on the import instrument, the MP instrument reveals no additional pre-trend, satisfying the exclusion restriction for the MP channel.

Table 17: Instrument Pre-Trend Test: 1994–2000 Employment Growth $\sim$ *O*-Country Instruments

	Instrument Pre-Trend: 1994–2000 Employment Growth $\sim$ ROW Instrument			
	(1) Import IV	(2) Export IV	(3) MP IV	(4) Joint
$\Delta IM^{CHN \rightarrow ROW}$ (Import IV)	0.185** (0.082)			0.170** (0.083)
$\Delta EX^{ROW \rightarrow CHN}$ (Export IV)		0.208 (0.129)		-0.245 (0.204)
$\Delta MP^{ROW \rightarrow CHN}$ (MP IV)			0.073** (0.030)	0.072 (0.055)
Observations	218	218	218	218
OLS. Outcome: $(E_{r,2000} - E_{r,1994})/E_{r,1994}$. Province FE included.				
Robust SEs. H_0 : instrument is uncorrelated with pre-period employment growth.				
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$				

The residual significance of the import instrument in the joint test represents a genuine limitation for causal inference on β_{IM} under the *O*-country analog, consistent with the broader China-shock literature (Autor et al., 2013). We address this directly: Appendix A.9 constructs an alternative IV for the import channel based on China’s pre-committed WTO accession tariff reductions, which are legally fixed before the sample period and therefore immune to this pre-trend concern. The MP complementarity result is unchanged under this alternative ($\hat{\beta}_{MP} = 0.081^{**}$), and the Hansen J over-identification test ($p = 0.810$) confirms that the WTO tariff IV and the *O*-country import analog are jointly valid instruments. This rules out the possibility that the pre-trend in the *O*-country import instrument materially biases the structural estimates; nonetheless, causal inference on β_{IM} in Tables 2–3 should be interpreted with this caveat.

For completeness, Table 18 reports the analogous pre-trend test for the endogenous exposure variables themselves, which shows the expected pattern: the MP exposure measure (ΔMP_r , reflecting Korean affiliates’ actual production activity in China) is strongly correlated with pre-trends ($p < 0.01$), confirming that Korean firms’ investment decisions were not randomly allocated across regions — precisely the endogeneity that IV estimation is designed to correct.

Table 18: Endogenous Exposure Pre-Trend Test: 1994–2000 Employment Growth $\sim$ Shift-Share Exposure

	(1) Import	(2) Export	(3) MP	(4) Full
Import exposure (ΔIP_r)	0.534* (0.308)			−0.255 (0.487)
Export exposure (ΔEP_r)		0.654 (0.407)		0.602 (0.511)
MP exposure (ΔMP_r)			0.070*** (0.026)	0.057 (0.044)
Province FE	Yes	Yes	Yes	Yes
N	218	218	218	218

Notes: OLS. Outcome: $(E_{r,2000} - E_{r,1994})/E_{r,1994}$. Exposures use 2000 base-year industry shares. Robust SEs. * $p < 0.10$, *** $p < 0.01$.

A.2 Shock-Level Inference (BHJ ssaggregate)

Following Borusyak et al. (2022, 2025), we partial out S_i and region fixed effects from both the outcome and exposures, form exposure-weighted shock aggregates, and run the equivalent shock-level IV regression. This provides exposure-robust standard errors. Table 19 reports the resulting coefficients $\hat{\beta}$ for each channel. The signs are consistent with the baseline specification in Table 2 (Import is negative, MP-out is positive). However, the Kleibergen–Paap F -statistic of 0.89 at the shock level indicates that the instruments are weak in this specification; we therefore interpret the MP coefficient of 0.011 with caution, and significance stars reflect point estimates rather than reliable inference under weak instruments. The tenfold difference between the regional-level estimate ($\hat{\beta}_{MP} = 0.092$) and the shock-level estimate ($\hat{\beta}_{MP} = 0.011$) reflects differences in normalization: the regional-level estimate scales by employment-weighted industry shares, whereas the shock-level estimate uses industry-level shock units directly. These two estimates are not directly comparable without a common normalization; we present both as complementary diagnostics rather than independent replications.

A.3 Rotemberg Weights

To assess the effective sample size and identify influential industries, we compute Rotemberg weights $\hat{\alpha}_k$ following Goldsmith-Pinkham et al. (2020). These weights quantify the leverage of specific industry shocks in the estimator.

Table 20 lists the top five industries by Rotemberg weight for the outward MP channel. The

Table 19: Shock-Level IV Estimates (BHJ 2022 Exposure-Robust Inference)

	Exposure-Robust Inference (BHJ 2022)
	(1) Shock-Level 2SLS
$\beta_{\text{CHN} \rightarrow \text{KOR}}^{\text{IM}}$	-0.143 (0.130)
$\beta_{\text{KOR} \rightarrow \text{CHN}}^{\text{EX}}$	0.085 (0.068)
$\beta_{\text{KOR} \rightarrow \text{CHN}}^{\text{MP-out}}$	0.011*** (0.004)
Number of Shocks	61
1/HHI of Weights	20.23
Kleibergen-Paap F	0.89
<i>Notes:</i> Estimates correspond to the shock-level representation of the shift-share IV design following Borusyak et al. (2025). Standard errors are exposure-robust. 1/HHI measures effective number of influential shocks.	

results reveal a high degree of concentration: the **Motor Vehicles, Trailers and Semi-Trailers (C29)** sector dominates the identification, carrying a weight of 0.864 (86.4% of the total). The remaining weight splits across Chemicals (C20, 9.9%), Machinery and Equipment (C28, 8.4%), Other Transport Equipment (C30, 8.2%), and Fabricated Metal Products (C25, 7.1%). These are signed Rotemberg weights (the signed total sums to 1 by construction); the top five account for 120% of the signed total, implying that industries outside this list collectively carry approximately -20% negative weight—consistent with sectors whose O -country analog and employment response move in opposite directions, which provides some offsetting identification.

Following Goldsmith-Pinkham et al. (2020), we report two additional bias diagnostics at the bottom of Table 20. The *effective number of industries* $N_{\text{eff}} = 1/\sum_k \hat{\alpha}_k^2 = 1.2$ confirms the high concentration: the estimator draws identification from approximately one effective industry. The *bias-weighted F -statistic* (the Rotemberg-weight-averaged first-stage F for the MP channel) is 37.0, well above the conventional threshold of 10, indicating that the identifying variation from C29 is strongly relevant. This concentration reflects Korea’s large-scale automotive supply-chain integration with China, which generated both outward FDI and upstream domestic employment. While $N_{\text{eff}} = 1.2$ implies limited generalizability beyond the automotive sector, the bias-weighted F confirms the identifying shock is strong. As additional checks, we re-estimate the baseline

specification after dropping C29 (Motor Vehicles) from the exposure measure; the MP coefficient remains positive and significant (Table 21), suggesting the complementarity mechanism is not an artifact of a single outlier industry. We also construct alternative O -country baskets that replace the auto-heavy United States and Japan with France, Switzerland, and Austria; the C29 Rotemberg weight declines from 86% to 72% and N_{eff} rises to 1.8, but the concentration remains substantial, confirming its structural origin in Chinese outward MP patterns (see Appendix A.10 for details).

Table 20: Top 5 Industries by Rotemberg Weight (MP-Out Channel, 2000–2020)

ISIC Code	Rotemberg Weight ($\hat{\alpha}_k$)	Global MP Shock (g_k)	Sum of Shares ($\sum_r s_{rk}$)
C29	86.4%	66.197	3.635
C20	9.9%	27.923	1.420
C28	8.4%	23.109	1.401
C30	8.2%	9.279	2.630
C25	7.1%	5.488	4.463
<i>GP (2020) diagnostics:</i> Effective $N = 1.2$; Bias-weighted F (MP channel) = 36.98			
<i>Notes:</i> Industries are ranked by their signed Rotemberg weight $\hat{\alpha}_k$, shown as a percentage. The signed weights sum to 1 by construction; industries outside this top-5 carry negative weights collectively summing to approximately -20% .			

Leave-one-out industry robustness. To generalize the C29 exclusion test, Table 22 systematically drops each of the 17 manufacturing ISIC sectors from the shift-share exposure construction and re-estimates the preferred specification. In the economy-wide panel, the MP coefficient $\hat{\beta}^{MP}$ remains positive across all 17 leave-one-out iterations, ranging from 0.035 (dropping C29) to 0.134 (dropping C31). For 14 of 17 sectors, the coefficient lies in the narrow interval $[0.066, 0.084]$ around the baseline value of 0.093, demonstrating that no single industry drives the positive MP–employment association. The three exceptions—C29 (Motor Vehicles, $\hat{\beta} = 0.035$), C30 (Other Transport, $\hat{\beta} = 0.289$), and C31 (Furniture, $\hat{\beta} = 0.134$)—reflect C29’s dominant Rotemberg weight: dropping C29 attenuates the coefficient but preserves the sign, while dropping C30 or C31 shifts identification weight toward C29, inflating the estimate. The manufacturing-only specification shows comparable stability: $\hat{\beta}^{MP} \in [0.025, 0.031]$ for 14 of 17 sectors, with only C26 (Electronics, weak instruments when dropped) and C29 producing outliers.

Table 21: Robustness: Excluding C29 (Motor Vehicles) from Exposure Construction

	Economy-wide		Manufacturing	
	(1) Baseline	(2) No C29	(3) Baseline	(4) No C29
β^{IM}	-0.933 (1.186)	-2.152 (1.577)		
β^{EX}	0.848 (1.415)	2.530 (2.575)		
$\beta^{\text{MP-out}}$	0.093*** (0.030)	0.035 (0.258)		
β^{IM} (Mfg)			0.101 (0.463)	1.311 (3.374)
β^{EX} (Mfg)			-0.217 (0.560)	-3.171 (6.802)
$\beta^{\text{MP-out}}$ (Mfg)			0.031*** (0.010)	0.471 (0.803)
Observations	230	230	230	230
Kleibergen-Paap F	2.35	2.48	1.66	0.10
SW F : MP-out	17.80	8.68	93.80	0.39
<i>Notes:</i> columns exclude Motor Vehicles and Trailers (C29) from the shift-share exposure construction, including from the denominator of employment shares. C29 carries a Rotemberg weight of 0.864 in the baseline MP channel (Table 20). Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$				

Table 22: Leave-One-Out Industry Robustness

Dropped Sector	Economy-Wide			Manufacturing-Only		
	β^{MP}	(SE)	KP F	β^{MP}	(SE)	KP F
Food products (C10)	0.078	(0.035)	1.92	0.025	(0.007)	10.40
Textiles & apparel (C13)	0.075	(0.036)	4.48	0.031	(0.010)	1.06
Wood products (C16)	0.078	(0.035)	1.51	0.031	(0.010)	1.65
Paper products (C17)	0.076	(0.036)	1.49	0.030	(0.010)	1.66
Coke & petroleum (C19)	0.075	(0.036)	1.50	0.030	(0.010)	1.63
Chemicals (C20)	0.066	(0.041)	2.13	0.026	(0.009)	2.54
Pharmaceuticals (C21)	0.074	(0.036)	1.56	0.031	(0.010)	1.66
Rubber & plastics (C22)	0.084	(0.037)	1.34	0.030	(0.009)	1.80
Non-metallic minerals (C23)	0.078	(0.036)	1.43	0.030	(0.009)	1.26
Basic metals (C24)	0.071	(0.036)	1.23	0.029	(0.009)	1.62
Fabricated metals (C25)	0.074	(0.031)	1.43	0.027	(0.009)	1.70
Electronics & computers (C26)	0.081	(0.032)	3.30	-0.006	(0.175)	0.01
Electrical equipment (C27)	0.072	(0.036)	1.51	0.030	(0.010)	1.55
Machinery & equipment (C28)	0.075	(0.037)	1.53	0.029	(0.010)	1.69
Motor vehicles (C29)	0.035	(0.258)	2.48	0.471	(0.803)	0.10
Other transport equip. (C30)	0.289	(0.188)	0.56	0.018	(0.016)	1.64
Furniture & other mfg. (C31)	0.134	(0.091)	0.76	0.029	(0.010)	2.55
<i>Baseline (none dropped)</i>	0.093	(0.030)		0.031	(0.010)	

Notes: Each row re-estimates the preferred 2SLS specification (Table 2, Column 4) after dropping the named ISIC sector from the shift-share exposure construction, including the denominator of employment shares. C29 (Motor Vehicles) is bolded as it carries the highest Rotemberg weight in the baseline MP channel (Table 20). Robust standard errors in parentheses.

A.4 First-Stage Diagnostics

The validity of our shift-share design relies on the relevance of the O -country instruments. Table 23 reports the industry-level first-stage regressions, where we regress the Korean shock (e.g., $\Delta \text{Import}^{\text{CHN} \rightarrow \text{KOR}}$) on the O -country instrument (e.g., $\Delta \text{Import}^{\text{CHN} \rightarrow \text{O}}$), weighted by the Rotemberg weights.

We find strong positive correlations across all three channels, confirming that our instruments capture global supply-side shocks in China rather than Korea-specific demand factors.

Table 23: First-Stage Regressions: Korean Shocks on O -Country Instruments

	Economy-wide Exposure			Manufacturing Exposure		
	(1) Import	(2) Export	(3) MP-out	(4) Import	(5) Export	(6) MP-out
$\Delta \text{IM}^{\text{CHN} \rightarrow \text{ROW}}$	0.056*** (0.017)	0.060*** (0.012)	-0.061 (0.071)			
$\Delta \text{EX}^{\text{ROW} \rightarrow \text{CHN}}$	0.041 (0.039)	0.429*** (0.023)	-1.399*** (0.166)			
$\Delta \text{MP}^{\text{ROW} \rightarrow \text{CHN}}$	0.072*** (0.024)	-0.021* (0.011)	1.268*** (0.088)			
$\Delta \text{IM}^{\text{CHN} \rightarrow \text{ROW}}$ (Mfg)				0.071*** (0.006)	0.011 (0.013)	0.194*** (0.055)
$\Delta \text{EX}^{\text{ROW} \rightarrow \text{CHN}}$ (Mfg)				-0.003 (0.012)	0.432*** (0.021)	-1.680*** (0.094)
$\Delta \text{MP}^{\text{ROW} \rightarrow \text{CHN}}$ (Mfg)				0.082*** (0.008)	-0.009 (0.007)	1.192*** (0.035)
Observations	465	465	465	465	465	465
F-statistic	165.69	111.31	41.32	271.76	79.70	101.38
R^2	0.908	0.964	0.848	0.901	0.928	0.851

Standard errors in parentheses, robust to heteroskedasticity.
All regressions include the incomplete-shares control (S_r) and province fixed effects.
Each column is the OLS first stage for one endogenous variable on all three ROW instruments.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.5 Anderson–Rubin Confidence Sets

The joint Kleibergen–Paap F -statistics for the full specifications (2.35 economy-wide, 1.66 manufacturing-only) are below the conventional threshold of 10, raising weak-instrument concerns for standard Wald inference. Table 24 reports 95% projection-based Anderson–Rubin (AR) confidence sets

for each endogenous variable, computed using **weakiv** (Finlay et al., 2009). These sets are valid regardless of instrument strength and widen to the entire grid when instruments are uninformative.

In the economy-wide specification, all three AR confidence sets span the entire search grid, confirming that the joint instrument strength is insufficient for weak-instrument-robust inference on any individual coefficient. In the manufacturing specification, the AR confidence set for the MP-out coefficient is $[0.022, 0.060]$, which excludes zero and is consistent with the positive Wald estimate of 0.037. The import and export AR intervals are one-sided or uninformative, reflecting the low Sanderson–Windmeijer partial F -statistics for those regressors. Taken together, the results support a positive and statistically robust MP effect in the manufacturing sector, though weak-instrument-robust inference on the trade channels remains infeasible in the current sample.

Table 24: Anderson–Rubin 95% Projection Confidence Sets

	Economy-wide Full		Manufacturing Full	
	2SLS	AR 95% CI	2SLS	AR 95% CI
β^{IM}	-1.882	entire grid	0.101	entire grid
β^{EX}	2.061	entire grid	-0.217	entire grid
$\beta^{\text{MP-out}}$	0.074	entire grid	0.031	[.007433, .05391]
<i>Notes:</i> 95% projection-based Anderson–Rubin (AR) confidence sets computed with weakiv (Finlay et al., 2009). Intervals are projected onto each endogenous variable separately (marginalizing over remaining coefficients). “Entire grid” denotes an uninformative set spanning the full search range $[-6.78, 3.92]$ (EW) or $[-3.89, 5.91]$ (Mfg). One-sided sets reaching the grid boundary are reported as open intervals. Both specifications use the 2000-base long-difference dataset, consistent with Table 2. Robust to heteroskedasticity.				

A.6 Heterogeneity by Investment Margin (Eximbank)

To explore the mechanisms of offshoring, we re-estimate the model, replacing the OECD outward MP measure with administrative microdata from the Export-Import Bank of Korea (Eximbank). This dataset tracks the universe of outward FDI transactions by Korean firms, allowing us to distinguish the *extensive margin* (number of new affiliates, investment count) from the *intensive margin* (gross and net investment value).

Table 25 reports the results. Columns (1)–(2) use count-based extensive-margin proxies (new firms and investment count), while Columns (3)–(4) use value-based intensive-margin proxies (gross and net investment). A striking divergence emerges across margins. The extensive-margin measures

yield positive and significant MP coefficients: the number of new affiliates gives $\beta_{MP} = 0.080^{**}$ and investment count gives $\beta_{MP} = 0.041^{**}$, confirming the vertical complementarity result using firm-level administrative data. By contrast, the value-based intensive-margin proxies yield near-zero and statistically insignificant coefficients, with very weak Kleibergen–Paap F statistics (below 2), indicating that continuous investment values are poorly instrumented in the shift-share framework.

These results suggest that the *entry* of new overseas affiliates—rather than the scale of capital flows—is the dimension of multinational expansion most strongly linked to domestic employment. Each new affiliate generates demand for headquarters services, management, and upstream intermediate inputs in the home region, consistent with the vertical complementarity mechanism.

Table 25: Employment Effects by Eximbank MP Margins (Extensive vs. Intensive)

	Extensive Margin		Intensive Margin	
	(1) New Firms	(2) Inv. Count	(3) Gross Inv.	(4) Net Inv.
$\beta_{\text{New Firms}}^{\text{MP}}$	0.039 (0.031)			
$\beta_{\text{CHN} \rightarrow \text{KOR}}^{\text{IM}}$	0.398 (1.198)	0.405 (1.161)	-0.606 (1.304)	0.253 (1.009)
$\beta_{\text{KOR} \rightarrow \text{CHN}}^{\text{EX}}$	-0.062 (1.413)	-0.010 (1.315)	-0.207 (1.688)	6.265 (8.146)
$\beta_{\text{Inv. Count}}^{\text{MP}}$		0.019 (0.016)		
$\beta_{\text{Investment}}^{\text{MP}}$			0.000 (0.000)	
$\beta_{\text{Net Investment}}^{\text{MP}}$				0.000 (0.000)
Observations	230	230	230	230
Number of Shocks				
Kleibergen-Paap F	4.42	4.96	2.62	0.38
SW F : Import				
SW F : Export				
SW F : MP-out				
Standard errors are robust. All regressions include region fixed effects.				
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$				

A.7 Separate Channel Estimates

To diagnose the potential omitted variable bias discussed in Section 2.4, we estimate the effects of trade and MP channels separately. Table 26 reports these univariate and multivariate results.

Notably, Column (1) shows that when we estimate import competition in isolation, the coefficient is positive (0.493*), mirroring the "puzzle" found in previous literature. However, this coefficient turns negative in the "Trade Only" specification (Column 4), which controls for exports ($\beta = -5.456^{**}$), and remains negative in the fully saturated model controlling for MP. This confirms that omitted-variable bias—specifically the omission of Exports and MP—produces the spurious positive import–employment correlation in naive specifications.

Table 26: Robustness: Separate Channel Estimates

	(1) IM Only	(2) EX Only	(3) MP Only	(4) Trade Only	(5) Full
β^{IM}	0.493* (0.277)			-5.456** (2.769)	-0.933 (1.186)
β^{EX}		0.546* (0.322)		6.864** (3.377)	0.848 (1.415)
$\beta^{\text{MP-out}}$			0.081*** (0.028)		0.093*** (0.030)
Observations	230	230	230	230	230
Kleibergen-Paap F	34.28	749.78	122.52	2.50	2.35
Hansen J				0.000	0.000
p -value				.	.

A.8 Robustness to Trade Data Source (UN Comtrade)

As an additional check on our trade shock measures, we replace the OECD bilateral trade flows with data from UN Comtrade and re-estimate the baseline specification. The Comtrade-based import and export shocks are instrumented with O -country analogs (China's trade with the same six high-income partners). Table 27 reports the results for economy-wide and manufacturing-only exposure.

The outward MP coefficient remains positive and significant across all specifications ($\beta_{MP} = 0.121^{***}$ in the economy-wide full model; $\beta_{MP} = 0.034^{***}$ in the manufacturing full model), confirming that the vertical complementarity result is robust to the source of the trade data. The

Comtrade-based trade coefficients are smaller in magnitude than the OECD baseline and less precisely estimated, consistent with known differences in the two datasets' industry classification coverage.

Table 27: Robustness: UN Comtrade Trade Flows

	Economy-wide Exposure		Manufacturing Exposure
	(1) Trade Only	(2) Full	(3) Mfg-Only
$\beta_{\text{CHN} \rightarrow \text{KOR}}^{\text{IM}}$ (Comtrade)	-0.028 (0.120)	-0.050 (0.127)	
$\beta_{\text{KOR} \rightarrow \text{CHN}}^{\text{EX}}$ (Comtrade)	0.094 (0.068)	-0.040 (0.050)	
$\beta_{\text{KOR} \rightarrow \text{CHN}}^{\text{MP-out}}$		0.121*** (0.035)	
$\beta_{\text{CHN} \rightarrow \text{KOR}}^{\text{IM}}$ (Comtrade, Mfg)			-0.021 (0.065)
$\beta_{\text{KOR} \rightarrow \text{CHN}}^{\text{EX}}$ (Comtrade, Mfg)			0.000 (0.025)
$\beta_{\text{KOR} \rightarrow \text{CHN}}^{\text{MP-out}}$ (Mfg)			0.034*** (0.010)
Observations	230	230	230
Number of Shocks			
Kleibergen-Paap F	182.45	139.22	125.23
SW F : Import			
SW F : Export			
SW F : MP-out			
Standard errors are robust. All regressions include region fixed effects.			
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$			

A.9 Alternative IV: China WTO Accession Tariff Reductions

A potential concern with the O -country import instrument is that it captures not only supply-side shocks to China's comparative advantage but also Chinese demand cycles: periods of rapid Chinese income growth may have simultaneously expanded China's exports to Korea *and* China's demand from third-country markets, inducing spurious correlation between the O -country analog and regional employment outcomes through demand rather than supply channels. We address this concern by constructing an alternative instrument based on China's pre-committed WTO accession tariff schedule.

China’s Protocol of Accession to the WTO (December 2001) required specific, legally binding reductions in applied tariff rates by sector, phased in over 2002–2010. These commitments were negotiated before our sample period and represent exogenous variation in China’s sector-specific openness to international competition—they reduce Chinese domestic production costs and therefore expand China’s export capacity for supply-side reasons, independent of Korean regional demand or Chinese income dynamics. Following Topalova (2010) and Autor et al. (2013), we construct regional Bartik exposures as $\sum_j \ell_{rj,2000} \times \Delta\tau_j$, where $\Delta\tau_j$ is the sector-level tariff reduction from the WTO protocol and $\ell_{rj,2000}$ is the regional initial employment share in sector j . We use this instrument in place of the O -country import analog for the import competition channel; the export and MP channels retain O -country analogs.

Table 28 reports three specifications. Column (1) uses the WTO tariff Bartik as the sole import IV in the economy-wide full specification. Column (2) applies the same to the manufacturing-only shares. Column (3) is over-identified, stacking both the WTO tariff Bartik and the O -country import analog as instruments for the import channel (four instruments for three endogenous variables), which allows a formal test of IV consistency via the Hansen J statistic.

The MP coefficient in Column (1) is $\hat{\beta}_{MP} = 0.081^{**}$ (s.e. 0.034), nearly identical to the baseline estimate of 0.092 under the O -country analog. Column (2) is not suitable for causal inference: the Kleibergen–Paap F -statistic is 0.02, indicating near-complete instrument failure. This arises because the WTO tariff instrument operates at the 10-sector aggregation level, meaning all manufacturing codes within a sector group receive the same tariff reduction. When shares are restricted to manufacturing employment only (the mfg-only construction), the cross-regional variation in the Bartik exposure collapses, as nearly all manufacturing sectors carry positive tariff reductions. We accordingly disregard the Column (2) coefficients.

The key result is in Column (3): the Hansen J p -value is 0.810, far from rejecting the null that the WTO tariff Bartik and the O -country import analog are jointly valid instruments. If the O -country analog were contaminated by Chinese demand cycles, the two instruments would disagree on the structural coefficients for the import and MP channels, producing a low p -value. The failure to reject confirms that demand-side factors do not materially confound the baseline identification strategy. The over-identified MP estimate, $\hat{\beta}_{MP} = 0.077^{***}$ (s.e. 0.029), is more precisely estimated than either instrument alone, consistent with both IVs measuring the same underlying supply-side

shock.

Table 28: Robustness: Alternative IV Using China WTO Accession Tariff Reductions

	Economy-Wide	Mfg-Only	Over-ID
$\widehat{\Delta IP}_{-r}$ (import comp., EW)	-1.955 (1.927)		-1.711 (1.280)
$\widehat{\Delta EP}_{-r}$ (export, EW)	2.235 (2.276)		1.960 (1.576)
$\widehat{\Delta MP}_{-r}$ (outward MP, EW)	0.081** (0.034)		0.077*** (0.029)
$\widehat{\Delta IP}_{-r}$ (import comp., mfg)		-0.259 (4.505)	
$\widehat{\Delta EP}_{-r}$ (export, mfg)		0.243 (5.578)	
$\widehat{\Delta MP}_{-r}$ (outward MP, mfg)		0.037*** (0.009)	
KP F -stat	4.33	0.02	4.73
Hansen J p -val	.	.	0.810

Notes: Dependent variable is 20-year regional employment growth rate, 2000–2020. All columns use the 20-year long-difference specification with province fixed effects and the incomplete-shares control S_i . The instrument for the import competition channel ($\widehat{\Delta IP}_{-r}$) is the regional Bartik exposure to China’s pre-committed WTO accession tariff reductions: $\sum -j(\ell_{-rj}, 2000 \times \Delta\tau_{-j})$, where $\Delta\tau_{-j}$ is the sector-level tariff reduction from China’s WTO Protocol of Accession (2001). Export and MP channels retain Rest-of-World analogs as instruments. Column (3) is over-identified (4 instruments for 3 endogenous variables); the Hansen J p -value tests whether the WTO tariff and ROW import IVs are jointly valid. Column (2) (mfg-only) has KP $F = 0.02$ and should not be interpreted causally. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.10 O-Country Instrument Composition

A potential concern with our identification strategy is that the six-country O basket—the United States, Japan, Germany, Taiwan, the Netherlands, and Singapore—includes the United States and Japan, whose business cycles are most closely synchronized with Korea’s. If common demand shocks to these two economies correlate with Korean regional outcomes, they may weaken the exclusion restriction. A related concern is that the Rotemberg weight concentration in C29 (Motor Vehicles)—which carries 86% of the identifying variation (Table 20)—may reflect the auto-heavy composition of the O group rather than a structural feature of Chinese outward MP.

To assess these concerns, we re-estimate the preferred manufacturing-only stacked specification under seven alternative O -country compositions. Columns (1)–(4) systematically remove the United

States, Japan, or both from the baseline basket. Columns (5)–(7) replace the auto-heavy United States and Japan with France, Switzerland, and Austria—countries whose multinational production in China is concentrated in chemicals, pharmaceuticals, and electronics rather than motor vehicles. In each case, only the *O*-country aggregate used in the export and MP instruments changes; the import instrument retains the baseline six-country *O* because the importer-side aggregate is pre-computed from the OECD ICIO bilateral matrix and cannot be decomposed post-hoc.²³

Table 29 reports the results. The outward MP coefficient ($\hat{\beta}_{MP}$) is positive and significant across all seven compositions, ranging from 0.022 to 0.055, confirming that demand-side contamination from the United States or Japan does not drive the baseline result. The diversified baskets (Columns 5–7) yield point estimates that are somewhat larger than the baseline, though with wider confidence intervals reflecting reduced instrument power.

The diversified baskets also partially address the Rotemberg concentration concern. Under the “Diversified” composition (Column 5), which replaces the United States and Japan with France and Switzerland, the C29 Rotemberg weight declines from 86% to 72% and the effective number of industries rises from $N_{\text{eff}} = 1.2$ to 1.8. That C29 remains dominant despite the removal of the two most auto-heavy *O* countries suggests the concentration is partly structural—reflecting the automotive sector’s outsized role in Chinese outward multinational production—rather than an artifact of the baseline country selection. The Kleibergen–Paap *F*-statistic declines for the diversified baskets (2.1–3.4 vs. 8.8 at baseline), reflecting the smaller total MP volumes of the replacement countries.

A.11 Migration Response: Full Estimates

Section 2.4.3 summarizes the reduced-form migration responses to trade and MP shocks. Here we report the full 2SLS estimates. Using Population Census 2% micro-data (2000, 2010, 2020), we construct an inter-province in-migration rate for each si-gun-gu from the five-year-ago residence question. The dependent variable is the 20-year change in the share of employed workers whose province of residence five years prior differs from their current province. Table 30 reports 2SLS estimates using the same instruments and province fixed effects as the main specification.

²³The WTO tariff IV in Appendix A.9 provides an independent robustness check for the import channel that does not rely on the *O*-country aggregate.

Table 29: Robustness: Alternative *O*-Country Instrument Compositions

	Baseline (1)	No USA (2)	No Japan (3)	No USA/JPN (4)	Diversified (5)	Broad (6)	European (7)
$\widehat{\Delta MP_r}$ (outward MP)	0.034*** (0.013)	0.041*** (0.012)	0.022 (0.013)	0.039*** (0.015)	0.055* (0.030)	0.041** (0.019)	0.055* (0.030)
$\widehat{\Delta IP_r}$ (import comp.)	-0.293 (0.119)	-0.389 (0.121)	-0.178 (0.102)	-0.338 (0.119)	-0.497 (0.257)	-0.380 (0.187)	-0.501 (0.259)
$\widehat{\Delta EP_r}$ (export)	0.113 (0.040)	0.151 (0.041)	0.072 (0.033)	0.130 (0.039)	0.188 (0.087)	0.147 (0.065)	0.190 (0.088)
KP <i>F</i> -stat	8.81	11.30	5.93	7.60	2.09	3.43	2.08
<i>N</i>	465	465	465	465	465	465	465

O-country basket composition:

(1) USA, JPN, DEU, TWN, NLD, SGP; (2) drop USA; (3) drop JPN; (4) drop USA & JPN;

(5) DEU, TWN, NLD, SGP, FRA, CHE; (6) all 9; (7) DEU, TWN, NLD, SGP, FRA, CHE, AUT.

Stacked panel (10-year intervals), manufacturing-only exposure shares.

Import IV uses baseline *O*-group (importer-side aggregate); see Appendix A.9 for alternative.

SEs clustered by region. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 30: Migration Response to Trade and MP Exposure

	Inter-Prov Migration (1) EW Trade	Mover Rate (2) EW Full	Census Emp (3) Mfg Full	(4) EW Full	(5) EW Full
Δ Imports from China	-0.1059 (0.0716)	-0.1106 (0.0911)		-0.0001 (0.0001)	0.3185 (0.7518)
Δ Imports from China (Mfg)			-0.0312 (0.0344)		
Δ Exports to China	0.1232 (0.0912)	0.1269 (0.1064)		0.0002 (0.0002)	-0.5579 (0.8870)
Δ Exports to China (Mfg)			0.0332 (0.0407)		
Δ Outward MP in China		0.0003 (0.0021)		-0.0000 (0.0000)	0.0634 (0.0210)
Δ Outward MP in China (Mfg)			-0.0005 (0.0006)		
Observations	234	234	234	234	234
KP rk Wald F	6.82	2.26	1.73	2.26	2.26

Notes: Province FE included. Robust standard errors in parentheses. “Inter-prov migration” is the change in the share of employed workers from a different province (Census five-year-ago residence question). “Mover rate” is the change in the total geographic mobility rate. “Census emp” is Census-based employment growth, included for validation against the establishment survey measure used in the main results.

Columns (1)–(3) report the inter-province in-migration regression. Import exposure enters with a negative coefficient and export exposure with a positive coefficient, consistent with directed reallocation along the trade margin; MP-out is close to zero and insignificant. Column (4) replaces the dependent variable with the total mover rate (any cross-*any*-geography move); all three coefficients are essentially zero, indicating that the trade and MP shocks redirect where workers move rather than change whether they move. Column (5) is a validation exercise that regresses Census-based employment growth on the same instruments; the MP-out coefficient is positive and significant, mirroring Table 2. First-stage diagnostics (KP $F = 1.73$ – 6.82) are below the conventional threshold of 10 in the migration specifications; we therefore read the estimates as suggestive patterns, with Column (5) the most sharply identified. Together, the pattern—trade-driven directed migration, a local (non-migratory) MP employment response, and unchanged total mobility—motivates the structural model in Section 3, which endogenizes cross-regional spillovers through the migration share μ_{fg}^j .

Appendix B Proofs

B.1 Proof of Proposition 1

We follow steps similar to those in Galle et al. (2023).

By (14),

$$\lambda_{ii}^j = \frac{T_i^j \left(\tilde{c}_{ii}^j \right)^{-\theta^j}}{\sum_k T_k^j \left(\tilde{c}_{ki}^j \right)^{-\theta^j}} = \frac{T_i^j \left(\tilde{c}_{ii}^j \right)^{-\theta^j}}{\left(\Gamma^j P_i^j \right)^{-\theta^j}}$$

This implies that

$$\begin{aligned}
P_i^j &= \Gamma^j \left(\frac{T_i^j}{\lambda_{ii}^j} \right)^{-1/\theta^j} \tilde{c}_{ii}^j \\
&= \Gamma^j \left(\frac{T_i^j}{\lambda_{ii}^j} \right)^{-1/\theta^j} c_{ii}^j \left(\lambda_{iii}^j \right)^{(1-\rho)/\theta^j} \\
&= \Gamma^j \left(\frac{T_i^j}{\lambda_{ii}^j} \right)^{-1/\theta^j} \left(\lambda_{iii}^j \right)^{(1-\rho)/\theta^j} \Upsilon_i^j (w_i^j)^{\gamma_i^j} \prod_{k=1}^J (P_i^k)^{\gamma_i^{kj}}
\end{aligned}$$

Then, we have

$$\ln P_i^j = \ln \left(\Gamma^j \left(\frac{T_i^j}{\lambda_{ii}^j} \right)^{-1/\theta^j} \left(\lambda_{iii}^j \right)^{(1-\rho)/\theta^j} \Upsilon_i^j (w_i^j)^{\gamma_i^j} \right) + \sum_k \gamma_i^{kj} \ln P_i^k$$

Let $\gamma_i \equiv \left\{ \gamma_i^{kj} \right\}_{k \in J, j \in J}$ be J by J matrix of IO coefficients, $B_i \equiv \left\{ \ln \left(\Gamma^j \left(\frac{T_i^j}{\lambda_{ii}^j} \right)^{-1/\theta^j} \left(\lambda_{iii}^j \right)^{(1-\rho)/\theta^j} \Upsilon_i^j (w_i^j)^{\gamma_i^j} \right) \right\}$ be J by 1 matrix, and $X_i \equiv \left\{ \ln P_i^j \right\}$ be a J by 1 matrix. Then, we have

$$X_i = (I - \gamma_i^T)^{-1} B_i$$

Let $\tilde{a}_i^{jk}$ be the typical element of matrix $(I - \gamma_i^T)^{-1}$. Then,

$$\ln P_i^j = \sum_k \tilde{a}_i^{jk} B_{ik}$$

$$P_i^j = \prod_k \left(\Gamma^k \left(\frac{T_i^k}{\lambda_{ii}^k} \right)^{-1/\theta^k} \left(\lambda_{iii}^k \right)^{(1-\rho)/\theta^k} \Upsilon_i^k (w_i^k)^{\gamma_i^k} \right)^{\tilde{a}_i^{jk}}$$

Then, we have

$$\hat{P}_i^j = \prod_k \left(\left(\hat{\lambda}_{ii}^k \right)^{1/\theta^k} \left(\hat{\lambda}_{iii}^k \right)^{(1-\rho)/\theta^k} (\hat{w}_i^k)^{\gamma_i^k} \right)^{\tilde{a}_i^{jk}}$$

The aggregate consumer price index is a Cobb-Douglas aggregator of sectoral price indices with

final consumption shares α_i^j :

$$\hat{P}_i = \prod_j \left(\hat{P}_i^j \right)^{\alpha_i^j} = \prod_{j,k} \left(\left(\hat{\lambda}_{ii}^k \right)^{1/\theta^k} \left(\hat{\lambda}_{iii}^k \right)^{(1-\rho)/\theta^k} \left(\hat{w}_i^k \right)^{\gamma_i^k} \right)^{\alpha_i^j \tilde{a}_i^{jk}}$$

From the worker's migration optimality condition, the indirect utility of a worker from group g who works in sector k of region g satisfies $\hat{\Phi}_{ig} = (\hat{\mu}_{igg}^k)^{-1/\kappa_g} \hat{w}_i^k$ for each sector k . Since workers are indifferent across sectors at the optimum, this equality holds for every k simultaneously, so $\hat{\Phi}_{ig}$ is a well-defined scalar independent of k . Substituting $\hat{I}_{ig} = \hat{\Phi}_{ig}$ and taking the product over sectors gives

$$\begin{aligned} \frac{\hat{I}_{ig}}{\hat{P}_i} &= \frac{\hat{\Phi}_{ig}}{\prod_{j,k} \left(\left(\hat{\lambda}_{ii}^k \right)^{1/\theta^k} \left(\hat{\lambda}_{iii}^k \right)^{(1-\rho)/\theta^k} \left(\hat{w}_i^k \right)^{\gamma_i^k} \right)^{\alpha_i^j \tilde{a}_i^{jk}}} \\ &= \frac{1}{\prod_{j,k} \left(\left(\hat{\lambda}_{ii}^k \right)^{1/\theta^k} \left(\hat{\lambda}_{iii}^k \right)^{(1-\rho)/\theta^k} \left(\hat{\Phi}_{ig} \right)^{-\gamma_i^k} \left(\hat{w}_i^k \right)^{\gamma_i^k} \right)^{\alpha_i^j \tilde{a}_i^{jk}}} \left(\because \sum_{j,k} \alpha_i^j \gamma_i^k \tilde{a}_i^{jk} = 1^{24} \right) \\ &= \frac{1}{\prod_{j,k} \left(\left(\hat{\lambda}_{ii}^k \right)^{1/\theta^k} \left(\hat{\lambda}_{iii}^k \right)^{(1-\rho)/\theta^k} \left(\hat{\mu}_{igg}^k \right)^{\gamma_i^k / \kappa_g} \right)^{\alpha_i^j \tilde{a}_i^{jk}}} \end{aligned}$$

Therefore, we have

$$\hat{W}_{ig} = \prod_{j,k} \left(\hat{\lambda}_{ii}^k \right)^{-\alpha_i^j \tilde{a}_i^{jk} / \theta^k} \prod_{j,k} \left(\hat{\lambda}_{iii}^k \right)^{-(1-\rho) \alpha_i^j \tilde{a}_i^{jk} / \theta^k} \prod_{j,k} \left(\hat{\mu}_{igg}^k \right)^{-\alpha_i^j \tilde{a}_i^{jk} \gamma_i^k / \kappa_g}$$

Appendix C Equilibrium Uniqueness

A potential concern with any multi-country, multi-sector general equilibrium model is that the system of nonlinear equations may admit multiple solutions, raising the question of whether counterfactual welfare conclusions depend on which equilibrium the solver selects. We address this

²⁴This identity follows from the Cobb-Douglas cost normalization. Let γ_i^j denote sector j 's labor cost share and γ_i^{kj} its intermediate-input cost share from sector k . Cost shares must sum to one: $\gamma_i^j + \sum_k \gamma_i^{kj} = 1$ for every j , which in matrix form reads $(I - \gamma_i^T) \mathbf{1} = \gamma_i^L$, where $[\gamma_i^L]_j = \gamma_i^j$. Pre-multiplying by $\tilde{A}_i \equiv (I - \gamma_i^T)^{-1}$ gives $\tilde{A}_i \gamma_i^L = \mathbf{1}$, i.e., $\sum_k \tilde{a}_i^{jk} \gamma_i^k = 1$ for every j . Weighting by the final-expenditure shares α_i^j (which satisfy $\sum_j \alpha_i^j = 1$) and summing over j yields $\sum_{j,k} \alpha_i^j \tilde{a}_i^{jk} \gamma_i^k = 1$.

concern through three complementary checks: a global check via random restarts, a local check via homotopy continuation, and analytical sufficient conditions.

Random restarts. We solve the baseline equilibrium from $K = 20$ random initial wage vectors, each drawn independently as $W_0 = \exp(0.5 \cdot Z)$ where $Z \sim \mathcal{N}(0, I_{N \times J})$. All 20 restarts converge. Aggregate Korean welfare agrees to five decimal places across restarts (range $[1859.836, 1859.848]$, relative spread below 10^{-5}), and the maximum pairwise log-wage deviation is $\max_{k,l} \|\log W_k^* - \log W_l^*\|_\infty = 6.6 \times 10^{-4}$, at the level of the solver’s convergence tolerance (10^{-5} in log-wages with damping). We interpret this as strong numerical evidence of a unique equilibrium within the domain of economically plausible initial conditions.

Homotopy trace. We construct a homotopy from the 2000 baseline to the 2007 counterfactual by setting trade and MP costs to $d(\alpha) = d_{2000} + \alpha \cdot (d_{2007} - d_{2000})$ for $\alpha \in \{0, 0.02, 0.04, \dots, 1\}$ (51 steps), with each step warm-started from the previous solution. Figure 6 shows the resulting welfare path: aggregate welfare declines smoothly and monotonically from 1859.89 at $\alpha = 0$ to 1829.24 at $\alpha = 1$. The maximum consecutive welfare change (0.73) is within 20% of the mean change (0.61), indicating no bifurcation or discontinuous jump. This provides evidence that along the observed shock path—the economically relevant domain—the equilibrium correspondence is single-valued and welfare is a smooth function of shock intensity.

Analytical conditions. We verify three analytical conditions relevant to uniqueness. First, the Leontief invertibility condition holds at the calibrated IO coefficients: the spectral radius of the country-specific input-output matrices is well below 1 for all $N = 10$ countries (maximum $\rho = 0.64$ for China; all others below 0.55). This guarantees existence of the price-index fixed point. Second, the Allen and Arkolakis (2014) sufficient condition—requiring the migration elasticity κ to exceed θ^j times the maximum IO column sum for every sector—is not satisfied at our calibrated $\kappa = 1.5$ and sectoral trade elasticities (required thresholds range from 2.1 to 9.6). Third, the spectral radius of the numerically computed Jacobian of the wage-update map at the fixed point is $\rho(J) \approx 10.6 > 1$. The failure of these analytical contraction conditions means that our fixed-point iteration converges via damping (damping factor 0.95) rather than via a strict contraction mapping.

Taken together, the analytical conditions do not rule out multiplicity of equilibria globally.

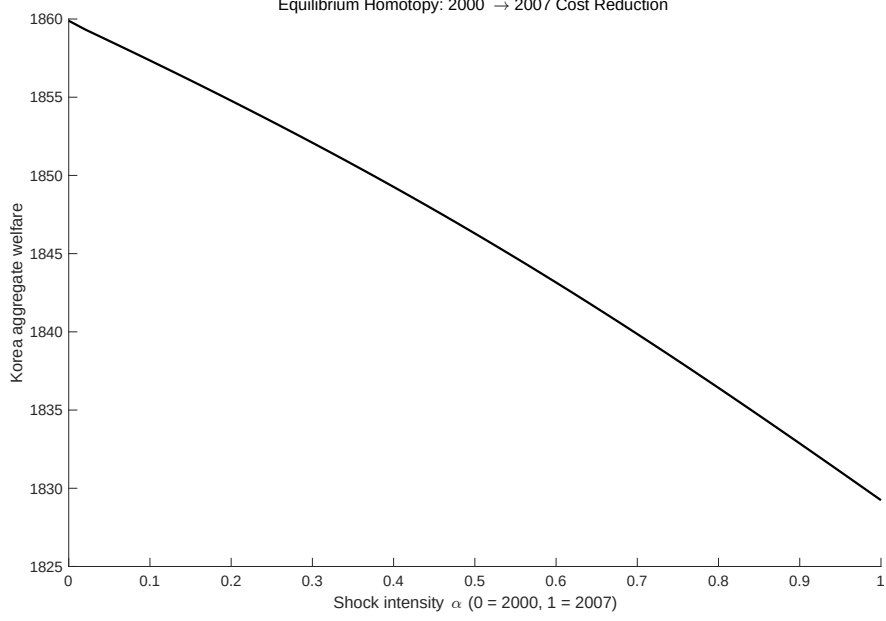


Figure 6: Homotopy trace: Korean welfare along the 2000–2007 shock path.

Notes: Horizontal axis is shock intensity $\alpha \in [0, 1]$, where $\alpha = 0$ corresponds to 2000 trade and MP costs and $\alpha = 1$ to 2007 costs. Vertical axis is aggregate Korean welfare. The smooth, monotone path rules out bifurcations along the observed shock.

However, the numerical evidence—universal convergence from random restarts within numerical precision and a smooth, monotone homotopy path—establishes uniqueness along the observed shock path. While we cannot rule out the existence of additional equilibria at extreme parameter configurations far from the calibrated region, the welfare conclusions in Section 5 are robust to equilibrium selection within the economically relevant domain.

Appendix D Model Discussion: Comparison with Related Frameworks

D.1 Comparison with Caliendo and Parro (2015)

In our model, country n 's imports from country l , M_{nl}^j , are given by equation (17). In contrast to Caliendo and Parro (2015), the right-hand side includes two additional terms from the presence of multinational production, specifically from the input bundle imports of foreign affiliates from their home country. This implies that an increase in country l 's MP in country n , captured by Y_{ln}^k , disproportionately raises country n 's imports from country l . Moreover, the input-output

coefficients of these bundles follow the home country l 's input-output structure. As a result, changes in MP costs lead to distinct trade patterns and welfare outcomes compared to changes driven by trade costs, productivity, or both (T_i^j).

D.2 Antràs and De Gortari (2020) and De Gortari (2019)

De Gortari (2019) provides evidence of “specialized inputs”: goods sold to different countries and industries are produced with different input mixes. The conventional roundabout model, as in Caliendo and Parro (2015), cannot explain this pattern, yet the pattern can significantly alter counterfactual results. Antràs and De Gortari (2020) develop a multi-stage general equilibrium model in which production functions depend on the downstream use of outputs, offering a theoretical foundation for “specialized inputs.”

Our model offers a parsimonious framework for rationalizing “specialized inputs.” It assumes that firms operating in the same country use different input mixes depending on their country of origin. Furthermore, as indicated by π_{iln}^j in equation (13), countries exhibit varying spending shares on goods from other countries, which also differ depending on whether domestic firms or foreign affiliates produce the goods. Consequently, for a given country-sector, the input mix depends on the destination country and industry. In this respect, our model aligns with their framework. However, while Antràs and De Gortari (2020) primarily studies how countries specialize within GVCs in a world with trade barriers, we study how changes in both trade and MP barriers affect regional economies.

D.3 Regional wage heterogeneity: model assumption and its limitations

The reduced-form evidence in Section 2 documents differential employment growth rates across Korean regions, consistent with heterogeneous regional exposure to China’s trade and MP integration. However, our model abstracts from region-specific wages within Korea: wages w_i^j are common across all regions of country i within sector j , and regional differences in workers’ realized incomes arise solely through migration costs ν_{igh} and idiosyncratic productivity draws governed by A_{if}^j . This assumption follows Galle et al. (2023) and is motivated by the law of one wage within a sector that holds in any model with integrated domestic goods markets and costless within-sector labor mobility. In the long run, with sufficiently elastic internal migration, regional wages are equalized

within a sector, and heterogeneous regional welfare outcomes arise through differences in labor reallocation $\hat{\mu}_{igg}^k$ (the third term in Proposition 1).

The shift-share regressions identify a causal effect of MP and trade exposure on regional employment growth, which captures the net flow of workers *into* or out of the exposed region. The model’s analog is $\hat{\mu}_{igg}^k$, the change in the share of workers from origin group g employed in sector k of their home region. These two objects are directionally consistent, and the model can qualitatively match the sign and relative ordering of regional employment responses. However, the model does not allow for the transitory wage differentials that likely arise within Korea during the adjustment period, particularly in manufacturing-intensive regions where MP restructuring is concentrated. A richer model with segmented regional labor markets—along the lines of Caliendo et al. (2019) or Adão et al. (2023)—would allow sector-region-specific wages and could more directly discipline the calibration against the reduced-form employment coefficients. We view this as a natural extension.

Appendix E Extended Calibration Methodology

This appendix provides details on three calibration procedures summarized in Section 4: the joint identification of trade and MP costs when $a^j > 0$ (Appendix E.1), the derivation of A_{if}^j from migration shares, and the source of the trade elasticity θ^j .

E.1 Joint Identification of Trade and MP Costs when $a^j > 0$

The identification challenge. When $a^j = 0$, trade costs d_{in}^j and MP costs h_{in}^j can be recovered from trade shares and affiliate production shares independently: the Novy (2013) formula inverts the trade-share equation holding MP terms fixed, and an analogous inversion recovers MP costs from affiliate production shares. When $a^j > 0$, this clean separation breaks down. The affiliate’s marginal cost in country l is

$$c_{il}^j = h_{il}^j \cdot \omega_{il}^j \cdot (w_l^j)^{(1-a^j)\gamma_l^j} \prod_k (P_l^k)^{(1-a^j)\gamma_l^{kj}}, \quad \omega_{il}^j = [a^j (c_i^j d_{il}^j / c_{il}^j)^{1-\xi} + (1-a^j)]^{1/(1-\xi)},$$

where ω_{il}^j depends on the home input bundle — and hence on trade costs d_{il}^j — through c_i^j . As a result, the Fréchet price index for country n includes contributions from affiliates whose costs

embed trade costs, so Novy applied to total ICIO trade shares (which include intra-firm flows) gives a composite of d and h rather than d alone.

Joint identification via simultaneous moment matching. The correct approach simultaneously fits both the bilateral trade share moments and the bilateral affiliate production share moments. Parameterize costs as gravity functions of distance, border, and language:

$$d_{in}^j = \exp(\delta_0^j + \delta_{dist}^j \text{dist}_{in} + \delta_{bord}^j b_{in} + \delta_{lang}^j l_{in}), \quad h_{in}^j = \exp(\zeta_0^j + \zeta_{dist}^j \text{dist}_{in} + \zeta_{bord}^j b_{in} + \zeta_{lang}^j l_{in}),$$

for $i \neq n$, with $d_{ii}^j = h_{ii}^j = 1$. Let $\Delta \equiv \{\delta^j, \zeta^j\}$ collect all gravity parameters. Given Δ and a^j , T_i^j is calibrated to match sectoral output or real income. The algorithm proceeds as follows:

- **Step 1.** Given Δ , compute d_{in}^j and h_{in}^j . Calibrate a^j to match OECD AAMNE-XVEM moments for the home-input expenditure share ω_{il}^j . Calibrate T_i^j to match sectoral output $\sum_n Y_{ni}^j$.
- **Step 2.** Given $[\Delta, a^j, T_i^j]$, solve for the equilibrium wages and prices. Compute bilateral trade and MP shares:

$$\lambda_{in}^{j, \text{Trade}} \equiv \frac{M_{in}^j}{\sum_l M_{ln}^j}, \quad \lambda_{in}^{j, \text{MP}} \equiv \frac{Y_{in}^j}{\sum_l Y_{ln}^j}.$$

Measure fit via normalized sum-of-squared deviations:

$$R^{\text{Trade}} \equiv 1 - \frac{\sum_{j,i,n: i \neq n} (\lambda_{in}^{j, \text{Trade}, \text{data}} - \lambda_{in}^{j, \text{Trade}, \text{model}})^2}{\sum_{j,i,n: i \neq n} (\lambda_{in}^{j, \text{Trade}, \text{data}})^2}, \quad R^{\text{MP}} \equiv 1 - \frac{\sum_{j,i,n: i \neq n} (\lambda_{in}^{j, \text{MP}, \text{data}} - \lambda_{in}^{j, \text{MP}, \text{model}})^2}{\sum_{j,i,n: i \neq n} (\lambda_{in}^{j, \text{MP}, \text{data}})^2}.$$

- **Step 3.** Choose Δ to minimize $(1 - R^{\text{Trade}}) + (1 - R^{\text{MP}})$. Because both objectives are minimized over the *same* parameter vector Δ — which jointly governs d and h — the procedure simultaneously identifies trade and MP costs from the combination of ICIO trade shares and AAMNE affiliate production shares. This resolves the identification problem: neither the trade share moments alone nor the affiliate production moments alone suffice when $a^j > 0$, but the joint system is identified under standard gravity regularity conditions.

Counterfactual cost changes across two equilibria. The two-equilibrium counterfactual (2000 vs. 2007) compares two solved equilibria that share the same non-cost fundamentals (A_{if}^j, T_i^j calibrated at 2000) but differ in trade and MP costs. Concretely, we apply Steps 1–3 above to each cross-section separately, obtaining $\{d_{in,2000}^j, h_{in,2000}^j\}$ and $\{d_{in,2007}^j, h_{in,2007}^j\}$; we then solve the full general-equilibrium system at both sets of costs and report welfare as the ratio of the 2007 equilibrium to the 2000 equilibrium, following Caliendo and Parro (2015) and Galle et al. (2023). The Proposition 1 decomposition is expressed in hat form $\hat{\lambda}_{ii}^k, \hat{\lambda}_{iii}^k, \hat{\mu}_{igg}^k$, and is computed post hoc from the two-equilibrium shares. Because non-cost fundamentals are held fixed, the T_i^j normalization cancels in the ratio.

Quantitative assessment. We assess the magnitude of the approximation error from using the standard Novy formula and the fit of the resulting cost calibration. Three findings emerge. First, the intra-firm sourcing share — the key driver of Novy misspecification — is modest: intra-firm inputs sourced from the Korean parent account for approximately **8.0%** of total Korean affiliate output in China. Given $a^j \in [0.04, 0.08]$ across sectors, the contamination of observed trade shares by intra-firm flows is small, so the approximation error in the Novy cost backout is of limited quantitative importance. Second, the domestic-diagonal fit statistics under the current (Novy-based) calibration are $R^{Trade} = 0.90$ and $R^{MP} = 0.79$, indicating that the single-step procedure already matches home-market trade and production shares reasonably well. Third, cost changes between 2000 and 2007 are mild in aggregate (mean $\hat{d} = 0.993$, mean $\hat{h} = 0.990$ across all bilateral pairs), with approximately 52% of country-pair observations showing trade cost reductions and 53% showing MP cost reductions. The Korea–China corridor exhibits larger decreases, particularly in Transportation ($\hat{d} = 0.71, \hat{h} = 0.66$) and Chemicals ($\hat{d} = 0.82, \hat{h} = 0.81$), consistent with the broad-based opening of China’s economy over this period. These magnitudes align with the welfare effects computed in Section 5.

Remark (simplified case: $a^j = 0$). When $a^j = 0$, the joint identification problem simplifies: trade costs can be recovered from trade shares alone via the standard Novy (2013) inversion, and MP costs from affiliate production shares independently. The algorithm above reduces to the procedure in Ramondo and Rodríguez-Clare (2013), with Steps 1–3 applied to the separable system. This

simplified case serves as the starting point for the joint identification; the current implementation uses the full algorithm with estimated $a^j > 0$.

E.2 Shape parameter of the productivity distribution: θ^j

We use sector-specific estimates of the trade elasticity θ^j from Caliendo and Parro (2015) (Table 1, 99% sample), which range from 2.62 (Food) to 13.53 (Mining). These estimates are obtained from triple-differenced gravity equations and are standard in the quantitative trade literature. In a single-sector context, Ramondo and Rodríguez-Clare (2013) and Arkolakis et al. (2018) choose θ so that the trade elasticity estimated from simulated data matches the value observed in real-world data. Our multi-sector setting requires sector-specific values, which the Caliendo–Parro estimates provide directly.

Appendix F Algorithm to solve the equilibrium

We solve the equilibrium using a nested fixed-point iteration, following Fan (2019) and Alvarez and Lucas (2007). The algorithm exploits the block structure of the equilibrium system: given wages, prices can be computed from the Fréchet price-index equation; given prices, excess labor demand determines the wage update.

1. **Initialization.** Set iteration counter $t = 0$. Initialize wages $\{w_i^{j,(0)}\}$ at their calibrated baseline values and price indices $\{P_i^{j,(0)}\}$ using the corresponding baseline values. Set convergence tolerance $\varepsilon = 10^{-8}$.
2. **Inner loop (price update).** Given current wages $\{w_i^{j,(t)}\}$, compute unit costs c_{il}^j from (9) and (11), expenditure shares π_{iln}^j from (13), and iterate the price-index equation (14) until convergence:

$$P_i^{j,(s+1)} = f(\{w_i^{j,(t)}\}, \{P_i^{j,(s)}\}), \quad s = 0, 1, 2, \dots$$

Continue until $\max_{i,j} |P_i^{j,(s+1)} - P_i^{j,(s)}| / P_i^{j,(s)} < \varepsilon$. Denote the converged prices $\{P_i^{j,(t)}\}$.

3. **Labor market evaluation.** Using $\{w_i^{j,(t)}\}$ and $\{P_i^{j,(t)}\}$, compute: (a) MP values Y_{il}^j from (15); (b) expenditures $P_n^j Q_n^j$ from (16); (c) labor demand LD_i^j from (20); (d) migration shares μ_{igf}^j and labor supply Z_{igf}^j from (21)–(22). Construct excess labor demand ELD_i^j from (24).

4. **Wage update.** Update wages using the excess-demand tatonnement of Alvarez and Lucas (2007):

$$w_i^{j,(t+1)} = w_i^{j,(t)} \cdot \left(1 + \delta \cdot \frac{ELD_i^{j,(t)}}{LD_i^{j,(t)}} \right)$$

where $\delta \in (0, 1)$ is a dampening parameter (we use $\delta = 0.3$). Re-normalize so that the US composite wage index equals unity.

5. **Convergence check.** If $\max_{i,j} |ELD_i^{j,(t)}|/LD_i^{j,(t)} < \varepsilon$, stop. Otherwise, set $t \leftarrow t + 1$ and return to Step 2.

For the two-equilibrium counterfactual (2000 vs. 2007), we apply the algorithm above separately to each cross-section—holding non-cost fundamentals (A_{if}^j, T_i^j) fixed at their 2000 calibrated values—and report welfare as the ratio W_{2007}/W_{2000} , following Caliendo and Parro (2015) and Galle et al. (2023). Under this ratio-of-equilibria design the T_i^j normalization cancels, so recovering its levels is not required; hat objects such as $\hat{\lambda}_{ii}^k = \lambda_{ii,2007}^k/\lambda_{ii,2000}^k$ are computed post hoc from the two-equilibrium solutions for the Proposition 1 decomposition.

Appendix G Calibration Validation: Model vs. Data

To validate the calibration, we compare the model’s equilibrium predictions with observed data along two dimensions: import penetration of Chinese goods and outward multinational production ratios.

Import penetration. Following Autor et al. (2013), we define country n ’s import penetration ratio for sector- j Chinese goods as $IP_n^j = \sum_i \pi_{i,\text{China},n}^j$, computed from the model’s equilibrium shares, and compare against values from the 2007 World Input-Output Table (WIOT). The pre-adjustment equilibrium overstates China’s import penetration; after the productivity adjustment described in Section 4, the model-implied ratios align more closely with the observed values.

Outward MP ratios. The outward MP ratio is $OMP_i^j = \sum_{l \neq i} Y_{il}/Y_{ii}$, with the China-specific variant $OMP_i^{j,\text{China}} = Y_{i,\text{China}}/Y_{ii}$, benchmarked against OECD AMNE data. As with import penetration, the productivity adjustment brings the model closer to observed values.

Appendix H Skill-Stratified Extension

The main text treats labor as homogeneous ($S = 1$). This appendix reports an extension in which labor in each country–sector (n, j) is a Cobb–Douglas bundle of high- and low-skill efficiency units. The extension preserves every structural element outside the wage block (trade costs, MP costs, IO coefficients, Fréchet origin–MP country draws, a^j , ρ , ξ) and introduces only two new primitives: sector-specific skill intensities $\alpha_H(j) \in (0, 1)$ and skill-specific migration elasticities κ_s . We first document reduced-form evidence of skill-heterogeneous responses to trade and MP exposure (§H.1), which motivates the structural extension developed in the remainder of the appendix.

H.1 Reduced-form skill heterogeneity

The main specification in Section 2.4 treats labor as homogeneous. However, if MP affiliates disproportionately demand skilled workers (engineers, managers) at the home headquarters while relocating production tasks abroad, the complementarity effect should be concentrated among high-skill workers. We test this prediction using two approaches, with skill defined as having completed a four-year university degree or above (Census education code ≥ 6).

Approach A: Skill-specific employment growth. We replace the establishment-survey employment growth rate with Census-based skill-specific employment growth as the dependent variable. Table 31 reports 2SLS estimates for high-skill and low-skill employment growth, using both economy-wide and manufacturing-only instruments.

Approach B: Subgroup analysis. We split the sample at the median base-year (2000) high-skill employment share and re-estimate the main specification separately for high-skill and low-skill regions (Table 32). If MP complementarity operates through skilled labor demand, the MP coefficient should be larger in regions with higher initial skill endowments.

The reduced-form evidence motivates the structural extension developed in the remainder of this appendix: if low- and high-skill workers face different labor demand shifts from China’s MP expansion, the welfare accounting must track the two groups separately.

Table 31: Skill-Specific Employment Growth: Trade and MP Exposure

	Economy-Wide (1) High-Skill	Manufacturing-Only (2) Low-Skill	(3) High-Skill	(4) Low-Skill
Δ Imports from China	-0.236 (3.617)	0.175 (0.575)		
Δ Imports from China (Mfg)			-0.279 (0.958)	0.062 (0.313)
Δ Exports to China	-0.651 (4.373)	-0.383 (0.673)		
Δ Exports to China (Mfg)			0.077 (1.168)	-0.117 (0.364)
Δ Outward MP in China	0.231** (0.106)	0.053*** (0.014)		
Δ Outward MP in China (Mfg)			0.038 (0.028)	0.014*** (0.005)
Observations	234	234	234	234
KP rk Wald F	2.26	2.26	1.73	1.73
SW F (Import)	11.54	11.54	5.26	5.26
SW F (Export)	13.20	13.20	5.20	5.20
SW F (MP)	15.90	15.90	89.01	89.01

Notes: Province FE included. Robust standard errors in parentheses. High-skill = four-year university degree or above (Census education code ≥ 6). Dependent variables are skill-specific employment growth rates computed from Population Census 2% micro-data.

Table 32: Subgroup Analysis by Regional Skill Composition

	(1) High-Skill Regions	(2) Low-Skill Regions
Δ Imports from China	-3.993 (3.785)	2.703 (2.247)
Δ Exports to China	4.835 (4.486)	-1.613 (1.797)
Δ Outward MP in China	0.048 (0.067)	0.082 (0.091)
Observations	114	116
KP rk Wald F	0.59	3.85
SW F (Import)	4.49	13.12
SW F (Export)	4.95	15.99
SW F (MP)	3.15	9.82

Notes: Split at median base-year (2000) high-skill employment share across si-gun-gu units. Province FE included. Robust standard errors in parentheses.

H.2 Model

Labor services in country–sector (n, j) are produced by a Cobb–Douglas aggregator of high- and low-skill efficiency units:

$$L_n^j = \left[Z_{H,n}^j \right]^{\alpha_H^j} \left[Z_{L,n}^j \right]^{1-\alpha_H^j}, \quad \alpha_H^j \in (0, 1), \quad (27)$$

with the intensity α_H^j common across countries (data parsimony; reasonable given OECD co-movement of sectoral skill intensity). The composite unit-cost of the labor bundle,

$$w_{\text{comp},n}^j = \left(\frac{w_{H,n}^j}{\alpha_H^j} \right)^{\alpha_H^j} \left(\frac{w_{L,n}^j}{1-\alpha_H^j} \right)^{1-\alpha_H^j}, \quad (28)$$

enters marginal costs, prices, and MP unit costs exactly as the scalar wage w_n^j did in the main text: the trade and MP blocks are untouched. Cobb–Douglas factor demand implies that total labor payments in (n, j) split $\alpha_H(j) : (1 - \alpha_H(j))$ between H and L , so the existing labor share $\gamma_{\text{labor},n}^k$ is unchanged.

Workers of skill $s \in \{H, L\}$ draw Fréchet-distributed idiosyncratic preferences over (origin region, sector) pairs with skill-specific scale parameter κ_s and amenities B_{gf} that are common across skills (distance-based; no data for skill-specific amenities; B_{gf} denotes deterministic amenities here and is distinct from the CES substitution elasticity ξ used in the MP nest). Workers do not switch skill; the endowments $L_{Korea,g}^H$ and $L_{Korea,g}^L$ are fixed at calibration from the 2000 Korean Census. The numeraire is $w_{H,1,1} = 1$ (Korea Seoul/Capital, sector Food, high-skill); $w_{L,1,1}$ floats and pins down the aggregate skill premium.

The price block operators inherited from the main text—the CES sectoral aggregator, the MP-nest pass-through, and the Leontief IO inverse—act on w_{comp} as a single scalar input rather than on w_H and w_L separately. Consequently, the total sensitivity of any sectoral price P_i^k to an individual skill wage factors via the chain rule as

$$\frac{\partial \log P_i^k}{\partial \log w_{s,n}^j} = \sum_{n', j'} \frac{\partial \log P_i^k}{\partial \log w_{\text{comp},n'}^{j'}} \cdot \frac{\partial \log w_{\text{comp},n'}^{j'}}{\partial \log w_{s,n}^j}. \quad (29)$$

Because $w_{\text{comp},n'}^{j'}$ depends on $w_{s,n}^j$ only at $(n', j') = (n, j)$, the sum collapses to a single term with

$\partial \log w_{\text{comp},n}^j / \partial \log w_{s,n}^j = \alpha_H^j$ for $s = H$ and $1 - \alpha_H^j$ for $s = L$. The trade and MP channels of the $S = 2$ model are therefore scalar rescalings of the $S = 1$ operator; skill heterogeneity enters the welfare decomposition only through the migration block, where the Fréchet stay-at-home share $\hat{\mu}_{igg,s}^k$ carries an explicit s -index.

Proposition 2 (Welfare Decomposition, skill-stratified) *Under the Cobb–Douglas skill aggregator, the percentage change in the real wage of group g of skill s in country i is, to first order,*

$$\hat{W}_{ig,s} = \prod_{j,k} (\hat{\lambda}_{ii}^k)^{-\alpha_i^j \tilde{a}_i^{jk} / \theta^k} \prod_{j,k} (\hat{\lambda}_{iii}^k)^{-(1-\rho) \alpha_i^j \tilde{a}_i^{jk} / \theta^k} \prod_{j,k} (\hat{\mu}_{igg,s}^k)^{-\alpha_i^j \tilde{a}_i^{jk} \gamma_i^k / \kappa_s}, \quad (30)$$

where $\hat{\mu}_{igg,s}^k$ is the stay-at-home migration share for skill s ; the trade and MP channels are common across skills because $\hat{\lambda}_{ii}^k$ and $\hat{\lambda}_{iii}^k$ depend only on the composite wage w_{comp} rather than w_s .

Equation (30) is exact for $S = 1$ (collapsing to Proposition 1) and first-order for $S = 2$. Two observations follow. First, the trade and MP channels are *common across skills*, so cross-skill differences in welfare are driven entirely by the migration channel. Second, because the migration coefficient scales inversely with κ_s and $\kappa_H > \kappa_L$ in the data, more-elastic (high-skill) workers generate smaller migration hat ratios $|\log \hat{\mu}_{igg,H}^k| < |\log \hat{\mu}_{igg,L}^k|$, consistent with the Fréchet structure.

Residual bound. The first-order nature of (30) in $S = 2$ reflects a wedge between the skill-specific and composite log wages,

$$\Delta \log w_{s,n}^k - \Delta \log w_{\text{comp},n}^k = \beta_{s,k} \Delta \log \left(\frac{w_{H,n}^k}{w_{L,n}^k} \right), \quad \beta_{H,k} = 1 - \alpha_H^k, \quad \beta_{L,k} = -\alpha_H^k. \quad (31)$$

Propagating (31) through the Leontief IO inverse that produces the trade and MP coefficients in (30) yields an explicit residual

$$R_{g,s} = \sum_{j,k} \alpha_n^j \tilde{a}_n^{jk} \gamma_{\text{labor},n}^k \beta_{s,k} \Delta \log \left(\frac{w_{H,n}^k}{w_{L,n}^k} \right), \quad (32)$$

where n indexes Korea. Two properties keep $|R_{g,s}|$ small. First, $|\beta_{s,k}| \leq \max(\alpha_H^k, 1 - \alpha_H^k) \leq 1$, and the Leontief weight $\alpha_i^j \tilde{a}_i^{jk}$ inherits the spectral-radius-controlled summability of the Caliendo–Parro inversion used in the proof of Proposition 1 (Appendix B). Second, a direct bound follows from the

triangle inequality:

$$|R_{g,s}| \leq \max_k |\Delta \log(w_H^k/w_L^k)| \cdot \sum_{j,k} \alpha_n^j \tilde{a}_n^{jk} \gamma_{\text{labor},n}^k, \quad (33)$$

where n indexes Korea. The first factor is the largest equilibrium skill-premium change across sectors; the second is a Leontief-weighted labor-cost share bounded by $\rho_{\max}(\tilde{A}\Gamma) < 1$. In our calibration, $\max_k |\Delta \log(w_H^k/w_L^k)| = 0.014$ and the Leontief sum equals 0.173, so $|R_{g,s}| < 2.5 \times 10^{-3}$ across all Korean region $\times$ skill cells, consistent with the a-posteriori numerical residual reported in Table 33.

H.3 Parameterization

We set skill intensities $\alpha_H(j)$ to the college+ employment share by 10-sector KSIC aggregation from the 2000 Korean Census. We set skill-specific migration elasticities to $\kappa_L = 1.20$ and $\kappa_H = 1.80$, which bracket the Roy–Fréchet shape parameter $\kappa \approx 1.5$ that Galle et al. (2023) estimate for U.S. commuting-zone-by-group cells; the ordering $\kappa_H > \kappa_L$ captures the well-documented higher spatial mobility of college-educated workers (see also Diamond, 2016, for skill-differentiated spatial sorting in the U.S.). The ordering is consistent with developing-economy estimates: Bryan and Morten (2019) find higher migration elasticities for more-educated workers in Indonesia, and Tombe and Zhu (2019) document similar patterns in China. KLIPS is too thin to identify κ_s separately for H . Regional skill endowments $L_{Korea,g}^s$ are taken from the 2000 Census. For ROW countries we assume the Korean $\alpha_H(j)$ applies uniformly (a homogeneity assumption imposed by ILOSTAT coverage gaps for Taiwan and Mexico). The remainder of the parameterization is identical to Section 4.

H.4 Skill-differentiated welfare gains

Table 33 reports the aggregate and regional welfare gains from the 2000–2007 counterfactual by skill. The aggregate Korean gain decomposes into +3.28% for low-skill and +1.94% for high-skill workers, a high-minus-low differential of -1.34 pp. The high-minus-low gap is most negative in the Southeast (-1.90 pp), the region with the highest automotive and electronics exposure, and least negative in Seoul/Capital (-1.00 pp). Intuitively, Korea’s 2000–2007 integration with China was concentrated in vertically fragmented manufacturing sectors whose Cobb–Douglas labor intensities skew toward low-skill inputs ($\alpha_H(j)$ below one-half), so the price and migration channels transmit

disproportionately to low-skill real wages. Figure 7 visualizes the Korean regional pattern. The pop-weighted $S = 2$ aggregate is +2.44% ($1 \times$ XVEM column of Table 34), defined as the geometric mean of cell-level welfare hats across the 10 Korean region $\times$ skill cells weighted by the 2000 Census population shares $L_{\text{Korea},g,s} / \sum_{g',s'} L_{\text{Korea},g',s'}$ (L/H split $\approx 75/25$ nationally). The skill-aggregate hats +3.28% (low-skill) and +1.94% (high-skill) in Table 33 are a different object: the ratio of Korea-wide skill-specific real income in 2007 vs. 2000 (an income-weighted arithmetic aggregate across regions), and enter the +2.44% only through their region $\times$ skill components. The $S = 1$ baseline of +2.02% in the main text is the analogous pop-weighted geometric mean across the 5 region cells under a representative-worker calibration. The +0.42 pp gap between +2.44% and +2.02% has two compositional sources: (i) the $S = 2$ model re-calibrates Korean technology A_{Korea}^j under the Cobb–Douglas labor aggregator, producing slightly different 2000 sectoral price levels—the $S = 1$ and $S = 2$ solutions therefore sit on different baseline equilibria; and (ii) $\kappa_L < \kappa_H$ amplifies the migration response of the majority low-skill group, whose cell-level hats then dominate the geometric mean. The finding qualifies the aggregate picture: China’s integration raised welfare for both skill groups in all Korean regions, but the gains were regressively distributed.

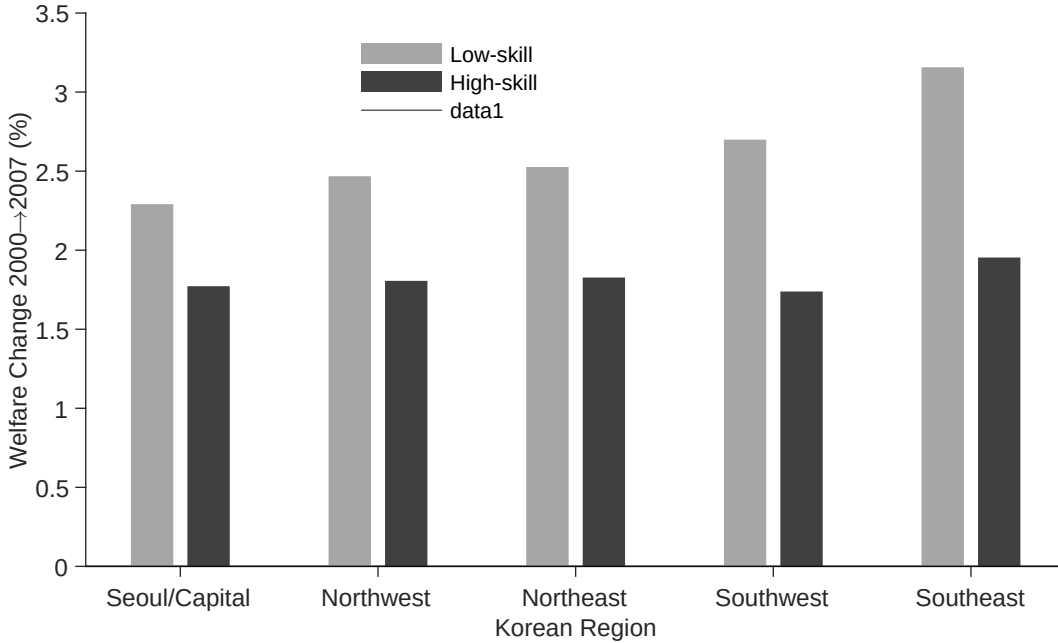


Figure 7: Welfare gains by Korean region $\times$ skill, 2000→2007.

Notes: Light gray: low-skill; dark gray: high-skill. Low-skill gains exceed high-skill gains in every region, with the largest differential in the Southeast (automotive/electronics manufacturing cluster). Source: $S = 2$ GE model; see Table 33.

	Low-skill (%)	High-skill (%)	Difference (H–L) (pp)
<i>South Korea</i>			
Aggregate	3.28	1.94	–1.34
Seoul/Capital	2.90	1.90	–1.00
Northwest	3.18	1.96	–1.21
Northeast	3.24	2.01	–1.23
Southwest	3.51	1.86	–1.65
Southeast	4.03	2.13	–1.90
<i>Other countries</i>			
USA	0.35	0.25	–0.10
Japan	2.29	1.74	–0.55
China	6.14	6.27	0.12
Taiwan	10.32	6.43	–3.89
Germany	1.85	1.53	–0.31
U.K.	0.15	–0.39	–0.54
Canada	0.41	–0.60	–1.01
Mexico	1.29	0.69	–0.60
EU/ROW	0.57	0.36	–0.21

Table 33: **Welfare gains by skill, 2000→2007 China integration.** Columns report percentage welfare change for low-skill (below college) and high-skill (college+) workers; the third column reports the high-minus-low differential in percentage points. Korean regions use the 5-region partition; the Korean aggregate row uses GE income aggregation across regions. Computed from the $S = 2$ GE model re-solved at 2000 and 2007 OECD–ICIO trade and MP costs with $\kappa_L = 1.20$, $\kappa_H = 1.80$, and sector-specific $\alpha_H(j)$ from the 2000 Korean Census. Proposition 2 residual $< 2.5 \times 10^{-3}$ across all Korean region $\times$ skill cells.

H.5 Sensitivity of skill-stratified gains to a^j

Table 34 reports the a^j sensitivity under the $S = 2$ model, spanning $0.5\times$ to $5\times$ the XVEM estimate ($\max a^j \approx 0.42$, far beyond any data-supported level).²⁵

Three patterns emerge. First, the Korean aggregate gain rises monotonically in a^j , from +2.23% at $0.5\times$ XVEM through +2.44% at baseline to +3.79% at $5\times$ XVEM, confirming that the baseline specification is conservative in its MP contribution. Second, the gain is strongly concave in a^j : moving from $1\times$ to $3\times$ XVEM adds +1.09 pp, whereas $3\times$ to $5\times$ adds only +0.26 pp, indicating that the vertical MP channel saturates as affiliate cost dependence on home inputs rises. The $3\times$ XVEM variant ($\max a^j \approx 0.25$) coincides with the upper end of home-input shares observed in OECD AMNE data for MP-intensive countries (8–15% at the sector level), so we view +3.53% as the data-supported upper bound on Korea’s aggregate welfare gain. Third, the skill-regressive incidence widens with a^j : the high-minus-low differential grows more negative from -0.53 pp at $0.5\times$ XVEM through -0.83 pp at baseline to -1.94 pp at $5\times$ XVEM, indicating that larger vertical fragmentation amplifies rather than attenuates the skill gap.

Table 34: Sensitivity to a^j under the skill-stratified ($S = 2$) model. Each column re-calibrates A_{Korea} and re-solves the full 2000–2007 two-equilibrium counterfactual at the indicated a^j . Aggregate and regional rows pool across skills using the 2000 Census regional skill shares; low- and high-skill rows pool across regions within each skill group.

	$0.5\times$ XVEM	XVEM	$2\times$ XVEM	$3\times$ XVEM	$4\times$ XVEM	$5\times$ XVEM
Korea (aggregate)	+2.23%	+2.44%	+3.03%	+3.53%	+3.75%	+3.79%
Seoul/Capital	+2.00%	+2.14%	+2.61%	+3.04%	+3.23%	+3.27%
Northwest	+2.11%	+2.31%	+2.89%	+3.38%	+3.60%	+3.64%
Northeast	+2.18%	+2.39%	+3.00%	+3.52%	+3.74%	+3.77%
Southwest	+2.24%	+2.48%	+3.13%	+3.66%	+3.90%	+3.94%
Southeast	+2.61%	+2.91%	+3.65%	+4.22%	+4.47%	+4.51%
Korea (Low-skill)	+2.36%	+2.65%	+3.39%	+3.99%	+4.24%	+4.27%
Korea (High-skill)	+1.83%	+1.82%	+1.96%	+2.15%	+2.27%	+2.33%
Range (max – min)	0.61 pp	0.78 pp	1.04 pp	1.18 pp	1.24 pp	1.24 pp

²⁵The $a^j = 0$ corner is excluded because the $S = 2$ calibration residual stalls near $\text{diff}_{Korea} \approx 0.019$ around iteration 400 and drifts upward thereafter (to 0.028 by iteration 800), never reaching the 0.012 tolerance required for acceptance. A dedicated low-corner probe (`probe_low_a_j_s2.m`) confirmed that the $0.5\times$ XVEM variant, by contrast, converges cleanly (Korea-target residual = 0.0120 at iteration 409; ROW residual = 0.1196), so it appears as the leftmost column. The analogous $S = 1$ number (+2.11% at $a^j = 0$) is reported in Table 15 of the main text.

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